
Information-based Invariance, Equivariance, and Expressivity for Graph Neural Networks

Anonymous Author(s)

Affiliation

Address

email

Abstract

Graph Neural Networks (GNNs) leverage the topology of graphs to learn informative representations. Since isomorphic graphs represent the same information, there has been a significant effort to develop GNNs that return the same results over isomorphic graphs (invariance/equivariance) and different results over non-isomorphic graphs (expressivity). However, it is known that the same information is often represented in nonisomorphic forms. In this paper, we postulate that GNNs should be invariant/equivariant and expressive with respect to these variations. We introduce the notions of invariance/equivariance of GNNs with respect to these variations and prove that current GNN architectures are not invariant/equivariant to these variations. We propose a general approach to modify a GNN to make it invariant/equivariant to these variations. We also generalize the current notion of expressivity to consider these variations and investigate the expressivity of the invariant/equivariant GNNs. Our empirical studies using multiple GNN architectures over real-world graphs indicate that our approach achieves perfect invariance/equivariance and improves the accuracy of current GNNs on average without imposing significant time overhead.

1 Introduction

Graph data from a wide variety of fields such as chemistry, biology, sociology, transportation, and computer science are now readily available. With this abundance of graph data, there has been great interest in analyzing it for useful patterns, leading to a flurry of activity in the area of graph learning. In particular, Graph Neural Networks (GNNs) are widely used to solve various predictive problems in graphs [25, 79]. They map the features and relationships of the nodes in a graph to a rich vector representation of the graph, which can be used for downstream ML tasks.

Learning representations for graphs is closely connected to the problem of graph isomorphism. Naturally, we want the learned representations to be invariant (or equivariant) under graph isomorphisms [44, 25, 27] and, ideally, to be fully expressive—assigning distinct representations to non-isomorphic graphs [49, 48, 4]. Consequently, many graph representation algorithms, such as GNNs, can be viewed as heuristics for graph isomorphism. In particular, message passing neural networks (MPNNs) are closely related to the well-known Weisfeiler-Leman (WL) test for graph isomorphism [47, 78, 18]. This resemblance guarantees invariance by construction and enables us to use the rich body of techniques developed in the study of the WL hierarchy for graph isomorphism to investigate the expressivity of graph representation algorithms.

However, in several types of graphs, such as knowledge graphs and biological and engineering networks, we need to work with a more general notion of equivalence than graph isomorphism—one that renders much of the theory above not immediately applicable [13, 2, 22, 56, 60, 65, 12, 66, 76].

Here, two graphs are considered equivalent if they represent the same information, even if they are far from being isomorphic. Figure 1 shows two non-isomorphic knowledge graphs that represent essentially the same information, i.e., relationships between a set of objects, in non-isomorphic structures. As another example, organic compounds, e.g., ADP and ATP, are sometimes represented as vertices and other times as edges in metabolic networks [60], resulting in non-isomorphic graphs with the same information. Similar problems appear in encode protein annotations [65]. In the field of circuit design where circuits are represented as graphs, certain transformations of circuit graphs, such as Δ -Y or generally star-mesh, result in non-isomorphic but equivalent circuit graphs [12, 66]. The AI and knowledge representation communities have long recognized that the same information may appear in non-isomorphic structures [9, 31, 24]. This type of equivalence includes isomorphism as a special case, but is often more challenging to define and study with precision.

The prevalence of non-isomorphic but informationally equivalent graphs are well established in knowledge and data management. For example, it is well-known that graph and relational data often contain redundancies, which can be removed (added) by a well-known non-isomorphic and bijective set of mappings in form of switchings called $\bowtie$ (denormalization) [2,9]. Due to their prevalence, these mappings are taught in undergraduate and graduate knowledge and data management courses. Another well-known example is “blank nodes” in knowledge graphs, that can/should often be removed without adding/losing information by bijective mappings ([22], A. Hogan et al, Everything you always wanted to know about blank nodes, J. Web Semantics 2014). A. Hogan et al show that about 45% of public knowledge graphs contain blank nodes. Also, it is shown that relations in graphs often have properties that lead to bijective mappings. For instance, many relations are transitive [e.g., Z. Sun et al, ROTATE: Knowledge Graph Embedding By Relational Rotation in Complex Space, ICLR 2019]. Each transitive property $(a,b), (b,c) \Rightarrow (a,c)$ induces a bijective mapping between the graph that does not have edge (a,c) and the one that has it.

The results of ML tasks must be the same for the input data (training and testing) that contain the same information. Thus, a GNN should learn the same representation over the data graphs that contain the same information. Since current GNNs usually use the topology in a graph to learn its representation, they can learn different representations over nonisomorphic graphs that contain the same information. Thus, GNNs may deliver accurate results in some styles of graph representation and perform poorly in others. As GNN applications grow rapidly, they will face more variations in graph representations, increasing the impact of this shortcoming on the effectiveness of GNNs.

In this paper, we introduce the notions of invariance/equivariance and expressivity for GNNs with respect to these information-based equivalences. We show that the current popular GNN architectures are not invariance/equivariance to this type of equivalence. Because graph learning spans diverse settings (e.g., knowledge vs. biological graphs) and types of architectures, designing invariant/equivariant architectures for each is impractical. Instead, we build theoretically grounded plug-in tools that make current GNNs invariant/equivariant with respect to these equivalences by relatively small changes to their architectures. We prove that these changes are the minimal possible changes to a GNN to achieve this goal. We study the expressivity of the resulting architectures and prove that our changes do not reduce the expressivity of GNNs and strictly increase their expressivity in some cases.

2 Information Equivalence

Basics. Let $\mathcal{V} \subset \mathbb{N}$ be countably finite sets of node ids. A *directed graph* G on $\mathcal{V}$ is a tuple (V, E) where $V \subset \mathcal{V}$ ($V \neq \emptyset$) and $E \subseteq V \times V$. By abuse of notation, we show the set of nodes and edges

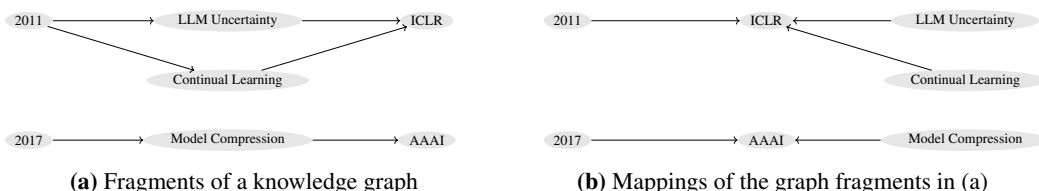


Figure 1: Example of two knowledge graphs whose node labels are *year*, *paper*, and *venue*.

of G as $V(G)$ and $E(G)$, respectively. Each node in a graph is associated with a single member in a finite set of *labels* $\mathcal{L}$. Such labels are commonly available in our applications of interest, particularly in knowledge and biological graphs. For the rest of the paper, we assume that the nodes have always labels (labeled graphs). Our results also apply to graphs without labels by assuming all nodes have the same label. We denote the edge from node u with the label L_1 to node v with the label L_2 as $(L_1(u), L_2(v))$. We also associate each node with a vector in $\mathbb{R}^d$, i.e., its features. The graph $G = (V, E)$ is undirected provided that if $(u, v) \in E$, then $(v, u) \in E$. In this paper, we assume that the graphs are directed. Our results extend to undirected graphs. Two graphs are **isomorphic** if there is a bijection between their nodes that preserves the features and labels of each node and the edges between nodes. Given the graphs P and G , a **homomorphism** from P to G is a mapping from nodes in P to the nodes G that preserves node labels and adjacency (edges). Each *occurrence* of P in G is the homomorphism image of P in G . We use $\text{hom}(P, G)$ to denote the set of occurrences of P in G .

Mappings & Equivalence. Let $\mathcal{G}$ denote the family of graphs and let $\tau : \mathcal{G} \rightarrow \mathcal{G}$ be a bijection. A graph $G \in \mathcal{G}$ is called **τ -equivalent** to a graph $H \in \mathcal{G}$ if $G = \tau(H)$. An example of bijection mapping is *permutation*: let $\mathcal{G}^n$ be the set of graphs with n nodes. Let $[n]$ denote the set $\{1 \dots n\}$. Without loss of generality, we assume that for graph $G \in \mathcal{G}^n$, we have $V = [n]$. The permutation $\pi : \mathcal{G}^n \rightarrow \mathcal{G}^n$ establishes a bijection from $[n]$ to $[n]$ on the node ids, which preserves the node labels, features, and edges between the nodes, i.e., G and $\pi(G)$ are isomorphic. The τ -equivalence captures equivalence relations beyond permutation. For example, let $\mathcal{G}$ be the set of all (finite) graphs with a single label l . Consider the mapping δ that is a bijection on nodes and preserves the node ids, labels, and features and for every two nodes $v, u \in V(G)$ if $(l(v), l(u)) \in E(G)$, we have $(l(u), l(v)) \in E(\delta(G))$, i.e., δ reverses the directions of the edges of G . δ is bijective since for each G there is one and only one $\tau(G)$ so G and $\delta(G)$ are δ -equivalent. Similarly, the mapping on the set of all graphs with labels $\{l_1, l_2\}$ which is a bijection on nodes and preserves their id/label/features and reverses the edges of nodes with the label l_1 is bijective.

Example 1. Figure 1 shows fragments of two graphs on the relationships between papers, venues, and their years. The graph in Figure 1(a) connects the year of a venue to the papers published in the venue, while the graph in Figure 1(b) connects these years directly to the venues. Intuitively, the information represented in each graph can be recovered from the other one. More precisely, let $\mathcal{G}$ be the set of all connected graphs with labels $\{\text{paper}, \text{venue}, \text{year}\}$ with at least one node from each label. Let $\beta : \mathcal{G} \rightarrow \mathcal{G}$ map each node $v \in V(G)$ to one and only one node $\beta(v) \in V(\beta(G))$ such that v and $\beta(v)$ have the same node id/label/features. For every $u, v, w \in V(G)$ if we have edges $(\text{year}(u), \text{paper}(v))$ and $(\text{paper}(v), \text{venue}(w))$ in G , β maps them to $(\text{year}(\beta(u)), \text{venue}(\beta(w)))$ and $(\text{paper}(\beta(v)), \text{venue}(\beta(w)))$. β preserves all other edges between $v \in V(G)$ and $\beta(v)$ that do not satisfy the preceding condition in G . It is easy to see that β is bijective.

Let $\mathcal{G}^n$ be the set of graphs with n nodes. Let $\mathcal{T}^n$ be a set of bijective mappings (mappings) from $\mathcal{G}^n \rightarrow \mathcal{G}^n$ that include identity mapping Id^n and for every $\tau \in \mathcal{T}^n$, $\tau^{-1} \in \mathcal{T}^n$. We define the group $\Gamma^{\mathcal{T}^n} = (\mathcal{T}^n, \circ)$ where $\circ$ is the composition of mappings. Let Π^n be the permutation group on $\mathcal{G}^n$ with composition operator $\circ$, specifically for $\pi \in \Pi^n$ and $G \in \mathcal{G}^n$, $\pi(G)$, just relabels the vertices of G without changing its edge. Let $\Delta^{\mathcal{T}^n}$ be the union of $\Gamma^{\mathcal{T}^n}$ and Π^n . $\Delta^{\mathcal{T}^n}$ is a group with $\circ$ whose identity element is Id^n . The action of group $\Delta^{\mathcal{T}^n}$ on the set $\mathcal{G}^n$ is a function from $\Delta^{\mathcal{T}^n}$ and $\mathcal{G}^n$ to $\mathcal{G}^n$. The orbit of a graph $G \in \mathcal{G}^n$ is the set of graphs obtained by the action of every member of $\Delta^{\mathcal{T}^n}$ to G . Graphs H and J are equivalent if they belong to the orbit of the same graph.

It is straightforward to check that the τ -equivalence is an equivalence relation in $\mathcal{G}$ and therefore; partitions $\mathcal{G}$ into equivalence classes. For the rest of the paper, we refer to bijective mappings simply as mappings. Let $\mathcal{T}$ be a set of mappings on $\mathcal{G}$. We say G is $\mathcal{T}$ -equivalent to H if there is a mapping $\tau \in \mathcal{T}$ such that G is τ -equivalent to H . The set of mappings is closed under composition $\circ$, i.e., the composition of bijective mappings is a bijective mapping. Hence, if G is τ -equivalent to H and H is ρ -equivalent to F , G is $\tau \circ \rho$ -equivalent to F .

Representing Mappings. Since mappings often modify the structure of the graph, we are interested in the case where mappings are defined via a collection of **switchings** specified by pairs of labeled (pattern) graphs, $\tau = \{(P_i, P'_i)\}_{i=1}^t$, where there is a mapping between the nodes in P_i and P'_i that preserves the node id and label. Then $\tau(G)$ is obtained by replacing all occurrences of P_i in G with P'_i for every $i \in [t]$. More precisely, graph G is mapped to graph H by switching (P_i, P'_i) if and only if there is bijection b_i between the sets of subgraphs $\text{hom}(P_i, G)$ and $\text{hom}(P'_i, H)$ such that

for every $g \in \text{hom}(P_i, G)$, there is a bijection between nodes of g and $b_i(g)$ that preserve node ids, labels, and features. Given mapping $\tau = \{(P_i, P'_i)\}_{i=1}^t$, the graph G is mapped to graph H using τ iff G is mapped to H using every (P_i, P'_i) and each node in G appears in at least one occurrence of a graph P_i . For example, the mapping δ explained above, which reverses the directions of the edges of a graph, is represented by a single switching (P_1, P'_1) where P_1 is $(l(v), l(u))$ and P'_1 is $(l(u), l(v))$.

Example 2. Consider the graphs in Figure 1 and the mapping β in Example 1. A switching in β is (P_1, P'_1) , where P_1 is the graph with edges $(\text{year}(u), \text{paper}(v))$ and $(\text{paper}(v), \text{venue}(w))$ and P'_1 is the graph with edges $(\text{year}(u), \text{venue}(w))$ and $(\text{paper}(v), \text{venue}(w))$.

Switchings arise frequently in graph applications and theory: Δ - Y transforms [12, 66], Whitney twisting [74, 75, 46, 61], Seidel switching [59, 67], Godsil-McKay switching (yielding cospectral graphs) [20, 23], and edge flips in triangulations.

Every invariant DNN, e.g., GNN, is designed for a priorly known group of transformations. Similarly, our algorithm is invariant to every member of the group of bijective mappings. Our approach introduces a novel and modular method to inject domain invariances to GNNs. These mappings can be both specified manually, and hence enable a principled approach to incorporating domain knowledge, or extracted from data. Specifically, the properties of relations in the graph, e.g., symmetry or transitivity, induce bijective mappings, e.g., given $(a,b) \Rightarrow (b,a)$ for a relation, there is a bijective mapping between the graph representation with (a,b) and the one with (b,a) . As another example, the mapping illustrated in Figure 1 of Page 2 is induced by rule $(\text{year}(v1), \text{paper}(v3))(\text{paper}(v3), \text{venue}(v4))(\text{paper}(v2), \text{venue}(v4)) \rightarrow (\text{year}(v1), \text{paper}(v2))$ on the left-hand side graph. These rules can be learned from the data, e.g., M. Qu et al, Rnnlogic: Learning logic rules for reasoning on knowledge graphs, ICLR-2021. We can construct mappings algorithmically based on the learned rules.

To ensure the bijection of τ , we have the following.

Proposition 1. Given bijective mapping $\tau = \{(P_i, P'_i)\}_{i=1}^t$, for all $i \neq j$, if P_i is isomorphic to P_j , then P'_i is isomorphic to P'_j .

If a set of switchings τ follows the condition stated in lines 111-116 (Page3) for a set of graphs $\mathcal{G}$, it defines a unique bijective mapping over $\mathcal{G}$. Intuitively, the condition of mappings under switching (lines 111-116), e.g., node id preservation, ensures that the nodes in to the occurrences of two distinct patterns P_i and P_j of τ in G are mapped to same nodes in $\tau(G)$ and avoids creating non-recoverable duplicates $\tau(G)$. P_i and P_j can have overlapping edges in G . We prove that this condition guarantees bijectivity of the mapping defined by the switching as follows.

■ Every set of switchings that satisfy the above condition defines a unique bijective mapping.

Proof: Given the switching $\tau = \{P_i, P'_i\}_{i=1}^t$, we use the Chase algorithm from the data/graph transformation literature [e.g., P. Barcelo et al, Schema Mappings and Data Exchange for Graph Databases, ICDT-13] to implement the mapping for a given set of switching τ . More precisely, to compute $\tau(G)$, Chase starts with an empty graph G'_k , $k = 0$. In step $k > 0$, for each $\{P_i, P'_i\} \in \tau$, Chase finds every homomorphism image of P_i in G (p_i), computes the corresponding isomorphic subgraph to P'_i based on switching $\{P_i, P'_i\}$, p'_i . The node id, label, and features of p'_i is set using node id, label, and features of p_i to follow the condition stated in lines 111-116 (Page 3). It sets $G'_k = G'_k \cup p'_i$. The algorithm ends when there is no application of any switching left with G_k as output. We represent each switching $\{P_i, P'_i\}$ using a first order logic formula over G in the form $\phi(\bar{x}) \rightarrow \psi(\bar{y})$, where ϕ (ψ) is a conjunction of binary predicates. Each predicate in ϕ (ψ) corresponds to an edges in P_i (P'_i) and expresses a relationship between two labeled nodes, e.g., $(R(x_1), S(x_2))$. Each variable $x \in \bar{x}$ and $y \in \bar{y}$ denotes a distinct node in P_i (P'_i). Due to the definition of mappings under switchings, $\bar{y} = \bar{x}$ and all variables in ϕ and ψ are universally quantified. The termination of the algorithm follows from Theorem 5.2 in [P. Barcelo et al, Schema Mappings and Data Exchange for Graph Databases, ICDT-13]. The uniqueness follows from [R. Fagin et al, Data Exchange: Semantics and Query Answering, Theoretical Computer Science-05]. Since the conditions of a mapping expressed using a set of switching is symmetric, these results also hold for the switching $\{P'_i, P_i\}_{i=1}^t$ over $\tau(G)$ as input. Hence, τ defines a unique bijective mapping ■

Switchings $\tau = \{(P_i, P'_i)\}_{i=1}^t$ is *minimal* if there is no switching $i \neq j$ where there are isomorphisms between P_i (P'_i) and P_j (P'_j) that preserve labels. In this paper, we consider only minimal switchings.

3 Invariance & Equivariance

Representation Learning. GNNs learn low-dimensional representations of graphs that should exhibit properties expressed in the graph. The learned representation is often expressed as $m > 0$ -dimension feature vectors on the nodes in the graph. Given the graph $G = (V, E)$ of n nodes with a matrix representation that encodes its label and feature information $A \in \mathbb{R}^{n \times n \times (d+1)}$, we denote the learning of the representation by the function f from $\mathbb{R}^{n \times n \times (d+1)}$ to $\mathbb{R}^{n \times m}$. For each node $v \in V$ with id i , the row i in $\mathbb{R}^{n \times m}$ contains the learned representation for v . Sometimes, GNNs learn a scalar value for the input, e.g., graph classification so GNN is a function from $\mathbb{R}^{n \times n \times (d+1)}$ to $\mathbb{R}$.

τ -Invariance & τ -Equivariance. The definition of mapping over graphs naturally extends to their matrix representations. Given the mapping τ , a function $f : \mathbb{R}^{n \times n \times (d+1)} \rightarrow \mathbb{R}^{n \times m}$ is **τ -invariant** if $f(\tau(A)) = f(A)$ for every $A \in \mathbb{R}^{n \times n \times (d+1)}$, i.e., the transformation of the input does not affect the learned representation. The definition naturally extends to the functions that map the input matrix to $\mathbb{R}$. Each permutation on the family of graphs with n nodes can be represented by a $1 \times n$ matrix [42]. Let matrix P represent the permutation mapping π . Let mapping $\tau = \pi \circ \rho$ where π and ρ are a permutation and a non-permutation mapping, respectively. The function $f : \mathbb{R}^{n \times n \times (d+1)} \rightarrow \mathbb{R}^{n \times m}$ is **τ -equivariant** if $f(\tau * A) = \pi * f(A)$ for every $A \in \mathbb{R}^{n \times n \times d}$, i.e., the output is transformed when the input is transformed. If f is both τ -invariant and τ -equivariant, we call f τ -invariant/equivariant. Given a set of mappings $\mathcal{T}$, a function f is $\mathcal{T}$ -invariant ($\mathcal{T}$ -equivariant) iff it is τ -invariant (τ -equivariant) for all $\tau \in \mathcal{T}$.

τ -Invariance/Equivariance under Compositions. Since compositions of mappings may generate new mappings, we are interested in invariance/equivariance under the composition of mappings. Also, as permutations and (non-permutation) mappings preserve the information in the graph, we would like to design architectures that are invariant/equivariant to both permutations and the mappings represented by switchings. Following the definition of mappings, if f is τ -invariant (τ -equivariant), it will be τ^{-1} -invariant (τ^{-1} -equivariant). Since mappings are bijective, τ -invariance and τ -equivariance are preserved under the composition of mappings. Given a set of graphs $\mathcal{G}$ and a set of mappings $\mathcal{T}$ on $\mathcal{G}$, $\mathcal{T}$ is closed in the inverse if for every $\tau \in \mathcal{T}$, we have $\tau^{-1} \in \mathcal{T}$. Given the set of mappings $\mathcal{T}$ that is closed under inverse, let $\Gamma^{\mathcal{T}}$ be a set of all mappings constructed from mappings in $\mathcal{T}$ using (finite) compositions. Let $\mathcal{G}^n$ denote the set of graphs with n nodes, and Π^n be the set of all permutations on $\mathcal{G}^n$. $\mathcal{G} \subseteq \mathcal{G}^n$ is *closed under isomorphism* if for every $G \in \mathcal{G}$ and for all H isomorphic to G , $H \in \mathcal{G}$.

Proposition 2. *Given an isomorphic-closed $\mathcal{G} \subseteq \mathcal{G}^n$ and a set of mappings $\mathcal{T}$ on $\mathcal{G}$, $\Gamma^{\mathcal{T}} \cup \Pi^n$ is a group with composition on $\mathcal{G}$.*

Let τ be a mapping on an isomorphism-closed set $\mathcal{G}$. Since permutations are bijective, if f is invariant (equivariant) with respect to permutation $\pi \in \Pi^n$ and τ -invariant (τ -equivariant), it is $\tau \circ \pi$ -invariant ($\tau \circ \pi$ -equivariant). Since prominent GNNs are invariant to permutations (Π^n) [25], using Proposition 2, it is sufficient to ensure that they are τ -invariant/equivariant to show that they are invariance/equivariant to compositions and permutations and (non-permutation) mappings.

τ -Invariance/Equivariance of Current Architectures. There are various neural architectures for learning graph representations [34, 25, 79, 10, 8]. *Message passing neural networks* (MPNN) are a prominent architecture for GNN's that use the graph structure to learn representations [25]. Let $h_v^{(l)}$ represent the feature of node v at layer l . Given GNN f , graph G , and $v \in V(G)$, let $\mathcal{M}^f(G, v)$ be a permutation equivariant function that returns a non-empty and proper subset of $V(G)$. Many node-centric architectures f extend Equation 3 by sharing information between each node $v \in V(G)$ and $\mathcal{M}^f(G, v)$ in each layer. Each layer of an MPNN is composed of two learnable functions, the permutation invariant aggregate function and the update function:

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{AGGREGATE} \left(\{h_u^{(l)} : u \in \mathcal{M}^f(G, v)\} \right) \right). \quad (1)$$

$\mathcal{M}^f(G, v)$ is often defined using the structural properties of the connections between v and $u \in \mathcal{M}^f(G, v)$, e.g., nodes on all simple or shortest paths from v whose lengths are at most $K > 0$ [45, 21, 62] or $k \leq K$ -hop neighbors of v where $K = 1$ recovers *basic MPNN* [51, 1, 14].

Given architecture f , mapping τ , graph G , and node $v \in V(G)$, let UPDATE and AGGREGATE return different results for $\mathcal{M}^f(G, v)$ and $\mathcal{M}^f(\tau(G), v)$ if $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(\tau(G), v)| \neq |\mathcal{M}^f(G, v)|$. These are mild conditions and are often satisfied by typical AGGREGATE and UPDATE functions. In this case, we have the following result.

Proposition 3. *Consider the architecture f with $\mathcal{M}^f(G, v)$ in Equation 1. There is a mapping τ such that f is not τ -invariant/equivariant.*

This result extends to MPNN architectures that learn representations for subgraphs of the input and aggregate the results (*Subgraph MPNN*) [54, 5, 15] and those that learn representations for tuples of nodes *Higher-order MPNN* [49, 48, 4, 39, 84, 29] (Details in Appendix D). In this paper, we use GNN and MPNN interchangeably.

4 τ -Invariant/Equivariant GNNs

Given mapping τ , to create a τ -invariance/equivariance GNN, one approach is to produce $\tau(G)$ and average the output of f on G and $\tau(G)$ [73]. This method quickly becomes costly, especially for rather large graphs and/or a set of mappings. An alternative is canonicalization: map equivalent graphs to a common form, e.g., symmetry in images [40]. It demands heavy (manual) pre-processing and it is unclear whether natural forms exist that preserve accuracy. Next, we explain our approach to achieve invariance/equivariance.

4.1 Structural Bias

We first present a general approach to defining hypothesis spaces based on the structure of the graph that generalizes how many (node-centric) GNNs define their hypothesis spaces. This abstraction enables us to investigate and instill invariance/equivariance for a large set of GNNs.

A *rooted graph* $G^v = (V, E)$ is a graph with a single distinguished node $v \in V$. The definitions of isomorphism and homomorphism for rooted graphs G^v and G^u extend their definitions for graphs by always mapping v to u . The **structural bias** of GNN f is a finite and non-empty set of rooted and connected graphs $\mathcal{S}^f$ such that there are no two graphs $S, S' \in \mathcal{S}^f$ where S and S' are isomorphic. Intuitively, f shares information between nodes in the occurrences of graphs in $\mathcal{S}^f$ in G [41, 50, 32].

Example 3. *Consider the MPNN in Equation 1 that shares the information of each node with its 1-hop neighbors [36, 78, 42, 49, 71]. The structural bias of the MPNN architecture (Equation 1) over directed graphs where all nodes share the same label l has a single rooted graph $(l(\mathbf{u}), l(w))$ with root $\mathbf{u}$. As another example, GNNs based on meta-paths [16, 72] learn the representation of a node by sharing and aggregating information over all nodes in a set of paths that start from the node. The structural bias for these methods is a set of rooted paths. For example, a structural bias for these methods in Figure 1(a) is $\{(year(\mathbf{u}), paper(w), venue(z))\}$ with root $\mathbf{u}$.*

The graphs in the structural bias of an algorithm are significantly smaller than G [41, 50, 6, 32, 3]. Given graph G , let G^v be a rooted version of G for a $v \in V(G)$. Each layer of a GNN with structural bias $\mathcal{S}^f$ is composed of three learnable permutation-invariant functions of pattern-aggregation (PAGG), total-aggregation (TAGG), and update such that for every $S^u \in \mathcal{S}^f$:

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{TAGG} \left(\{ \{ (\text{PAGG}(\text{hom}(S^u, G^v), S^u), S^u) \} \} \right) \right). \quad (2)$$

The PAGG function generalizes the aggregation function in MPNNs (Equation 1): it maps the information of the set of occurrences of graph $S^u \in \mathcal{S}^f$ in G^v for each node $v \in V(G)$ and S^u to a vector in $\mathbb{R}^m$. The TAGG function maps the results of PAGG for all patterns in $\mathcal{S}^f$ (a multi-set of vectors in $\mathbb{R}^m$) to a vector in $\mathbb{R}^m$. The UPDATE function is similar to that in Equation 1. The exact instantiations of $\mathcal{S}^f$, PAGG, TAGG, and UPDATE depend on the specific GNN.

The definition of structural bias captures the inductive bias of using the structure of the graph to share information in many GNNs [41, 50, 6, 32, 3]. Consider again the MPNN in Example 3 that shares information of a node with its 1-hop neighbors. For each node $v \in G$, the set of subgraphs $\text{hom}(S^u, G^v)$ contains every path of one edge from v to all its neighbors, which enables PAGG to access the information of 1-hop neighbors of v . The PAGG function is the AGGREGATE function in Equation 1. Since $\mathcal{S}^f$ has only one graph, TAGG maps a set with a single vector $\mathbf{v}$ to $\mathbf{v}$. The

UPDATE function is the same as the one in Equation 3. Details on how Equation 2 generalizes other types of GNNs are in Appendix F. We do not use this abstraction to implement GNNs as more efficient implementations for a fixed structural bias are often possible e.g., MPNNs in Example 3.

Because mapping τ modifies the structure of the graph G , the pattern of information propagation of a GNN f may vary between G and $\tau(G)$, therefore, f may not be τ -invariant. Intuitively, we would like f to have a structural bias that for every pattern of information propagation in G there is an equivalent pattern in $\tau(G)$. Formally, let the mapping τ for rooted graphs naturally extend its definition for graphs by mapping the roots of the graphs, i.e., $\tau(G^v) = \tau(G)^{\tau(v)}$. Given mapping τ , bijection $b_\tau : S^f \rightarrow S^f$ is **isomorphism-preserving** such that for all G and $v \in V(G)$, there is a bijection between $\text{hom}(S^u, G^v)$ and $\text{hom}(b_\tau(S^u), \tau(G^v))$ that maps every member of $\text{hom}(S^u, G^v)$ to an isomorphic member in $\text{hom}(b_\tau(S^u), \tau(G^v))$ where the mapping preserves node ids/labels/features.

Lemma 1. *Given mapping τ , a GNN f with pattern aggregation function PAGG is τ -invariant if there is an isomorphism-preserving mapping b_τ such that $\text{PAGG}(\text{hom}(S^u, G^v), S^u) = \text{PAGG}(\text{hom}(b_\tau(S^u), \tau(G^v)), b_\tau(S^u))$ for every G^v .*

If we have an isomorphism-preserving mapping on S^f , following Lemma 1, the PAGG function will get essentially the same information. Thus, without significantly restricting possible instantiations of PAGG, we can achieve invariance. However, since τ modifies the topology of the graph, it is not possible to achieve τ -invariance with the current approaches to structural bias in GNNs without using PAGG functions that return the same results for non-equivalent patterns, which is often not desirable.

Proposition 4. *For every GNN f with structural bias S^f , there exists a mapping τ that admits no isomorphism-preserving mapping ($b_\tau : S^f \rightarrow S^f$).*

4.2 Extending Structural Bias with Conditional Graphs

Following Proposition 4, we should modify the current approach to structural bias to achieve invariance/equivariance. Ideally, we should change the structural bias in such a way that preserves its original one so as not to miss an accurate hypothesis. We should also not expand its structural bias significantly to keep the time overhead limited. To do so, we take into account how a mapping modifies a graph and reflect that information in the structural bias. That is, for each $S^u \in S^f$, for each $h \in \text{hom}(S^u, G^v)$, we find a h' in $\tau(G^v)$ such that h and h' are isomorphic to have exactly the same pattern of information propagation. We then find a structure R^w where $\text{hom}(R^w, G^v)$ represents exactly the set of all such h' . If we add R^w to S^f for every S^u we obtain an isomorphism-preserving mapping. There are two challenges to address in this approach: first, we should identify R^w and then ensure that this algorithm converges without adding too many members to S^f . Since our results apply to both invariance and equivariance properties, for brevity, we explain them to obtain invariance.

Conditional Graphs. To address the first challenge, we generalize the idea of structural bias based on exact pattern homomorphism. More precisely, for each $S^u : (V^S, E^S) \in S^f$, let E^S have two mutually exclusive and collectively exhaustive subsets: *context edges* (E_C^S) and *propagation edges* (E_P^S). We call the rooted subgraphs of S^u whose sets of edges are E_C^S and E_P^S the *context subgraph* (S_C^u) and the *propagation subgraph* (S_P^u), respectively. S_P^u and S_C^u share the same root, and $V(S_P^u) \subseteq V(S_C^u)$. In this case, we call S^u a *conditional graph* (CG). For the rest of this section, we assume that a structural bias is a set of CGs.

Information Propagation Using CG. Given GNN f , we modify Equation 2 so that for the rooted graph G^v and CG S^u , f checks the existence of the context subgraph S_C^u in G^v and uses the propagation subgraph S_P^u to share information between v and other nodes in G^v . More precisely, given the rooted graph G^v and the CG S^u , let $g_C \in \text{hom}(S_C^u, G^v)$ be an occurrence of S_C^u in G^v using the homomorphism mapping $\mathbf{h}$. Let g_P be a graph homomorphic to S_P^u created by $\mathbf{h}$. Since $V(S_P^u) \subseteq V(S_C^u)$, we can always generate such a g_P for every $\mathbf{h}$. We denote the set of generated graphs using all $\mathbf{h}$ from S_C^u to G^v as $\text{pom}(S^u, G^v)$. We use $\text{pom}(S^u, G^v)$ instead of $\text{hom}(S^u, G^v)$ as the first argument of PAGG in Equation 2. In current structural biases, we have $S^u = S_C^u = S_P^u$ for $S^u \in S^f$, therefore, structural biases with CGs generalize current structural biases.

Example 4. *A CG for a GNN that shares information between 2-hop neighbors in Figure 1(b) is $R_C^u : (\text{year}(\mathbf{u}), \text{venue}(v)), (\text{paper}(w), \text{venue}(v))$ and $R_P^u : (\text{year}(\mathbf{u}), \text{paper}(w))$. For every homomorphism of R_C^u in Figure 1(b), the GNN propagates information between the nodes in R_P^u .*

4.3 Minimally Modifying Structural Bias

Given the mapping τ , we propose an algorithm that modifies the structural bias of a GNN f to a set of CGs $\mathcal{S}_\tau^f$. For each switching $(P_i, P'_i) \in \tau$, $1 \leq i \leq t$, and each $S^u \in \mathcal{S}^f$, it follows three main steps.

1. Mapping Structural Bias Using Switchings. Assume that there is an isomorphism mapping from a switching P_i to a connected subgraph of $S^u \in \mathcal{S}^f$ that preserves the node labels. In this case, we can capture how (P_i, P'_i) modifies S^u by treating S^u as an input graph and applying (P_i, P'_i) to S^u to generate a new graph R_i^u . The algorithm adds S^u and R_i^u to $\mathcal{S}_\tau^f$, where all edges in S^u and R_i^u are both context and propagation edges. For example, consider a switching that reverses the direction of all edges in an input graph (ignoring labels for simplicity): $P_1 = (x, y)$ and $P'_1 = (y, x)$ and the structural bias for the MPNN that shares the information of each node with its 1-hop neighbors: $S^u = (\mathbf{u}, w)$. Applying the switching in this case, we will add $S^u = (\mathbf{u}, w)$ and $R_1^u = (w, \mathbf{u})$ to $\mathcal{S}_\tau^f$.

2. Using Switchings as Context Subgraphs. If there is no isomorphism mapping from P_i to any subgraph of S^u , the above step cannot accurately represent how a mapping modifies structural bias. For example, consider the switching in Figure 1 and the structural bias explained in the preceding paragraph. Because there is no isomorphism mapping from the switching to a subgraph of the member in the structural bias, we cannot use the previous step to represent how the input structural bias must be modified based on the switching. Hence, we use the switchings as context subgraphs and represent the input structural biases as propagation subgraphs in these contexts. More precisely, we construct the set of structural biases $\mathbf{R}_i^f = \bigcup_{v \in V(P_i)} \{P_i^v, P_i'^v\}$. The new structural bias must preserve the original method of propagating information represented by S^u . For every $R^w \in \mathbf{R}_i^f$, let b be an isomorphism mapping from S^u to R^w that preserves the node labels. We mark the edge (y, z) in R^w as propagation edges if $(b^{-1}(y), b^{-1}(z))$ exists in S^u . All edges in S^u and members of $\mathbf{R}_i^f$ are marked as context edges. After this step, each $\mathbf{R}_i^f$ contains the original method of propagating information in S^u . We add $\{S^u\} \cup \mathbf{R}_i^f$ to $\mathcal{S}_\tau^f$ and mark all edges in S^u as context and propagation.

3. Mapping Information Propagation Patterns. In the preceding step, $\mathbf{R}_i^f$ has not yet captured how information propagation should be modified according to switching (P_i, P'_i) . Thus, we add new rooted graphs to $\mathbf{R}_i^f$ as follows. Let p_i be the node id and label preserving mapping from P_i to P'_i (explained in Section 2). For every $R^u \in \mathbf{R}_i^f$ with the context graph P_i^u , we add a rooted graph to $\mathcal{S}_\tau^f$ with the context graph P_i^u and propagation edges (y, z) iff there is a propagation edge $(p_i(y), p_i(z))$ in R^u . This step ensures that the modified structural bias contains the propagation patterns represented by P_i^u for graphs that contain P_i^u . Similarly, for each $R^u \in \mathbf{R}_i^f$ with the context graph P_i^u , we add a rooted graph to $\mathcal{S}_\tau^f$ whose context graph is $P_i'^u$ and has a propagation edge (y, z) iff there is a propagation edge $(p_i^{-1}(y), p_i^{-1}(z))$ in R^u .

The definition of isomorphic CGs extends the definition of isomorphism for rooted graphs that preserves context and propagation types of edges.

Theorem 1. $\mathcal{S}_\tau^f$ does not have isomorphic CGs and GNN f with structural bias $\mathcal{S}_\tau^f$ is τ -invariant.

$\mathcal{S}_\tau^f$ is also *minimal*, ensuring the scalability of the training. Formally, given structural biases $\mathcal{S}$ and $\mathcal{S}'$, we define $\mathcal{S} \preceq \mathcal{S}'$ if for every $S^u \in \mathcal{S}$ there is a $S'^w \in \mathcal{S}'$ such that there is an isomorphism mapping from S^u to a subgraph of S'^w that preserves the node labels and the type of edges (context/propagation). We have $\mathcal{S} \prec \mathcal{S}'$ if at least one isomorphism mapping is from a $S^u \in \mathcal{S}$ to a proper subgraph of a $S'^w \in \mathcal{S}'$.

Proposition 5. Given GNN f and mapping τ , there is no structural bias $\mathcal{S}'$ such that f is τ -invariant using $\mathcal{S}'$ and $\mathcal{S}' \prec \mathcal{S}_\tau^f$.

The resulting structural bias is also complete: if the original GNN f learns representation of every node in graph G , f using $\mathcal{S}_\tau^f$ also learns the representation of every node in G (Details in Appendix G).

The algorithm requires to compute sub-graph isomorphism between graphs used in the structural bias and switchings. These graphs usually have a very few nodes and often in rather simple forms, e.g., paths. Hence, these isomorphisms can be computed efficiently. Ignoring those steps, the time complexity of the algorithm for GNN f and mapping τ is quadratic in the cardinality of $\mathcal{S}^f$ and the number of switchings in τ . These are also often quite small. For example, the cardinality of structural bias (ignoring labels) in MPNN that shares information between 1-hop neighbors is one.

5 τ -Expressivity

Another important desirable property with respect to an equivalence relation between graphs is for GNN f to learn distinct representations over non-equivalent graphs, i.e., *expressivity* [78, 49]. Current works define this property by considering two graphs equivalent if they are isomorphic [78, 49, 18]. Since both isomorphism and bijective mappings preserve the information in graphs, we define expressivity based on both. Formally, let $\mathcal{G}^n$ denote the set of graphs with n nodes and $\mathcal{G} \subseteq \mathcal{G}^n$ be an isomorphic-closed set of graphs. Let τ be a mapping on $\mathcal{G}$, Π^n denote the set of all permutations on $\mathcal{G}$, and Γ^τ be the set of all mappings constructed by (finite) compositions of τ and τ^{-1} . Following Proposition 2, $\Phi_n^\tau = \Gamma_n^\tau \cup \Pi^n$ is a group with composition on $\mathcal{G}$. f is **more τ -expressive** than f' ($f' \preceq_\tau f$) if $f'(G) \neq f'(H)$, then $f(G) \neq f(H)$ for all graphs $G \neq \rho(H)$ for all $\rho \in \Phi_n^\tau$. f is **strictly more τ -expressive** than f' ($f' \prec_\tau f$) if f is more τ -expressive than f' and there are G and $H \neq \rho(G)$ for all $\rho \in \Phi_n^\tau$ such that $f(G) \neq f(H)$ and $f'(G) = f'(H)$.

Maintaining Expressivity. It might reduce the τ -expressivity of a GNN to modify its structural bias. For example, if we modify the MPNN in Equation 2 to ignore the structure of the graph and aggregate the representation of all nodes in the input graph with v in each layer, the resulting GNN is invariant to all mappings τ . Nevertheless, this GNN will have poor τ -expressivity as it learns the same representation for all non-equivalent graphs. Hence, it is important to ensure that the modifications to the structural bias in the algorithm of Section 4.3 of a given GNN do not reduce its expressivity. Given a GNN f with structural bias S^f , we would like to know whether the GNN f_τ (f with structural bias S_τ^f) is (at least) as expressive as f for every mapping τ . Following assumptions in the theoretical analysis of expressivity with regard to isomorphism [78, 47, 18], we assume that the functions PAGG, TAGG, and UPDATE in Equation 2 in the input GNN to the algorithm in Section 4.3 are injective. This is a standard and established assumption in theoretical analysis of GNN expressivity [49, 78]. Since we compare our notion of expressivity with the state-of-the-art, we make the same assumption.

Theorem 2. For all mappings τ and GNN f , we have $f \preceq_\tau f_\tau$. Also, there are mapping τ and GNN f such that $f \prec_\tau f_\tau$.

The structural bias of our architecture encodes the minimal set of shared information across all variations so its learned representation is at least as informative as the best-performing variant. The accuracy gains often occur when the conditional graph makes task-relevant evidence consistently accessible across graph realizations. For example, a dependency that requires several message-passing layers in one representation may be exposed as a direct relation in another. This explains larger gains for architectures whose native structural bias is sensitive to path length or local topology, such as RGCN on DBLP/IMDb.

Maintaining Relative Expressivity. Another important question is whether τ -invariance maintains the relative expressivity of f . More precisely, let $f \preceq f'$ denote that f' is more expressive than f only with respect to isomorphism (permutations). Given mapping τ and GNNs f and f' , we investigate the relative expressivity of f_τ and f'_τ based on their relative expressivity with respect to the isomorphism.

Proposition 6. Given mapping τ and GNNs f and f' such that $f \preceq f'$, we have $f_\tau \preceq_\tau f'_\tau$.

6 Empirical Study

Datasets. We use the DBLP and IMDb datasets from [16]. DBLP contains 4,057 authors, 14,328 papers, 7,723 terms, and 20 venues, with a 4-class author research-area classification task. IMDb contains 4,278 movies, 5,257 actors, and 2,081 directors, with a 3-class movie genre classification task (Action, Comedy, Drama).

Dataset Variants. To evaluate invariance, we construct equivalent mapping variants (Figure 2). For DBLP, we add explicit *Area* nodes to create three variants and evaluate Paper–Venue link prediction. For IMDb, we add a *Link* node type to create four variants and evaluate the original movie classification task together with Movie–Link and Movie–Director link prediction.

Methods. We evaluate five baselines spanning diverse types of structural biases: local node-type-pair message passing (RGCN [57]), source-conditioned path propagation (CMPNN [29]), local node-type-aware message passing (SlotGAT [83]), metapath-based message passing (MAGNN [16]), and metapath-based feature propagation (SeHGNN [72]). This follows recent causal evidence [80] that

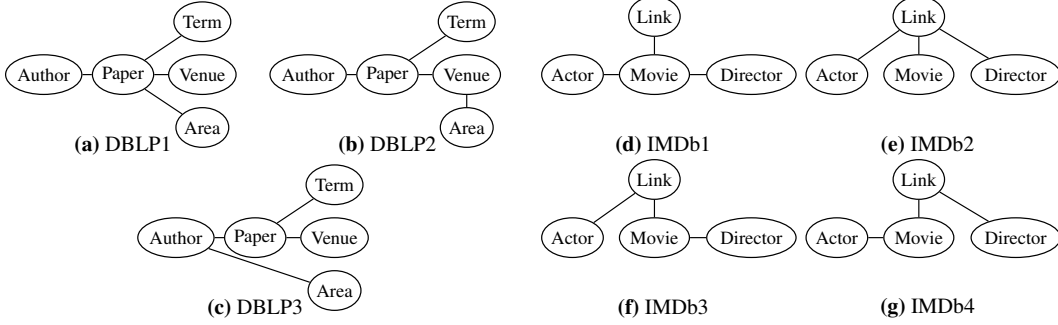


Figure 2: Mappings for DBLP and IMDb.

Task	Model	Micro-F1 $\uparrow$	Δ Micro-F1 $\downarrow$	Macro-F1 $\uparrow$	Δ Macro-F1 $\downarrow$	Kendall- τ $\uparrow$	Time $\downarrow$
		Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.
IMDb-NC	RGCN	55.70 $\pm$ 0.90 $\rightarrow$ 64.50 $\pm$ 1.20	3.50 $\rightarrow$ 0.00	55.40 $\pm$ 0.80 $\rightarrow$ 64.70 $\pm$ 1.30	3.70 $\rightarrow$ 0.00	0.43 $\rightarrow$ 1.00	0.08 $\rightarrow$ 0.22
IMDb-NC	SlotGAT	64.70 $\pm$ 1.00 $\rightarrow$ 64.80 $\pm$ 1.20	1.80 $\rightarrow$ 0.00	64.60 $\pm$ 1.00 $\rightarrow$ 64.50 $\pm$ 1.60	2.00 $\rightarrow$ 0.00	0.68 $\rightarrow$ 1.00	0.14 $\rightarrow$ 0.12
IMDb-NC	MAGNN	61.50 $\pm$ 1.30 $\rightarrow$ 64.70 $\pm$ 1.10	6.40 $\rightarrow$ 0.00	61.20 $\pm$ 1.70 $\rightarrow$ 64.80 $\pm$ 1.00	7.80 $\rightarrow$ 0.00	0.64 $\rightarrow$ 1.00	0.49 $\rightarrow$ 0.85
IMDb-NC	SeHGNN	63.60 $\pm$ 0.80 $\rightarrow$ 64.00 $\pm$ 0.50	1.20 $\rightarrow$ 0.00	63.60 $\pm$ 0.80 $\rightarrow$ 64.10 $\pm$ 0.60	1.30 $\rightarrow$ 0.00	0.63 $\rightarrow$ 1.00	0.79 $\rightarrow$ 1.69
Task	Model	Hits@3 $\uparrow$	Δ Hits@3 $\downarrow$	MRR $\uparrow$	Δ MRR $\downarrow$	Kendall- τ $\uparrow$	Time $\downarrow$
		Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.	Orig. $\rightarrow$ Inv.
DBLP-LP	RGCN	74.10 $\pm$ 6.10 $\rightarrow$ 89.80 $\pm$ 14.30	15.80 $\rightarrow$ 0.00	63.40 $\pm$ 12.00 $\rightarrow$ 84.70 $\pm$ 21.70	22.20 $\rightarrow$ 0.00	0.14 $\rightarrow$ 1.00	0.49 $\rightarrow$ 0.45
DBLP-LP	MAGNN	91.20 $\pm$ 2.00 $\rightarrow$ 90.90 $\pm$ 2.70	0.80 $\rightarrow$ 0.00	85.70 $\pm$ 2.10 $\rightarrow$ 85.80 $\pm$ 3.10	1.90 $\rightarrow$ 0.00	0.53 $\rightarrow$ 1.00	252.83 $\rightarrow$ 215.93
DBLP-LP	CMPNN	88.50 $\pm$ 4.70 $\rightarrow$ 90.80 $\pm$ 1.20	7.80 $\rightarrow$ 0.00	64.70 $\pm$ 4.80 $\rightarrow$ 66.70 $\pm$ 0.90	6.00 $\rightarrow$ 0.00	0.51 $\rightarrow$ 1.00	116.50 $\rightarrow$ 100.95

Table 1: Differences between original and invariant mapping of the same model. Each entry is original baseline $\rightarrow$ invariant adaptation. Micro-F1/Macro-F1 are reported for IMDb-NC, and Hits@3/MRR are reported for DBLP-LP. Performance values are averaged over dataset variants and reported with the average standard deviation over three runs. Δ values are the max-min spread across run-averaged variant scores. τ is the average pairwise Kendall- τ over output-score rankings across variants. Time is average training time in minutes.

performance gains are better explained by structural information than by architectural complexity alone. For link prediction, we include CMPNN because its source-conditioned path propagation mechanism better aligns with link prediction tasks than standard GNN architectures. Following Section 4.3, we construct invariant structural biases for these algorithms. In table 1, Org. denotes the original baseline, while Inv. denotes our invariant mapping of the same model. Additional implementation and mapping details are deferred to Appendix H.

Metrics. We report degree of invariance, effectiveness, and efficiency. For node classification, we report Macro-F1 and Micro-F1; for link prediction, we report MRR and Hits@3. Task performance is averaged over dataset variants and three runs, with the reported standard deviation computed as the average run-to-run standard deviation within each variant. Δ denotes the max-min spread across variant-level means, where each variant mean is computed over three runs. We use Kendall- τ to measure the degree of invariance in model outputs across dataset variants: a value of 1 indicates identical output-score rankings, whereas a value of -1 indicates completely reversed rankings. Kendall- τ is computed over output-score rankings for every pair of dataset variants and averaged over variant pairs, runs, and test instances. Time is average training time in minutes over variants and runs. Full metric definitions are provided in Appendix H.

Results Reporting. Table 1 reports the main results for invariance, effectiveness, and efficiency, with a representative set of models chosen to cover the primary structural inductive biases considered in this work. The full evaluation includes IMDb node classification, IMDb link prediction, and DBLP link prediction across all baselines and invariant adaptations. For brevity, we defer the IMDb link prediction results, along with all remaining model-task combinations, per-variant scores, additional metrics, and expanded analysis, to Appendix H.

Degree of Invariance. Our method consistently remove sensitivity to the non-isomorphic dataset variants under the reported aggregate. For every model and task, the performance spread drops to 0.00 for both metrics, and the average pairwise Kendall- τ rises to 1.00. This indicates that the invariant mappings produce stable output-score rankings across variants, whereas the original models can vary substantially; for example, DBLP-LP RGCN improves from Kendall- τ = 0.14 to 1.00, and IMDb-NC RGCN improves from 0.43 to 1.00. These results support the claim that the original architectures

are affected by transformation-induced structural differences, while the invariant adaptations make predictions stable across variants.

Effectiveness. The invariant adaptations generally preserve or improve predictive performance. On IMDb-NC, the largest gains are for RGCN and MAGNN: RGCN improves from 55.70 to 64.50 Micro-F1 and from 55.40 to 64.70 Macro-F1, while MAGNN improves from 61.50 to 64.70 and from 61.20 to 64.80. SlotGAT and SeHGNN are already competitive on the original variants, and their invariant mappings maintain comparable performance. On DBLP-LP, RGCN improves substantially, from 74.10 to 89.80 MRR and from 63.40 to 84.70 Hits@3. CMPNN also improves modestly, while MAGNN remains essentially unchanged, suggesting that the invariant bias can improve stability without sacrificing accuracy.

Efficiency. Runtime effects are architecture and dataset dependent. On IMDb-NC, invariant RGCN, MAGNN, and SeHGNN incur a small additional cost, while SlotGAT is slightly faster. On DBLP-LP, however, all invariant adaptations are faster than their originals, including MAGNN decreasing from 252.83 to 215.93 minutes and CMPNN from 116.50 to 100.95 minutes. Thus, invariance does not necessarily require extra training time; depending on the induced representation and preprocessing behavior, it can either add overhead or reduce runtime.

References

- [1] Sami Abu-El-Haija, Bryan Perozzi, Amol Kapoor, Nazanin Alipourfard, Kristina Lerman, Hrayr Harutyunyan, Greg Ver Steeg, and Aram Galstyan. 2019. MixHop: Higher-Order Graph Convolutional Architectures via Sparsified Neighborhood Mixing. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Kamalika Chaudhuri and Ruslan Salakhutdinov (Eds.), Vol. 97. PMLR, 21–29. <https://proceedings.mlr.press/v97/abu-el-haija19a.html>
- [2] Marcelo Arenas. 2006. Normalization Theory for XML. *SIGMOD Rec.* 35, 4 (Dec. 2006), 57–64. <https://doi.org/10.1145/1228268.1228284>
- [3] Pablo Barcelo, Floris Geerts, Juan L Reutter, and Maksimilian Ryschkov. 2021. Graph Neural Networks with Local Graph Parameters. In *Advances in Neural Information Processing Systems*, A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan (Eds.). <https://openreview.net/forum?id=yGKkl8wyV>
- [4] Maciej Besta, Florian Scheidl, Lukas Gianinazzi, Grzegorz Kwasniewski, Shachar Klaiman, Jurgen Muller, and Torsten Hoefler. 2026. Demystifying Higher-Order Graph Neural Networks. *IEEE Transactions on Pattern Analysis & Machine Intelligence* 48, 03 (March 2026), 2544–2565. <https://doi.org/10.1109/TPAMI.2025.3637114>
- [5] Beatrice Bevilacqua, Fabrizio Frasca, Derek Lim, Balasubramaniam Srinivasan, Chen Cai, G. Balamurugan, Michael M. Bronstein, and Haggai Maron. 2021. Equivariant Subgraph Aggregation Networks. *ArXiv abs/2110.02910* (2021). <https://api.semanticscholar.org/CorpusID:238407774>
- [6] Giorgos Bouritsas, Fabrizio Frasca, Stefanos Zafeiriou, and Michael M. Bronstein. 2023. Improving Graph Neural Network Expressivity via Subgraph Isomorphism Counting. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 1 (Jan. 2023), 657–668. <https://doi.org/10.1109/tpami.2022.3154319>
- [7] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. 2021. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges. *arXiv:2104.13478 [cs.LG]* <https://arxiv.org/abs/2104.13478>
- [8] Jiahang Cao, Jinyuan Fang, Zaiqiao Meng, and Shangsong Liang. 2022. Knowledge Graph Embedding: A Survey from the Perspective of Representation Spaces. *arXiv preprint arXiv:2211.03536* (2022).
- [9] E. F. Codd. 1971. Normalized data base structure: a brief tutorial. In *Proceedings of the 1971 ACM SIGFIDET (Now SIGMOD) Workshop on Data Description, Access and Control* (San Diego, California) (*SIGFIDET '71*). Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/1734714.1734716>

- [10] Yuanfei Dai, Shiping Wang, Neal N Xiong, and Wenzhong Guo. 2020. A survey on knowledge graph embedding: Approaches, applications and benchmarks. *Electronics* 9, 5 (2020), 750.
- [11] Pim de Haan, Taco Cohen, and Max Welling. 2020. Natural graph networks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada) (*NIPS '20*). Curran Associates Inc., Red Hook, NY, USA, Article 307, 11 pages.
- [12] P. G. Doyle and J. L. Snell. 1984. *Random Walks and Electric Networks*. Mathematical Association of America.
- [13] Wenfei Fan and Philip Bohannon. 2008. Information Preserving XML Schema Embedding. *TODS* 33, 1 (2008).
- [14] Jiarui Feng, Yixin Chen, Fuhai Li, Anindya Sarkar, and Muhan Zhang. 2022. How Powerful are K-hop Message Passing Graph Neural Networks?. In *Advances in Neural Information Processing Systems*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35. Curran Associates, Inc., 23537–23550. https://papers.neurips.cc/paper_files/paper/2022/file/1ece70d2259b8e9510e2d4ca8754cecf-Paper-Conference.pdf
- [15] Fabrizio Frasca, Beatrice Bevilacqua, Michael M. Bronstein, and Haggai Maron. 2022. Understanding and extending subgraph GNNs by rethinking their symmetries. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '22*). Curran Associates Inc., Red Hook, NY, USA, Article 2275, 15 pages.
- [16] Xinyu Fu, Jiani Zhang, Ziqiao Meng, and Irwin King. 2020. Magnn: Metapath aggregated graph neural network for heterogeneous graph embedding. In *Proceedings of the web conference 2020*. 2331–2341.
- [17] Yue Gao, Yilun Feng, Shimin Ji, and Zhenhui Ruan. 2019. Hypergraph Neural Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 3515–3522.
- [18] Floris Geerts and Juan L Reutter. 2022. Expressiveness and Approximation Properties of Graph Neural Networks. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=wIzUeM3TAU>
- [19] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. 2017. Neural Message Passing for Quantum Chemistry. In *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*. 1263–1272. <http://proceedings.mlr.press>
- [20] C. D. Godsil and B. D. McKay. 1982. Constructing cospectral graphs. *Aequationes Mathematicae* 25 (1982), 257–268.
- [21] Fabrizio Graziani et al. 2024. The Expressive Power of Path-Based Graph Neural Networks. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. <https://proceedings.mlr.press/v235/graziani24a.html>
- [22] Claudio Gutierrez, Carlos Hurtado, and Alberto O. Mendelzon. 2004. Foundations of semantic web databases (*PODS '04*). Association for Computing Machinery, New York, NY, USA, 95–106. <https://doi.org/10.1145/1055558.1055573>
- [23] W. H. Haemers. 1995. Interlacing eigenvalues and graphs. In *Linear Algebra in Combinatorics*. American Mathematical Society, 199–206.
- [24] Joseph Y. Halpern and Daphne Koller. 2004. Representation dependence in probabilistic inference. *J. Artif. Int. Res.* 21, 1 (March 2004), 319–356.
- [25] William L Hamilton. 2020. *Graph representation learning*. Morgan & Claypool Publishers.
- [26] Xiaolong Zheng Hu Tian, Bowei Ye and Desheng Dash Wu. 2020. Robust Graph Neural Networks Against Adversarial Attacks via Jointly Adversarial Training. *IFAC-PapersOnLine* 53, 5 (2020), 420–425. <https://doi.org/10.1016/j.ifacol.2021.04.225> 3rd IFAC Workshop on Cyber-Physical & Human Systems CPHS 2020.

- [27] Ningyuan (Teresa) Huang, Ron Levie, and Soledad Villar. 2023. Approximately equivariant graph networks. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '23*). Curran Associates Inc., Red Hook, NY, USA, Article 1505, 34 pages.
- [28] Xingyue Huang, Miguel Romero Orth, Pablo Barceló, Michael M. Bronstein, and İsmail İlkan Ceylan. 2024. Link Prediction with Relational Hypergraphs. arXiv:2402.04062 [cs.LG] <https://arxiv.org/abs/2402.04062>
- [29] Xingyue Huang, Miguel Romero, undefinedsmaill undefinedlkan Ceylan, and Pablo Barceló. 2023. A theory of link prediction via relational weisfeiler-leman on knowledge graphs. In *Proceedings of the 37th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '23*). Curran Associates Inc., Red Hook, NY, USA, Article 866, 35 pages.
- [30] Truong Son Hy, Shubhendu Trivedi, Horace Pan, Brandon M Anderson, and Risi Kondor. 2018. Predicting molecular properties with covariant compositional networks. *The Journal of Chemical Physics* 148, 24 (2018), 241745. <https://doi.org/10.1063/1.5024797>
- [31] Manfred Jaeger. 1996. Representation independence of Nonmonotonic Inference Relations. In *Proceedings of the Fifth International Conference on Principles of Knowledge Representation and Reasoning (KR'96)*. 461–474.
- [32] Emily Jin, Michael Bronstein, undefinedsmaill undefinedlkan Ceylan, and Matthias Lanzinger. 2024. Homomorphism counts for graph neural networks: all about that basis. In *Proceedings of the 41st International Conference on Machine Learning* (Vienna, Austria) (*ICML'24*). JMLR.org, Article 887, 24 pages.
- [33] Chaitanya K Joshi, Cristian Bodnar, Simon V Mathis, Taco Cohen, and Pietro Liò. 2023. On the Expressive Power of Geometric Graph Neural Networks. In *International Conference on Machine Learning (ICML)*. PMLR, 15809–15830.
- [34] Wei Ju, Siyu Yi, Yifan Wang, Zhiping Xiao, Zhengyang Mao, Hourun Li, Yiyang Gu, Yifang Qin, Nan Yin, Senzhang Wang, Xinwang Liu, Philip S. Yu, and Ming Zhang. 2025. A Survey of Graph Neural Networks in Real world: Imbalance, Noise, Privacy and OOD Challenges. arXiv:2403.04468 [cs.LG] <https://arxiv.org/abs/2403.04468>
- [35] Nicolas Keriven and Gabriel Peyré. 2019. *Universal invariant and equivariant graph neural networks*. Curran Associates Inc., Red Hook, NY, USA.
- [36] Thomas N. Kipf and Max Welling. 2016. Semi-Supervised Classification with Graph Convolutional Networks. *CoRR* abs/1609.02907 (2016). arXiv:1609.02907 <http://arxiv.org/abs/1609.02907>
- [37] Thomas N Kipf and Max Welling. 2016. Variational Graph Auto-Encoders. In *arXiv preprint arXiv:1611.07308*.
- [38] Johannes Klicpera, Janek Groß, and Stephan Günnemann. 2020. Directional Message Passing for Molecular Graphs. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=B1eWbxStPH>
- [39] Matthias Lanzinger and Pablo Barcelo. 2024. On the Power of the Weisfeiler-Leman Test for Graph Motif Parameters. In *The Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=FddFxi08J3>
- [40] George Ma, Yifei Wang, Derek Lim, Stefanie Jegelka, and Yisen Wang. 2025. A canonicalization perspective on invariant and equivariant learning. In *Proceedings of the 38th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada). Curran Associates Inc., Red Hook, NY, USA, Article 1948, 44 pages.
- [41] Takanori Maehara and Hoang NT. 2025. Deep homomorphism networks. In *Proceedings of the 38th International Conference on Neural Information Processing Systems* (Vancouver, BC, Canada) (*NIPS '24*). Curran Associates Inc., Red Hook, NY, USA, Article 1784, 32 pages.

- [42] Haggai Maron, Heli Ben-Hamu, Nadav Shami, and Yaron Lipman. 2018. Invariant and equivariant graph networks. *arXiv preprint arXiv:1812.09902* (2018).
- [43] Haggai Maron, Ethan Fetaya, Nimrod Segol, and Yaron Lipman. 2019. On the universality of invariant networks. In *International conference on machine learning*. PMLR, 4363–4371.
- [44] Jonathan Masci, Emanuele Rodolà, Davide Boscaini, Michael M Bronstein, and Hao Li. 2016. Geometric deep learning. In *SIGGRAPH ASIA 2016 Courses*. 1–50.
- [45] Gaspard Michel, Giannis Nikolentzos, Johannes Lutzeyer, and Michalis Vazirgiannis. 2023. Path Neural Networks: Expressive and Accurate Graph Neural Networks. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Vol. 202. PMLR, 24737–24755. <https://proceedings.mlr.press/v202/michel23a.html>
- [46] Iain Moffatt and Jaeseong Oh. 2021. A 2-isomorphism theorem for delta-matroids. *Advances in Applied Mathematics* 126 (2021), 102133. <https://doi.org/10.1016/j.aam.2020.102133> Special Issue: Dedicated to Joseph Kung.
- [47] Christopher Morris, Kristian Kersting, Petra Mutzel, and Marianne Neumann. 2019. Weisfeiler and Leman Go Neural: Higher-Order Graph Neural Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- [48] Christopher Morris, Gaurav Rattan, and Petra Mutzel. 2020. Weisfeiler and Leman go sparse: Towards scalable higher-order graph embeddings. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33. 21824–21840.
- [49] Christopher Morris, Martin Ritzert, Matthias Fey, William L. Hamilton, Jan Eric Lenssen, Gaurav Rattan, and Martin Grohe. 2019. Weisfeiler and leman go neural: higher-order graph neural networks. In *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence* (Honolulu, Hawaii, USA) (AAAI’19/IAAI’19/EAAI’19). AAAI Press, Article 565, 8 pages. <https://doi.org/10.1609/aaai.v33i01.33014602>
- [50] Hoang Nguyen and Takanori Maehara. 2020. Graph Homomorphism Convolution. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Hal Daumé III and Aarti Singh (Eds.), Vol. 119. PMLR, 7306–7316. <https://proceedings.mlr.press/v119/nguyen20c.html>
- [51] Giannis Nikolentzos, George Dasoulas, and Michalis Vazirgiannis. 2020. k-hop graph neural networks. *Neural Networks* 130 (2020), 319–330.
- [52] Theodore Papamarkou, Tolga Birdal, Michael M. Bronstein, Gunnar E. Carlsson, Justin Curry, Yue Gao, Mustafa Hajij, Roland Kwitt, Pietro Lio, Paolo Di Lorenzo, Vasileios Maroulas, Nina Miolane, Farzana Nasrin, Karthikeyan Natesan Ramamurthy, Bastian Rieck, Simone Scardapane, Michael T Schaub, Petar Veličković, Bei Wang, Yusu Wang, Guowei Wei, and Ghada Zamzmi. 2024. Position: Topological Deep Learning is the New Frontier for Relational Learning. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (Eds.), Vol. 235. PMLR, 39529–39555. <https://proceedings.mlr.press/v235/papamarkou24a.html>
- [53] Mathilde Papillon, Mustafa Hajij, Fabrizio Frasca, and Mikko Kivelä. 2023. Architectures of Topological Deep Learning: A Survey of Message-Passing Topological Neural Networks. *arXiv preprint arXiv:2304.10031* (2023).
- [54] Pál András Papp, Karolis Martinkus, Lukas Faber, and Roger Wattenhofer. 2021. DropGNN: Random Dropouts Increase the Expressiveness of Graph Neural Networks. In *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (Eds.), Vol. 34. Curran Associates, Inc., 21997–22009. https://proceedings.neurips.cc/paper_files/paper/2021/file/b8b2926bd27d4307569ad119b6025f94-Paper.pdf

- [55] Pal Andras Papp and Roger Wattenhofer. 2022. A Theoretical Comparison of Graph Neural Network Extensions. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (Eds.), Vol. 162. PMLR, 17323–17345. <https://proceedings.mlr.press/v162/papp22a.html>
- [56] Jose Picado, Arash Termehchy, Alan Fern, and Parisa Ataei. 2017. Schema Independent Relational Learning. In *Proceedings of the 2017 ACM International Conference on Management of Data (Chicago, Illinois, USA) (SIGMOD '17)*. Association for Computing Machinery, New York, NY, USA, 929–944. <https://doi.org/10.1145/3035918.3035923>
- [57] Michael Schlichtkrull, Thomas N Kipf, Peter Bloem, Rianne Van Den Berg, Ivan Titov, and Max Welling. 2018. Modeling relational data with graph convolutional networks. In *The semantic web: 15th international conference, ESWC 2018, Heraklion, Crete, Greece, June 3–7, 2018, proceedings 15*. Springer, 593–607.
- [58] Kristof T Schütt, Pieter-Jan Kindermans, Huziel E Saucedo, Stefan Chmiela, Alexandre Tkatchenko, and Klaus-Robert Müller. 2017. SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. In *Advances in Neural Information Processing Systems 30*. Curran Associates, Inc., 991–1001.
- [59] J. J. Seidel. 1976. A survey of two-graphs. *Colloquio Internazionale sulle Teorie Combinatorie* (1976), 481–511.
- [60] {Márcio Rosa Da} Silva, Jibin Sun, Hongwu Ma, Feng He, and {An Ping} Zeng. 2007. *Metabolic Networks*. John Wiley & Sons, Inc., 233–253. <https://doi.org/10.1002/9780470253489.ch10>
- [61] Derek A. Smith, Lorenzo Traldi, and William Watkins. 2019. A note on 2-isomorphisms and the signed Laplacian matrix of a graph. *Linear Algebra Appl.* 563 (2019), 277–286. <https://doi.org/10.1016/j.laa.2018.11.002>
- [62] Yifei Sun, Haoran Deng, Yang Yang, Chunping Wang, Jiarong Xu, Renhong Huang, and ... Linfeng. 2022. Beyond Homophily: Structure-aware Path Aggregation Graph Neural Network. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence, IJCAI-22*.
- [63] Erik Henning Thiede, Wenda Zhou, and Risi Kondor. 2021. Autobahn: Automorphism-based Graph Neural Nets. *arXiv:2103.01710 [cs.LG]*
- [64] Nathaniel Thomas, Tess Smidt, Steven Kearnes, Lusann Yang, Li Li, and Patrick Kohlhoff, Kai and. 2018. Tensor field networks: Rotation- and translation-equivariant neural networks for 3d point clouds. *arXiv preprint arXiv:1802.08219* (2018).
- [65] K Trivodaliev, A Bogojaska, and L Kocarev. 2014. Exploring Function Prediction in Protein Interaction Networks via Clustering Methods. *PLoS ONE* 9, 6 (2014), 8. <https://doi.org/10.1371/journal.pone.0099755>
- [66] K. Truemper. 1989. On the delta-wye reduction for planar graphs. *Journal of Graph Theory* 13, 2 (1989), 141–148. <https://doi.org/10.1002/jgt.3190130202> [arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/jgt.3190130202](https://onlinelibrary.wiley.com/doi/pdf/10.1002/jgt.3190130202)
- [67] J. H. van Lint and J. J. Seidel. 1966. Equilateral point sets in elliptic geometry. *Indagationes Mathematicae* 28 (1966), 335–348.
- [68] Ameya Velingker, Ali Sinop, Ira Ktena, Petar Veličković, and Sreenivas Gollapudi. 2023. Affinity-Aware Graph Networks. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*. https://proceedings.neurips.cc/paper_files/paper/2023/hash/d642b0633afad94f660554e05b40608e-Abstract-Conference.html
- [69] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2017. Graph Attention Networks. *6th International Conference on Learning Representations* (2017).

- [70] Guangtao Wang, Rex Ying, Jing Ren, Mehran Liang, and Jure Leskovec. 2021. Multi-hop Attention Graph Neural Networks. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence (IJCAI-21)*.
- [71] Xiao Wang, Houye Ji, Chuan Shi, Bai Wang, Yanfang Ye, Peng Cui, and Philip S Yu. 2019. Heterogeneous graph attention network. In *The world wide web conference*. 2022–2032.
- [72] Xiuming Wang, Xindong Zhou, Jinghui Cheng, Zhaowei Ren, and Jianxin Li. 2023. Simple and Efficient Heterogeneous Graph Neural Network. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 12297–12305. <https://ojs.aaai.org/index.php/AAAI/article/view/26283>
- [73] Zaitian Wang, Pengfei Wang, Kunpeng Liu, Pengyang Wang, Yanjie Fu, Chang-Tien Lu, Charu C. Aggarwal, Jian Pei, and Yuanchun Zhou. 2025. A Comprehensive Survey on Data Augmentation. arXiv:2405.09591 [cs.LG] <https://arxiv.org/abs/2405.09591>
- [74] H. Whitney. 1933. 2-isomorphic graphs. *American Journal of Mathematics* 55 (1933), 245–254.
- [75] H. Whitney. 1933. On the Classification of Graphs. *American Journal of Mathematics* 55, 1 (1933), 236–244.
- [76] E. C. Wood, Amy K. Glen, Lindsey G. Kvarfordt, Finn Womack, Liliana Acevedo, Timothy S. Yoon, Chunyu Ma, Veronica Flores, Meghamala Sinha, Yodsawalai Chodpathumwan, Arash Termehchy, Jared C. Roach, Luis Mendoza, Andrew S. Hoffman, Eric W. Deutsch, David Koslicki, and Stephen A. Ramsey. 2021. RTX-KG2: a system for building a semantically standardized knowledge graph for translational biomedicine. *bioRxiv* (2021). <https://doi.org/10.1101/2021.10.17.464747> arXiv:<https://www.biorxiv.org/content/early/2021/10/25/2021.10.17.464747.full.pdf>
- [77] Donglin Xia, Xiao Wang, Nian Liu, and Chuan Shi. 2023. Learning Invariant Representations of Graph Neural Networks via Cluster Generalization. In *Thirty-seventh Conference on Neural Information Processing Systems*. <https://openreview.net/forum?id=zrCmeqV3Sz>
- [78] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. 2019. How Powerful are Graph Neural Networks?. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=ryGs6iA5Km>
- [79] Carl Yang, Yuxin Xiao, Yu Zhang, Yizhou Sun, and Jiawei Han. 2020. Heterogeneous network representation learning: A unified framework with survey and benchmark. *IEEE Transactions on Knowledge and Data Engineering* 34, 10 (2020), 4854–4873.
- [80] Xiao Yang, Xuejiao Zhao, and Zhiqi Shen. 2025. Are Heterogeneous Graph Neural Networks Truly Effective? A Causal Perspective. *arXiv preprint arXiv:2510.05750* (2025).
- [81] Bohang Zhang, Shengjie Luo, Liwei Wang, and Di He. 2023. Rethinking the Expressive Power of GNNs via Graph Biconnectivity. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. <https://openreview.net/forum?id=kFUZ2-BJa3H>
- [82] Jianan Zhao, Xiao Wang, Chuan Shi, Binbin Hu, Guojie Song, and Yanfang Ye. 2021. Heterogeneous Graph Structure Learning for Graph Neural Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 4685–4693.
- [83] Ziang Zhou, Jieming Shi, Renchi Yang, Yuanhang Zou, and Qing Li. 2023. SlotGAT: slot-based message passing for heterogeneous graphs. In *International Conference on Machine Learning*. PMLR, 42644–42657.
- [84] Zhaocheng Zhu, Zuobai Zhang, Louis-Pascal Xhonneux, and Jian Tang. 2021. Neural bellman-ford networks: a general graph neural network framework for link prediction. In *Proceedings of the 35th International Conference on Neural Information Processing Systems (NIPS '21)*. Curran Associates Inc., Red Hook, NY, USA, Article 2256, 15 pages.

A Limitations

Our approach has two main limitations. First, we assume that the relevant graph transformations are given explicitly through the mapping τ . That is, the algorithm modifies the structural bias of a GNN with respect to a specified set of switchings, rather than discovering those switchings from data or from the task itself. This makes the method effective when the desired invariance is known in advance, but it does not address the complementary problem of identifying which transformations should be treated as invariance-inducing. There are bijective mappings that are expressed using more complex languages than switchings, e.g., graph languages based on first order logic with/out recursion [13,22]. Since they are bijective and constitute a group, our framework extends to these mappings by redefining conditional graphs using a more complex language. In this paper, we focus on the ones expressed using switchings because we believe that there are frequent graph variations that can be expressed in the form of switching, e.g., data normalization or examples in [2,60,65,66,76]. Moreover, it is generally hard to define/check the bijectivity conditions for complex mappings on relational structures, e.g., R.Fagin “Inverting Schema Mappings”, PODS 2006. More importantly, as the resulting conditional graphs become too expressive, the training may be inefficient. Hence, it requires more than the space and scope of a single paper to address the challenges of these mappings. Our work introduces a new direction for informational equivalence and opens the door for a much broader scope of future work. As a result, the quality of the resulting invariant adaptation depends on the choice of τ : missing transformations may leave unwanted sensitivity in the model, while overly broad transformations may remove useful task-relevant information.

Second, the current formulation targets exact invariance. The modified structural bias $\mathcal{S}_\tau^f$ is constructed so that the resulting GNN is invariant to the specified mapping τ , while remaining minimal with respect to the proposed ordering over structural biases. However, in some applications, exact invariance may be stronger than necessary. Certain transformations may preserve most, but not all, task-relevant information; enforcing full invariance to them could therefore reduce expressivity or obscure meaningful structural differences. A promising direction is to extend the framework to approximately invariant mappings, where the model is encouraged to behave similarly across transformed variants without requiring identical behavior. Our method can use incomplete and partial mappings to encode the given information in the structural bias. We expect that the resulting method outperforms the baseline GNN method over the same dataset. If the mappings are approximate, the induced structural bias by our method will propagate information across nodes as if there are noises in the input graph. The robustness of the resulting methods depend on the robustness of the underlying GNN to edge noises. Since our methods can be used in any GNN, we can use our method with GNN that are robust to noise, e.g., Y. In, Training Robust Graph Neural Network By Modeling Noise Dependencies, NeurIPS-25. We expect the resulting model to outperform its underlying GNN as it uses additional information about some variations of the data in learning. Such an extension could provide a tunable trade-off between invariance and expressivity, allowing the model to retain useful sensitivity to transformations that are only partially semantics-preserving.

B Broader Impacts

This work contributes to more reliable and reproducible graph learning by reducing dependence on incidental choices in graph representation. As graph data are increasingly used in scientific, engineering, social, and information systems, the same underlying information may be encoded using different schemas, node types, edge types, or relational structures. If model behavior changes substantially across such encodings, downstream conclusions may reflect representation artifacts rather than meaningful properties of the data. By inducing invariance with respect to specified mappings, our approach can make GNN-based predictions more stable across equivalent graph representations and support more consistent comparisons between models, datasets, and preprocessing pipelines.

The broader societal benefit is most relevant in domains where graph schemas are designed by humans or integrated from heterogeneous sources. In biomedical and scientific knowledge graphs, invariant mappings may reduce sensitivity to database-specific modeling conventions and improve the portability of learned models across resources. In infrastructure, transportation, and engineering

networks, they may support more consistent analysis when systems are represented at different levels of abstraction. In recommender systems, fraud detection, and information retrieval, they may reduce unintended variation caused by schema design choices, making deployed systems easier to audit and reproduce.

Potential negative impacts may arise when invariant mappings are mis-specified or applied without sufficient domain validation. If a mapping is too broad, the resulting model may suppress task-relevant distinctions; if invariance metrics are treated as evidence of general fairness or correctness, they may obscure other sources of error or bias. This is particularly important in high-stakes graph applications involving biomedical, social, economic, or institutional data. Invariant mappings should therefore be documented, domain-justified, and evaluated together with task performance and subgroup-level analyses.

More generally, this work encourages a shift from evaluating GNNs only on fixed benchmark representations toward evaluating whether their predictions are stable under information-preserving changes in representation. Such evaluation can improve transparency by making explicit which mappings are assumed to preserve task-relevant information. The intended societal impact is therefore methodological: to support graph learning systems whose behavior is less tied to arbitrary encoding decisions and more aligned with the underlying information that the graph is meant to represent.

C Related Work

GNN Invariance & Equivariance. In addition to the invariance/equivariance of GNNs with respect to the isomorphism [43, 42, 35], researchers have investigated the invariance / equivariance of GNNs to other types of permutations/isomorphisms. For example, the authors in [63] and [11] have investigated the invariance/equivariance of GNNs with respect to automorphisms and isomorphisms in the local structures in the graph, respectively. Researchers have also investigated the properties of GNN that are approximately equivariant to a set of permutations [27].

Invariance & Equivariance of Higher-Order Networks. There have also been investigations of the GNN invariance/equivariance with respect to permutations in higher-order networks, e.g., hyper-graphs where each edge may connect more than two nodes [17, 30, 53, 52, 28]. Our work differs from this line of research in that we explore the invariance/equivariance of GNNs to variations other than permutations over graphs.

Geometric Graphs. Sometimes, graphs contain information about geometric objects, e.g., positions of objects in a 3D space (geometric graphs). Researchers have proposed deep learning architectures for geometric graphs that are invariant/equivariant to equivalence-preserving geometrical variations, such as rotation or translation [7, 58, 38, 33, 64]. These variations are specific to geometrical domains, whereas we consider mappings that are applied in general graphs. Moreover, as opposed to our mappings, the aforementioned geometrical mappings do not modify the relationships between nodes/objects.

GNN Robustness Against Noise. Several GNN architectures have been proposed to improve robustness against both node-level and graph-level noise [77, 26]. Prior work primarily addresses robustness to adversarial attacks or out-of-distribution generalization, often under the assumption that one graph contains more or less information than another. In contrast, our work examines scenarios in which two graphs encode the *same* information but differ in their topological structure.

D τ -Invariance/Equivariance of Current Methods

D.1 Message Passing Neural Networks

There are various neural architectures for learning graph representations [34, 25, 79, 10, 8]. **Message passing neural networks (MPNN)** are a prominent architecture for GNN’s that use the graph structure to learn representations [25]. Let $h_v^{(l)}$ represent the feature of node v at layer l . Each layer of an MPNN is composed of two learnable functions, the permutation invariant aggregate function and the update function:

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{AGGREGATE} \left(\{h_u^{(l)} : u \in \mathcal{N}(v)\} \right) \right). \quad (3)$$

Repeated layers allow information to propagate through the graph. As (basic) MPNN has some limits in distinguishing non-isomorphic graphs (e.g., [78]), researchers incorporate additional information on the structure of the graph, e.g., sets of paths [16, 68, 81]. or add a higher-order structure [47] to MPNNs. Due to the popularity of MPNNs, in this paper, we focus on these architectures.

D.2 τ -Invariance/Equivariance of MPNNs

Current MPNN-based architectures use the graph structure to learn the graph representation [25]. For example, the MPNN architecture in Equation 3 shares the information of each node v with its neighbors in each layer [69, 57, 37, 19, 78]. Given GNN f , graph G , and $v \in V(G)$, let $\mathcal{M}^f(G, v)$ be a permutation equivariant function that returns a non-empty and proper subset of $V(G)$, i.e., $\mathcal{M}^f(\pi(G), \pi(v)) = \pi(\mathcal{M}^f(G, v))$. Many node-centric architectures f extend Equation 3 by sharing information between each node $v \in V(G)$ and $\mathcal{M}^f(G, v)$ in each layer.

$$h_v^{(l+1)} = \text{UPDATE} \left(h_v^{(l)}, \text{AGGREGATE} \left(\{h_u^{(l)} : u \in \mathcal{M}^f(G, v)\} \right) \right). \quad (4)$$

$\mathcal{M}^f(G, v)$ is often defined using the structural properties of the connections between v and $u \in \mathcal{M}^f(G, v)$. For example, some architectures set $\mathcal{M}^f(G, v)$ to (all) nodes on all simple or shortest paths from v whose lengths are at most $K > 0$ [45, 21, 62] and some methods set $\mathcal{M}^f(G, v)$ to $k \leq K$ -hop neighbors of v where $K = 1$ recovers the basic MPNN [51, 1, 70, 55, 14]. There are also architectures that define $\mathcal{M}^f(G, v)$ using homomorphism or isomorphism from a fixed set of (labeled) graphs $\mathbf{F}$: $v \cup \mathcal{M}^f(G, v)$ are the nodes of homomorphism (isomorphism) images from $H \in \mathbf{H}$ in G [41, 16, 82, 72]. These methods may aggregate additional structural information about the nodes in $\mathcal{M}^f(G, v)$, e.g., their distances to v [21]. Since a bijective mapping τ modifies the structure of the graphs, e.g., the neighborhood structure of some nodes as in Figure 1, $\mathcal{M}^f(G, v)$ and $\mathcal{M}^f(\tau(G), v)$ are often different. Hence, these architectures will share information between different sets of nodes in some G and $\tau(G)$ and may not be τ -invariant/equivariant. Given architecture f , mapping τ , graph G , and node $v \in V(G)$, let UPDATE and AGGREGATE return different results for $\mathcal{M}^f(G, v)$ and $\mathcal{M}^f(\tau(G), v)$ if $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(G, v)| \neq |\mathcal{M}^f(\tau(G), v)|$. These are mild conditions and are often satisfied by typical AGGREGATE and UPDATE functions. In this case, we have the following result.

Proposition 7. *Consider the architecture f with $\mathcal{M}^f(G, v)$ in Equation 4. There is a bijective mapping τ such that f is not τ -invariant or τ -equivariant.*

Proof. If the graph is directed, we set τ to the bijective mapping that reverses the direction of the edges. We can choose G so that we have $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(G, v)| \neq |\mathcal{M}^f(\tau(G), v)|$. f will learn different representations for G and $\tau(G)$ in each layer. Assume that G is not directed, i.e., if $(u, v) \in E(G)$ we have $(v, u) \in E(G)$. The mapping τ that complements the graph G , i.e., $(u, v) \in E(G)$ iff $(u, v) \notin E(\tau(G))$, is bijective. We can choose G so that the degree of each v in G is not equal to its degree in $\tau(G)$. Since $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(G, v)| \neq |\mathcal{M}^f(\tau(G), v)|$, f will learn different representations over G and $\tau(G)$. $\square$

The proof of Proposition 7 holds for both directed and undirected graphs, as well as labeled and unlabeled ones (all nodes share the same label).

Subgraph MPNNs. To improve the accuracy of node-centric GNNs, some GNNs partition the input graph into multiple subgraphs based on a fixed policy, learn the representation for each subgraph using the methods described in the preceding paragraph, and aggregate these representations [54, 5, 15]. Since they use node-centric MPNNs to learn the representation of each subgraph, the result of Proposition 3 extends to these architectures.

D.3 τ -Invariance/Equivariance of Higher-order MPNN

Higher-order MPNN (HO-MPNN) learn representations for higher order structures [49, 48, 4, 39, 84, 29]. Most notably, they extend the MPNN architecture to learn the representation for every tuple of $k > 1$ nodes in $V(G)$, $\bar{t}_k \in V(G)^k$. The initial representation of $\bar{t}_k$ encodes the relationship between its nodes in the graph. That is, let $G[\bar{t}_k]$ denote the induced subgraph of nodes in $\bar{t}_k$ in G . Tuples $\bar{t}_k$ and $\bar{t}'_k$ have identical initial representation iff there is an isomorphism mapping between $G[\bar{t}_k]$ and $G[\bar{t}'_k]$. Let $C(\bar{t}_k, u, j)$ denote the tuple in which node $v_j \in \bar{t}_k$ is replaced by node $u \in V(G)$,

1 $\leq j \leq k$ and let $\mathcal{C}(\bar{t}_k)$ be the set of all $C(\bar{t}_k, u, j)$. In each layer, the HO-MPNN shares the information between the tuple $\bar{t}_k$ and the tuples in $\mathcal{C}(\bar{t}_k)$ using the UPDATE and AGGREGATION functions similar to Equation 3.

Since it often takes too long to learn representations using this approach due to the large number of tuples, some HO-MPNN share the information of each tuple with a subset of $\mathcal{C}(\bar{t}_k)$ where u and v_j are neighbors in G , i.e., **local HO-MPNN** [84, 29]. In this paper, we refer to both variations as HO-MPNN.

Since the initial values of the tuples $\bar{t}_k$ reflect the structure of the graph and since bijective mappings τ modify this structure, these GNNs are not τ -invariant. More precisely, assume that the UPDATE function returns a different value if at least one of its input elements is different. Also, assume that the AGGREGATION function returns a distinct value if at least one of the aggregated tuples is different. This is a mild condition that is satisfied by typical UPDATE and AVERAGE functions. We will have the following results for the HO-MPNN described above.

Proposition 8. *There is a bijective mapping τ such that HO-MPNN f is not τ -invariant or τ -equivariant.*

Proof. Let $\bar{t}_k^G$ ($k > 1$) denote a tuple of k nodes on G . We assume that τ does not permute the node ids. Our proof will extend to the cases where τ also permutes the node ids. Since τ does not permute nodes, there is an identity mapping between tuples of k nodes on G and $\tau(G)$, i.e., for each $\bar{t}_k^G$ there is one and only one $\bar{t}_k^{\tau(G)}$ where $\bar{t}_k^G = \bar{t}_k^{\tau(G)}$. We will use the same mapping used in the proof of Proposition 3. In both mappings, for every $\bar{t}_k^G$ $\bar{t}^\tau(G)_k$, $G[\bar{t}_k^G]$ is not isomorphic to $\tau(G)[\bar{t}_k^{\tau(G)}]$. Thus, in each layer, f will learn a distinct representation for $G[\bar{t}_k^G]$ and $\tau(G)[\bar{t}_k^{\tau(G)}]$. $\square$

The proof of Proposition 8 covers graphs with/without directed edges and labels.

E Proofs

E.1 Proofs for Section 2

Proof of Proposition 1. If the conclusion does not hold, then the mapping τ will not be bijective. $\square$

E.2 Proofs for Section 3

Proof of Proposition 2.

We prove the theorem for both cases that the mapping τ can be expressed as a set of switching and the case when it may not be expressed using switchings. We first prove the following lemma that indicates that we can construct a set of switchings $\iota = \{(P_i, P'_i)\}_{i=1}^t$ that implement identity mapping: the permutation that preserves node ids.

Lemma 2. *Given a family of graphs $\mathcal{G}$, there is a set of switchings ι equivalent to the permutation that preserves node ids on $\mathcal{G}$.*

Proof. The set of switchings $\tau = \{(P_i, P'_i)\}_{i=1}^t$ maps each graph to itself (identity) if every P_i is either a graph with a single node or a simple path with two nodes, every node and edge in the graph is isomorphic to a single P_i and P_i is isomorphic to P'_i where the isomorphism mapping preserves labels. Since every set of switchings preserves the node ids, ι is equivalent to the permutation that preserves node ids. $\square$

Next, we prove Proposition 2:

Proof. Given a family of graphs $\mathcal{G}$ and the set of switchings τ on $\mathcal{G}$, $\Gamma^\tau \cup \iota$ forms a group with composition operation with ι as the identity element. Generally, given the family of graphs $\mathcal{G}$, let Φ be the set of all mappings created by compositions of the members in a set of distinct switchings $\mathcal{T} = \{\tau_i\}_{i=1}^k$ or their inverses. Since switchings are bijective, the inverse of the composition of two switchings $(\tau \circ \rho)^{-1}$ is the composition of their inverses $\rho^{-1} \circ \tau^{-1}$. Hence, $\Gamma^\tau \cup \iota$ is a group with composition operation on $\mathcal{G}$. Let $\mathcal{G}^n$ be the family of graphs with n nodes. Given an isomorphism-closed set of graphs $\mathcal{G} \subseteq \mathcal{G}^n$, let Π^n be the group of permutations with composition on $\mathcal{G}$. Let π_I be

the permutation mapping on $\mathcal{G}$ that maps the id of each node to itself, i.e., the identity element in Π^n . Since ι and π_I express the same mapping on $\mathcal{G}$, $\Gamma^T \cup \Pi^n$ is a group with composition on $\mathcal{G}$.

If the mapping τ cannot be expressed as switching, we use the fact that an identity permutation is a bijection. The rest of the proof is similar to the case where the mapping is expressed as a set of switchings.

□

Proof of Proposition 3. If the graph is directed, we set τ to the bijective mapping that reverses the direction of the edges. We can choose G so that we have $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(G, v)| \neq |\mathcal{M}^f(\tau(G), v)|$. f will learn different representations for G and $\tau(G)$ in each layer. Assume that G is not directed, i.e., if $(u, v) \in E(G)$ we have $(v, u) \in E(G)$. The mapping τ that complements the graph G , i.e., $(u, v) \in E(G)$ iff $(u, v) \notin E(\tau(G))$, is bijective. We can choose G so that the degree of each v in G is not equal to its degree in $\tau(G)$. Since $\mathcal{M}^f(G, v) \cap \mathcal{M}^f(\tau(G), v) = \emptyset$ and $|\mathcal{M}^f(G, v)| \neq |\mathcal{M}^f(\tau(G), v)|$, f will learn different representations over G and $\tau(G)$. □

E.3 Proofs for Section 4

Proof of the Lemma 1. Since the PAGG function in Equation 2 receives equal inputs on G and $\tau(G)$, given the same initializations, in each layer, f will learn the same representation on G and $\tau(G)$. □

Proof of Proposition 4. Given GNN f , we construct a mapping τ where there is a switching (P_i, P'_i) , $1 \leq i \leq t$, where P_i is not isomorphic to P'_i and P_i and P'_i do not belong to $\mathcal{S}^f$. An example of τ is the mapping shown in Figure 1 and MPNN that shares the information between 1-hop neighbors where P_i and P'_i are not isomorphic and do not belong to $\mathcal{S}^f$ of MPNN. If the structural bias contains P_i and P'_i , we create another mapping by adding a single node to P_i and P'_i . □

Proof of Theorem 1. Since τ is minimal, there are no two switchings (P_i, P'_i) (P_j, P'_j) , $i \neq j$ such that P_i (P'_i) and P_j (P'_j) are isomorphic. There are no two isomorphic graphs in $\mathcal{S}^f$. Thus, the steps of the algorithms are applied to each $S^u \in \mathcal{S}^f$ separately, and the algorithm does not generate any two isomorphic CGs.

The mapping b_τ is isomorphism-preserving for CG S^u if it is an isomorphism-preserving mapping with respect to S_C^u as defined before and there is a bijection between $\text{pom}(S^u, G^v)$ and $\text{pom}(b_\tau(S^u), \tau(G^v))$ that maps every member of $\text{pom}(S^u, G^v)$ to an isomorphic member in $\text{pom}(b_\tau(S^u), \tau(G^v))$ where the isomorphism mapping preserves node ids/labels/features. For every S^u in the input structural bias $\mathcal{S}^f$, the algorithm places one and only one CG S^u where every edge is context and propagation. For every such CG S^u , the algorithm produces a CG such that there is an isomorphism-preserving mappings between them. Also, for each newly generated CG in step 2 S^w , the algorithm produces one and only one CG S'^w in step 3 such that there is an isomorphism-preserving mapping between S^w and S'^w . Since the algorithm does not generate any other types of CG, there is a isomorphism-preserving for every CG in $\mathcal{S}_\tau^f$. Thus, f using $\mathcal{S}_\tau^f$ will be τ -invariant. □

Proof of Proposition 5. Suppose we remove a CG S^u from $\mathcal{S}_\tau^f$, then there will be no isomorphism-preserving mapping in $\mathcal{S}_\tau^f$. Let G be a graph that has at least one occurrence of S_C^u . Then, the GNN will not be τ -invariant on G and $\tau(G)$. Additionally, if we remove any context or propagation edge from a $S \in \mathcal{S}_\tau^f$ there will be no $\mathcal{S}_\tau^f$. □

E.4 Proofs for Section 5

Proof of Theorem 2. To simplify our proof, we assume that all nodes in the graph share the same label and that none have any features. Our proof extends to other cases. Given the non-isomorphism mapping τ , we have $\mathcal{S}^f \subset \mathcal{S}_\tau^f$. Thus, for a graph G and each node $v \in V(G)$, the inputs to PAGG, TAGG, and UPDATE in each layer of f_τ are superset or equal to their inputs for v in the same layer in f . Consider graph G' where the results of TAGG and PAGG are different on G and G' in f . Thus, since PAGG, TAGG, and UPDATE are injective, for the same node initializations, their results will be different for G and G' in f_τ . Thus, f_τ is at least as expressive as f .

1004 To prove the second part, we also show that improving the structural bias using this algorithm enables
 1005 f_τ to access more information about the graph that is not present in the original f bias. That is, there
 1006 are GNN f and mapping τ such that f_τ is strictly more expressive than f . Consider the mapping
 1007 δ that reverses the direction of all edges in the input graph. Assume that we use the algorithm in
 1008 Section 4.3 to produce a δ -invariant $f(f_\delta)$. f_δ will propagate the information in both directions of
 1009 an edge. For example, consider the graphs G and H where $V(G) = V(H) = 1, 2, 3, 4$, $E(G) =$
 1010 $\{(1, 2), (3, 2), \}$ and $E(H) = \{(1, 2), (3, 4), \}$. The MPNN that shares information only between
 1011 1-hop neighbors (1-WL) does not distinguish G and H , but MPNN_δ does. $\square$

1012 **Proof of Proposition 6.** If $f \preceq f'$, we have $S^f \subseteq S^{f'}$. Given a mapping τ , we will have $S^{f_\tau} \subseteq S^{f'_\tau}$,
 1013 therefore; $f_\tau \preceq_\tau f'_\tau$. $\square$

1014 F More Details on Structural Bias

1015 We explain how MPNN based on structural bias in Equation 2 generalizes multiple families of
 1016 popular (node-centric) MPNN architectures. Some of these families cover only unlabeled graphs.
 1017 Our formulation covers these cases by assigning the same label to all nodes. Some of them consider
 1018 undirected graphs. Our abstraction generalizes these cases by directed edges in both directions
 1019 between all two neighboring nodes in the graph.

1020 **K-hop Message Passing.** Some MPNN architectures share information between node $v \in V(G)$
 1021 and all its $0 < k \leq K$ -hop neighbors in G [51, 1, 14, 70, 55]. Since these methods ignore labels on
 1022 the nodes, the structural bias for these architectures contains only one simple path of length (number
 1023 of edges) K . In this case, the occurrences of the graph of the structural bias in the input graph are
 1024 computed without the restriction of label-preservation. The PAGG function varies according to the
 1025 structural information shared between v and its k -hop neighbors. For example, some methods share
 1026 with v the shortest distance of its k -hop neighbor, which might be less than k . The PAGG function
 1027 has access to this information using its input. The exact function PAGG varies on the basis of the
 1028 underlying architecture. The TAGG function is a function that maps a set of a single vector to the
 1029 vector, and the UPDATE function will be the same as UPDATE in these architectures.

1030 **Path-based MPNNs.** Some MPNNs share information of $v \in V(G)$ with all nodes in $V(G)$ whose
 1031 shortest path to v has a length shorter than or equal to $K > 0$ [45, 21, 62]. The structural bias in
 1032 these methods is similar to the preceding family. They have PAGG functions different from those of
 1033 the preceding family and extract different types of information from these paths.

1034 **MPNNs Based on Meta-Paths.** Some MPNNs graphs share information between $v \in V(G)$ and
 1035 all other nodes on a set of simple paths that start in v with a given label l_1 and end with a node
 1036 with a label l_1 , and the node labels on each path follow a given sequence of labels $l_2, \dots, l_k, \dots, l_2$
 1037 (symmetric paths) [16, 72]. The structural bias for these families contains a set of paths where the
 1038 labels at the nodes of each path follow the given sequence of labels $(l_1, l_2, \dots, l_k, \dots, l_2, l_1)$.

1039 **Homomorphism-based MPNNs** Like our abstraction for structural bias, homomorphism-based
 1040 MPNNs propagate and aggregate information by computing the occurrences of a set of fixed rooted
 1041 graphs in the input graph [41]. For each occurrence of a graph in the structural bias in the input
 1042 graph, the PAGG functions in these methods may require both the occurrence and the homomorphism
 1043 mapping used to produce the occurrence. We can slightly modify Equation 2 and include both
 1044 occurrences and their homomorphism mappings as the input. Our results in the paper extend to these
 1045 cases.

1046 **Different Structural Bias per Layer.** Some algorithms use a different S in each layer [41]. Our
 1047 definition of structural bias covers these cases by a distinct set of rooted pattern graphs for each layer
 1048 in S .

1049 G Completeness of Structural Bias

1050 We call a structural bias S^f *complete* if for every graph G and every node $v \in G$ there is at least one
 1051 $S^u \in S^f$ such that $\text{hom}(S^u, G^v)$ is not empty. Intuitively, a complete structural bias for G ensures
 1052 that the information for each node in the input graph G is shared with some other nodes in G . It is
 1053 often desirable for a structural bias to be complete, so GNN learns the representation for every node

Table 2: Statistics of IMDb dataset variations for Node Classification.

Dataset Variation	Node	NC: Movie
IMDb1	# movie (M): 4180 # director (D): 2081 # actor (A): 5258 # link (L): 4180	M-L: 4180 M-A: 12534 M-D: 4180
IMDb2	# movie (M): 4180 # director (D): 2081 # actor (A): 5258 # link (L): 4180	D-L: 4180 A-L: 12534 M-L: 4180
IMDb3	# movie (M): 4180 # director (D): 2081 # actor (A): 5258 # link (L): 4180	A-L: 12534 M-D: 4180 M-L: 4180
IMDb4	# movie (M): 4180 # director (D): 2081 # actor (A): 5258 # link (L): 4180	M-A: 8454 D-L: 4180 M-L: 4180
IMDb*	# movie (M): 4180 # director (D): 2081 # actor (A): 5258 # link (L): 4180	M-L: 4180 M-A: 12534 M-D: 4180 A-L: 12534 D-L: 4180

in the graph. If the original structural bias is complete, the algorithm generates a complete structural bias.

Proposition 9. *If $\mathcal{S}^f$ is complete, $\mathcal{S}_\tau^f$ is complete.*

Proof. Since the algorithm preserves all original structural biases $\mathcal{S}^f$ (first step), and $\mathcal{S}^f$ is complete, $\mathcal{S}_\tau^f$ is complete. $\square$

H Experiment Details - use old version for Arxiv

Code to run experiments can be found here: <https://github.com/JMJMJM1230/INV-GNN>

Datasets. Statistics of datasets can be found on Tables 2, 3, and 4.

Methods. We evaluate six GNN baselines that cover different ways of using graph structure to define information propagation patterns: RGCN [57], SlotGAT [83], CMPNN [29], MAGNN [16], SeHGNN [72], and DHN [41]. These models differ in the structural object that determines how information is shared. RGCN uses local node-type-pair relations, SlotGAT uses local node-type-aware aggregation channels, CMPNN uses source-conditioned paths for link prediction, MAGNN uses metapath instances for message passing, and SeHGNN uses metapath-derived feature channels. Together, these baselines cover local, path-conditioned, metapath-instance, and metapath-feature structural biases.

RGCN is a local node-type-pair message passing model for heterogeneous or relational graphs. In our setting, edge relations are interpreted as being induced by the labels/types of the incident nodes, so RGCN aggregates messages according to node-type-pair relations. At each layer, a node receives normalized messages from its immediate neighbors, with relation-specific transformations applied before aggregation. This provides a simple but strong local heterogeneous message-passing baseline.

SlotGAT is a local node-type-aware message passing model for heterogeneous graphs. Instead of directly mixing information from all neighboring nodes, SlotGAT maintains separate representation channels for different node types. Neighbor features are aggregated into the channel associated with their node type, which helps preserve type-specific feature spaces and avoid semantic mixing. Attention is then used to combine the node-type-aware representations for downstream prediction.

Table 3: Statistics of IMDb dataset variations for Link Prediction.

Dataset Variation	Node	Edge (task dependent)	
		LP: Movie–Link	LP: Movie–Director
IMDb1	# movie (M): 4180 # director (D): 2081 # actor (A): 1739 # link (L): 4180	M–L: 2926 M–A: 4180 M–D: 4180	M–L: 2926 M–A: 4180 M–D: 4180
IMDb2	# movie (M): 4180 # director (D): 2081 # actor (A): 1739 # link (L): 4180	M–L: 2926 D–L: 4180 A–L: 2926	–
IMDb3	# movie (M): 4180 # director (D): 2081 # actor (A): 1739 # link (L): 4180	A–L: 12534 M–L: 2926 M–D: 4180	A–L: 12534 M–L: 2926 M–D: 4180
IMDb4	# movie (M): 4180 # director (D): 2081 # actor (A): 1739 # link (L): 4180	M–L: 2926 M–A: 4180 D–L: 2926	–
IMDb*	# movie (M): 4180 # director (D): 2081 # actor (A): 1739 # link (L): 4180	M–L: 2926 M–A: 4180 M–D: 4180 A–L: 2926 D–L: 2926	M–L: 4180 M–A: 12534 M–D: 2926 A–L: 12534 D–L: 4180

Table 4: Statistics of DBLP dataset variations for Link Prediction.

Dataset Variation	Node	Edge (task dependent)
		LP: Paper–Area
DBLP1	# Author (A): 4057 # Paper (P): 14328 # Terms (T): 8898 # Venues (V): 20 # Areas (Ar): 4	A–P: 19645 P–T: 114273 P–V: 10029 P–Ar: 15213
DBLP2	# Author (A): 4057 # Paper (P): 14328 # Terms (T): 8898 # Venues (V): 20 # Areas (Ar): 4	A–P: 19645 P–T: 114273 P–V: 10029 V–Ar: 78
DBLP3	# Author (A): 4057 # Paper (P): 14328 # Terms (T): 8898 # Venues (V): 20 # Areas (Ar): 4	A–P: 19645 P–T: 114273 P–V: 10029 A–Ar: 4057
DBLP*	# Author (A): 3948 # Paper (P): 13462 # Terms (T): 8642 # Venues (V): 20 # Areas (Ar): 4	A–P: 17568 P–T: 107154 P–V: 9448 P–Ar: 13462

1080 CMPNN is a source-conditioned path propagation model for relational link prediction. Given a
 1081 candidate link prediction query, CMPNN propagates information through the graph conditioned on
 1082 the source node and queried relation. This allows representations to capture path-based evidence
 1083 relevant to the candidate target, rather than relying only on node embeddings computed independently
 1084 of the link being scored. We include CMPNN as a link-prediction-specific baseline because its
 1085 conditional propagation mechanism is naturally aligned with predicting missing relational edges.

1086 MAGNN is a metapath-based message passing model for heterogeneous graphs. It uses predefined
 1087 metapaths to guide aggregation over multi-hop heterogeneous neighborhoods. For each metapath,
 1088 MAGNN encodes metapath instances by incorporating information from the intermediate nodes
 1089 along the path. It then applies attention across metapaths to learn the relative importance of different
 1090 semantic paths for prediction. This makes MAGNN a representative baseline for metapath-instance-
 1091 based heterogeneous message passing.

1092 SeHGNN is a metapath-based feature propagation model for heterogeneous graphs. Rather than
 1093 repeatedly performing neighbor aggregation during training, SeHGNN first constructs features
 1094 propagated along selected metapaths using lightweight aggregation. These metapath-propagated
 1095 features are then fused with a transformer-based module that learns interactions across different
 1096 semantic channels. This makes SeHGNN a representative baseline for models that use metapath-
 1097 derived features rather than explicit metapath-instance message passing.

1098 DHN is a homomorphism-pattern-based graph neural network. Unlike message-passing models that
 1099 primarily aggregate information from local neighborhoods or predefined metapaths, DHN represents
 1100 graphs through homomorphism counts from a chosen family of pattern graphs. These pattern
 1101 homomorphism features can capture higher-order structural information such as motifs, repeated
 1102 substructures, and other global graph patterns that may not be easily represented by local aggregation
 1103 alone. In our benchmark, DHN therefore serves as a structurally expressive baseline whose inductive
 1104 bias is based on pattern homomorphisms rather than node-type-pair relations, attention over typed
 1105 channels, source-conditioned paths, or metapath-guided message passing.

1106 **Hyper-parameters.** We tune hyper-parameters separately for each model and dataset. For MAGNN
 1107 on IMDB, we use feature type 0, 2 layers, hidden dimension 64, 8 attention heads, attention vector
 1108 dimension 128, and the RotatE0 aggregator. Models are trained for up to 300 epochs with early
 1109 stopping (patience = 30). For MAGNN on DBLP link prediction, we use feature type 0, hidden
 1110 dimension 64, 8 attention heads, attention vector dimension 128, and RotatE0. Models are trained
 1111 for up to 100 epochs with early stopping (patience = 5). For RGCN on IMDB, we use input, hidden,
 1112 and output dimensions of 128, 2 layers, 8 bases, dropout 0.3, learning rate 2×10^{-3} , and weight
 1113 decay 10^{-5} . Models are trained for up to 200 epochs with early stopping (patience = 20). For RGCN
 1114 on DBLP, we use input dimension 128, hidden and output dimensions 256, 3 layers, 16 bases, dropout
 1115 0.1, learning rate 2×10^{-3} , weight decay 10^{-5} , and embedding regularization 10^{-6} . Training is
 1116 performed for up to 200 epochs with early stopping (patience = 20). For CMPNN, we use input
 1117 dimension 32, hidden dimension 32, and 6 message-passing layers. Models are trained for up to 100
 1118 epochs with early stopping (patience = 5). In our setting, both the additional readout component
 1119 and the history function are disabled by default. For SlotGAT on IMDB, we use feature type 0, 2
 1120 layers, hidden dimension 64, 8 attention heads, learning rate 5×10^{-3} , weight decay 1×10^{-4} , and
 1121 the slot attention aggregator with attention dimension 3. Models are trained for up to 300 epochs
 1122 with early stopping (patience = 40). For SeHGNN on IMDB, we use embedding and hidden size 512,
 1123 max feature propagation hops 4, max label propagation hops 0, 2 feature projection layers, a single
 1124 task-classification layer, learning rate 5×10^{-3} , and weight decay 0. Models are trained for up to
 1125 200 epochs with early stopping (patience = 40). For DHN, we use two homomorphism-convolution
 1126 layers with rooted patterns $\{p_1, c_2\}$, output dimension 64 per pattern, and ReLU activation. For
 1127 IMDB node classification, we use the provided node features and AdamW with learning rate 5×10^{-3}
 1128 and weight decay 10^{-3} . We train for 200 epochs with an early stopping patience of 40 and retain
 1129 the checkpoint with the best validation accuracy. For IMDB link prediction, we use 64-dimensional
 1130 learned node embeddings and a dot-product decoder with the same AdamW settings, training for up
 1131 to 200 epochs with early stopping (patience = 15). For DBLP link prediction, we use 64-dimensional
 1132 learned node embeddings and Adam with learning rate 5×10^{-3} and weight decay 10^{-3} , training for
 1133 up to 100 epochs with early stopping (patience = 5). For DHN on WordNet link prediction, we use
 1134 64-dimensional learned entity and relation embeddings with a DistMult decoder. We optimize using
 1135 Adam with learning rate 10^{-3} and weight decay 10^{-4} , using 10 balanced head- and tail-corrupted

Table 5: SeHGNN Metapath-feature mapping statistics

Variant	#Metapaths
IMDb1	25
IMDb2	19
IMDb3	17
IMDb4	19
IMDb* + RestrictedK	39
IMDb* + FullK	77

Table 6: Metapaths for DBLP dataset variations (LP: Paper–Venue). Skip nodes are denoted by (X).

Dataset	Original	Invariant Mapping
DBLP1	A: PAP , VPAPV T: PTP Ar: PArP V: PVP , VPV	A: PAP , PA(P)Ar(P)AP , VPAPV T: PTP Ar: PArP V: PV(P)Ar(P)VP , PVP , VPV
DBLP2	A: PAP , VPAPV T: PTP Ar: PVA rVP V: PVP , VPV	A: PAP , PA(P)(V)Ar(V)(P)AP , VPAPV T: PTP Ar: PVA rVP V: P(V)Ar(V)P , PVP , VPV
DBLP3	A: PAP , PAArAP , VPAPV T: PTP Ar: PAArAP V: PVP , VPV	A: PAP , VPAPV T: PTP Ar: PAArAP , P(A)Ar(A)P V: PV(P)(A)Ar(A)(P)VP , PVP , VPV

negatives per positive training triple. Models are trained for up to 200 epochs with early stopping (patience = 10). For all experiments, results are reported over three runs using three different seeds.

Hardware. For RGCN, CMPNN, MAGNN and DHN architectures, we run experiments on one NVIDIA V100 GPU with 32GB of VRAM. For SlotGAT and SeHGNN architectures, we run experiments on one NVIDIA H100 GPU with 80 GB of VRAM.

H.1 Invariant Mappings

Our theoretical framework defines invariance over transformation-equivalent graph semantics, rather than over a particular GNN architecture. Accordingly, an invariant baseline must be constructed by applying the transformation-invariant mapping to the structural object on which the model operates. Since different architectures consume heterogeneous structure in different forms, we instantiate the same theoretical mapping in a model-specific way. We evaluate each baseline in its original form, denoted as the base model name (e.g RGCN), and in one or more invariant forms, denoted by appending an asterisk to the model name (e.g. RGCN*). In all cases, the invariant instantiation modifies only the model-specific structural input, while keeping the architecture, loss function, training objective, and training protocol unchanged. This directly implements the transformation-invariant mapping defined in Section 4.3.

For graph-level mappings, the structural object is the typed graph consumed by the model. This includes RGCN, CMPNN, and SlotGAT, which correspond to local node-type-pair message passing, source-conditioned path propagation, and local node-type-aware message passing, respectively. For these architectures, we instantiate the invariant mapping by constructing the union graph over transformation-equivalent variants, denoted IMDb* and DBLP*. The union graph exposes relational evidence that may appear under different non-isomorphic realizations of the same semantic structure, thereby providing a transformation-invariant graph-level input without changing the message-passing architecture.

For MAGNN, the structural object is the metapath set supplied to the model. Since MAGNN performs metapath-based message passing, we instantiate the invariant mapping as a metapath-set mapping. We construct invariant metapath sets from the metapaths induced by the transformation-equivalent variants. We consider two instantiations. UNION includes all metapaths observed across variants and adds reconstruction metapaths with skip nodes when needed so that cross-variant semantic

Table 7: Metapaths for IMDB dataset variations (NC: Movie). Skip nodes are denoted by (X).

Dataset	Original	Invariant Mapping
IMDb1	M: MDM , MAM , MLM D: DMD , DMAMD , DMLMD A: AMA , AMDMA , AMLMA L: LML , LMDML , LMAML	M: MDM , ML(M)D(M)LM , MAM , ML(M)A(M)LM , MLM D: DMD , D(M)LML(M)D , DMAMD , D(M)L(M)A(M)L(M)D , D(M)LMAML(M)D , DML(M)A(M)LMD , DMLMD , D(M)L(M)D A: AMA , A(M)LML(M)A , AMDMA , AML(M)D(M)LMA , A(M)LMDML(M)A , A(M)L(M)D(M)L(M)A , AMLMA , A(M)L(M)A L: LML , LMDML , L(M)D(M)L , LMAML , L(M)A(M)L
IMDb2	M: MLDLM , MLALM , MLM D: DLMLD , DLALD , DLD A: ALMLA , ALDLA , ALA L: LML , LDL , LAL	M: MLDLM , M(L)D(L)M , MLALM , M(L)A(L)M , MLM D: DLMLD , D(L)M(L)D , DLALD , D(L)M(L)A(L)M(L)D , DLM(L)A(L)MLD , D(L)MLALM(L)D , DLD , D(L)MLM(L)D A: ALMLA , A(L)M(L)A , ALDLA , A(L)M(L)D(L)M(L)A , ALM(L)D(L)MLA , A(L)MLDLM(L)A , ALA , A(L)MLM(L)A L: LML , LM(L)D(L)ML , LDL , LAL , LM(L)A(L)ML
IMDb3	M: MDM , MLALM , MLM D: DMD , DMLALMD , DMLMD A: ALMLA , ALMDMLA , ALA L: LML , LMDML , LAL	M: MDM , ML(M)D(M)LM , MLALM , M(L)A(L)M , MLM D: DMD , D(M)LML(M)D , DMLALMD , DM(L)A(L)MD , D(M)LAL(M)D , D(M)LM(L)A(L)ML(M)D , DMLMD , D(M)L(M)D A: ALMLA , A(L)M(L)A , ALMDMLA , AL(M)D(M)LA , A(L)MDM(L)A , A(L)ML(M)D(M)LML(L)A , ALA , A(L)MLM(L)A L: LML , LMDML , L(M)D(M)L , LAL , LM(L)A(L)ML
IMDb4	M: MLDLM , MAM , MLM D: DLMLD , DLMAMLD , DLD A: AMA , AMLDLMA , AMLMA L: LML , LDL , LMAML	M: MLDLM , M(L)D(L)M , MAM , ML(M)A(M)LM , MLM D: DLMLD , D(L)M(L)D , DLMAMLD , D(L)MAM(L)D , DL(M)A(M)LD , D(L)ML(M)A(M)LML(L)D , DLD , D(L)MLM(L)D A: AMA , A(M)LML(M)A , AMLDLMA , A(M)LM(L)D(L)ML(M)A , A(M)LDL(M)A , AM(L)D(L)MA , AMLMA , A(M)L(M)A L: LML , LDL , LM(L)D(L)ML , LMAML , L(M)A(M)L

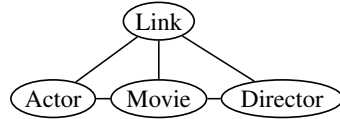
dependencies remain expressible. INTERSECT keeps only the shortest reconstruction required to preserve each cross-variant dependency. These variants test whether transformation invariance requires high-recall metapath coverage or whether a compact invariant metapath set is sufficient.

For SeHGNN, the structural object is the set of metapath-based feature channels generated during preprocessing. Since SeHGNN performs metapath-based feature propagation rather than direct metapath-instance message passing, we instantiate the invariant mapping as a metapath-feature mapping applied to the feature-propagation pipeline. In FULLK, we run the original SeHGNN feature-propagation procedure on the invariant union graph and generate all semantic channels up to length k . In RESTRICTEDK, we restrict the generated semantic channels to those appearing in the original transformed variants. This separates the effect of invariant structural augmentation from the effect of expanding the semantic feature space.

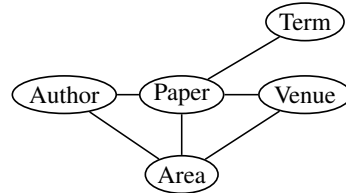
For DHN, the structural object is the set of rooted homomorphism-pattern occurrences supplied to the homomorphism-convolution layers. Since DHN aggregates information over occurrences of predefined rooted patterns, we instantiate the invariant mapping as a homomorphism-pattern mapping. We first construct the canonical union graph over transformation-equivalent variants and then recompute the same rooted patterns, $(\{p_1, c_2\})$, on this graph. Consequently, every equivalent

Table 8: Metapaths for IMDB dataset variations (LP). Skip nodes are denoted by (X) .

Task	Dataset	Original	Invariant Mapping
LP: Movie-Link	IMDb1	M: LML D: MDM , LMDML A: MAM , LMAML L: MLM	M: LML D: MDM , L(M)D(M)L , ML(M)D(M)LM , LMDML A: MAM , LMAML , ML(M)A(M)LM , L(M)A(M)L L: MLM
	IMDb2	M: LML D: MLDLM , LDL A: MLALM , LAL L: MLM	M: LML D: MLDLM , LDL , M(L)D(L)M , LM(L)D(L)ML A: MLALM , LAL , M(L)A(L)M , LM(L)A(L)ML L: MLM
	IMDb3	M: LML D: MDM , LMDML A: MLALM , LAL L: MLM	M: LML D: MDM , LMDML , ML(M)D(M)LM , L(M)D(M)L A: MLALM , LAL , M(L)A(L)M , LM(L)A(L)ML L: MLM
	IMDb4	M: LML D: MLDLM , LDL A: MAM , LMAML L: MLM	M: LML D: MLDLM , LDL , M(L)D(L)M , LM(L)D(L)ML A: MAM , LMAML , ML(M)A(M)LM , L(M)A(M)L L: MLM
LP: Movie-Director	IMDb1	M: DMD D: MDM , DMLMD A: MAM , DMAMD L: MLM	M: DMD , D(M)LML(M)D D: MDM , ML(M)D(M)LM , DMLMD A: MAM , DMAMD , DML(M)A(M)LMD , D(M)LMAML(M)D , ML(M)A(M)LM , D(M)L(M)A(M)L(M)D L: MLM , D(M)L(M)D
	IMDb3	M: DMD D: MDM A: MLALM , DMLALMD L: MLM , DMLMD	M: DMD , D(M)LML(M)D D: MDM , ML(M)D(M)LM A: MLALM , DMLALMD , M(L)A(L)M , DM(L)A(L)MD , D(M)LAL(M)D , D(M)LM(L)A(L)ML(M)D L: MLM , DMLMD , D(M)L(M)D



(a) IMDb*



(b) DBLP*

Figure 3: Graph-level mappings for IMDb and DBLP.

variant is mapped to an identical set of structural pattern occurrences, providing a transformation-invariant input without modifying the underlying homomorphism-convolution layers.

The graph mappings are displayed in Figure 3. Original metapaths and their mappings used in MAGNN can be found in Tables 6, 8, and 7. SeHGNN metapath-feature mapping statistics can be found in Table 5.

Architecture	Structural bias	Mapping type
RGCN	Local node-type-pair message passing	Graph mapping
CMPNN	Source-conditioned path propagation	Graph mapping
SlotGAT	Local node-type-aware message passing	Graph mapping
MAGNN	Metapath-based message passing	Metapath-set mapping
SeHGNN	Metapath-based feature propagation	Metapath-feature mapping
DHN	Rooted homomorphism-pattern aggregation	Homomorphism-pattern mapping

Table 9: Model-specific invariant mapping instantiations for each baseline. Each instantiation applies the transformation-invariant mapping to the structural object used by the model while keeping the core architecture unchanged.

1186 H.2 Metrics

1187 Here we define how we compute the following metrics: Accuracy, Precision, Recall, Micro-F1,
 1188 Macro-F1, Hits@1, Hits@3, MRR, Kendall- τ , Kendall- τ @1, and Kendall- τ @3. For all Kendall-
 1189 based metrics, scores are computed over output-score rankings across graph variants and averaged
 1190 over variant pairs, test instances, and runs.

1191 **Accuracy:** The accuracy score measures the fraction of correctly classified nodes. For each node i ,
 1192 denote y_i the ground-truth class label and $\hat{y}_i$ the predicted class label. We have

$$\text{Accuracy} = \frac{1}{|S|} \sum_{i \in S} \mathbf{1}(\hat{y}_i = y_i)$$

1193 where S denotes the evaluation set.

1194 **Precision:** Precision is calculated as the ratio of true positive predictions to total positive predictions.
 1195 For a class c , let TP_c denote the number of instances correctly predicted as class c and let FP_c denote
 1196 the number of instances incorrectly predicted as class c . For a multiclass setting with C total classes,
 1197 we calculate the macro-averaged precision by taking the mean per-class precision across all classes.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad \text{Precision} = \frac{1}{C} \sum_{c=1}^C \text{Precision}_c$$

1198 **Recall:** Recall is calculated as the ratio of true positive predictions to the total number of ground-truth
 1199 positive instances of a given class. Let FN_c denote the number of instances whose true label is c but
 1200 are incorrectly predicted as another class. For a multiclass setting with C total classes, we calculate
 1201 the macro-averaged recall by taking the mean per-class recall across all classes.

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad \text{Recall} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c$$

1202 **Macro-F1:** Macro-F1 is the mean of per-class F1 scores, where each per-class F1 score is the
 1203 harmonic mean of the per-class Precision $_c$ and Recall $_c$ scores defined above. For a multiclass setting
 1204 with C total classes, we have

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

1205 **Micro-F1:** Micro-F1 is the harmonic mean of Micro-Precision and Micro-Recall. For a multiclass
 1206 setting with C classes we have

$$\text{Micro-Precision} = \frac{\sum_{c=1}^C TP_c}{\sum_{c=1}^C (TP_c + FP_c)} \quad \text{Micro-Recall} = \frac{\sum_{c=1}^C TP_c}{\sum_{c=1}^C (TP_c + FN_c)}$$

1207 and the corresponding Micro-F1 score is

$$\text{Micro-F1} = \frac{2 \cdot \text{Micro-Precision} \cdot \text{Micro-Recall}}{\text{Micro-Precision} + \text{Micro-Recall}}$$

1208 **Hits@k:** Hits@k is a ranking metric used for link prediction that measures whether the correct
1209 answer is within the top k positions of a ranked candidate list. For each query node i , a true positive
1210 edge is ranked among K sampled negative edges, giving $1 + K$ candidates total. Denote r_i as the
1211 rank of the true positive among all candidates and Q as the set of all query nodes. We have

$$\text{Hits}@k_i = \mathbf{1}(r_i \leq k) \quad \text{Hits}@k = \frac{1}{|Q|} \sum_{i \in Q} \mathbf{1}(r_i \leq k)$$

1212 We report Hits@1 and Hits@3, i.e. $k \in \{1, 3\}$.

1213 **MRR:** Mean Reciprocal Rank (MRR) is a metric used for link prediction that measures the average
1214 reciprocal rank of the true positive edge across all query nodes. For each query node i , a true positive
1215 edge is ranked among K sampled negative edges. Denote r_i as the rank of the true positive among
1216 all $1 + K$ candidates and Q as the set of all query nodes. We have

$$\text{MRR} = \frac{1}{|Q|} \sum_{i \in Q} \frac{1}{r_i}$$

1217 **Kendall- τ :** Kendall- τ measures the similarity between two rankings by comparing the number of
1218 concordant and discordant pairs. In our setting, the rankings are induced by the model’s output
1219 scores across graph variants. Let $\mathcal{V} = \{G^{(1)}, \dots, G^{(m)}\}$ denote the set of dataset variants, and let
1220 $s^{(v,r)}(x, c)$ be the score assigned to candidate output c for instance x under variant $G^{(v)}$ and run r .
1221 These scores induce a ranking

$$\pi_x^{(v,r)} = \text{rank}_{c \in \mathcal{C}(x)} s^{(v,r)}(x, c),$$

1222 where $\mathcal{C}(x)$ is the candidate output set for instance x . For two variants $G^{(u)}$ and $G^{(v)}$, Kendall- τ
1223 compares the corresponding rankings $\pi_x^{(u,r)}$ and $\pi_x^{(v,r)}$. Let P_x be the set of unordered candidate
1224 pairs in $\mathcal{C}(x)$. A pair $(a, b) \in P_x$ is concordant if both rankings order a and b in the same way, and
1225 discordant otherwise. Denoting the number of concordant and discordant pairs by $C_x^{u,v,r}$ and $D_x^{u,v,r}$,
1226 respectively, we compute

$$\tau_x^{u,v,r} = \frac{C_x^{u,v,r} - D_x^{u,v,r}}{C_x^{u,v,r} + D_x^{u,v,r}}.$$

1227 The reported Kendall- τ score averages over all test instances, all variant pairs, and all runs:

$$\text{Kendall-}\tau = \frac{1}{R} \sum_{r=1}^R \frac{2}{m(m-1)} \sum_{u < v} \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \tau_x^{u,v,r}.$$

1228 Higher values indicate that the model produces more similar output rankings across graph variants.

1229 **Kendall- τ @K:** Kendall- τ @K measures rank agreement among the K top-ranked predictions across
1230 graph variants. For instance x , run r , and variants $G^{(u)}$ and $G^{(v)}$, let

$$\mathcal{C}_{u,v,r}^3(x) = \text{Top}_K(\pi_x^{(u,r)}) \cup \text{Top}_K(\pi_x^{(v,r)})$$

1231 be the union of the top-K candidates from the two rankings. We then restrict both rankings to this
1232 shared candidate set and compute Kendall- τ :

$$\tau @ K_x^{u,v,r} = \tau \left(\pi_x^{(u,r)}|_{\mathcal{C}_{u,v,r}^K(x)}, \pi_x^{(v,r)}|_{\mathcal{C}_{u,v,r}^K(x)} \right).$$

1233 The reported Kendall- τ @K scores is averaged over all test instances, all variant pairs, and all runs:

$$\text{Kendall-}\tau @ K = \frac{1}{R} \sum_{r=1}^R \frac{2}{m(m-1)} \sum_{u < v} \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \tau @ K_x^{u,v,r}.$$

1234 We report Kendall- τ @K for link prediction to evaluate whether the top-ranked venue predictions
1235 remain stable across graph variants.

1236 For the data-augmentation experiments, the prefix before a colon identifies the variants used to
1237 train the shared model, while the text after the colon identifies the two input-variant rankings being
1238 compared. For example, *Data Augmentation: IMDb1 vs IMDb2* means that one model was jointly
1239 trained on IMDb1–4 and its test rankings on IMDb1 and IMDb2 were compared. The joint training
1240 regime does not itself produce a separate “IMDb1–4” test ranking.

Table 10: IMDb node classification results

Model	Method	Variant	Accuracy	Precision	Recall	Micro-F1	Macro-F1
RGCN	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
SlotGAT	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
DHN	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
MAGNN	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant + Union	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant + Intersect	IMDb1 IMDb2 IMDb3 IMDb4					
Model	Method	Variant	Accuracy	Precision	Recall	Micro-F1	Macro-F1
PathNet	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
MAGNA	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
PAIN	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					
SeHGNN	Original	IMDb1 IMDb2 IMDb3 IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1 IMDb2 IMDb3 IMDb4					
	Invariant	IMDb1 IMDb2 IMDb3 IMDb4					

Table 11: IMDb node classification invariance

[illegible]

Table 12: IMDb node classification scalability

Model	Method	Variant	Train Time (s)	Epochs	Parameter (MiB)	Peak GPU (MiB)	
						Train	Infer.
RGCN	Original	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
SlotGAT	Augmentation	IMDb1-4					
	Invariant	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
DHN	Original	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
MAGNN	Augmentation	IMDb1-4					
	Invariant + Union	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Original	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
PathNet	Augmentation	IMDb1-4					
	Invariant	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
MAGNA	Original	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
PAIN	Augmentation	IMDb1-4					
	Invariant	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
SeHGNN	Original	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					
	Canonical $\bigcup_{i \in \{1,2,3,4\}} \text{IMDb}_i$	$\bigcup_i \text{IMDb}_i$					
	Augmentation	IMDb1-4					
	Invariant	IMDb1					
		IMDb2					
		IMDb3					
		IMDb4					

Table 13: IMDb link prediction results

Model	Method	Task	Variant	Precision	Recall	F1	Hits@1	Hits@3	MRR
RGCN	Original	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$						
		Movie – Director $i \in \{1, 3\}$	$\bigcup_i \text{IMDb}_i$						
	Augmentation	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
MAGNN	Original	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$						
		Movie – Director $i \in \{1, 3\}$	$\bigcup_i \text{IMDb}_i$						
	Augmentation	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant + Union	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant + Intersect	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
CMPNN	Original	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$						
		Movie – Director $i \in \{1, 3\}$	$\bigcup_i \text{IMDb}_i$						
	Augmentation	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
DHN	Original	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$						
		Movie – Director $i \in \{1, 3\}$	$\bigcup_i \text{IMDb}_i$						
	Augmentation	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
MAGNA	Original	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$						
		Movie – Director $i \in \{1, 3\}$	$\bigcup_i \text{IMDb}_i$						
	Augmentation	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						
	Invariant	Movie – Year	IMDb1 IMDb2 IMDb3 IMDb4						
		Movie – Director	IMDb1 IMDb3						

Table 14: IMDb link prediction invariance

[illegible]

Table 15: IMDb link prediction scalability

Model	Method	Task	Variant	Train Time (s)	Epochs	Parameter (MIB)	Peak GPU (MIB)	
							Train	Infer.
RGCN	Original	Movie – Year	IMDb1					
			IMDb2					
			IMDb3					
			IMDb4					
	Movie – Director		IMDb1					
			IMDb3					
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$					
	Movie – Director $i \in \{1, 3\}$		$\bigcup_i \text{IMDb}_i$					
MAGNN	Augmentation	Movie – Year	IMDb1-4					
			IMDb1,3					
	Movie – Director		IMDb1					
			IMDb3					
	Invariant	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Invariant + Union	Movie – Year	IMDb1					
			IMDb2					
MAGNA	Augmentation	Movie – Year	IMDb3					
			IMDb4					
	Movie – Director		IMDb1					
			IMDb3					
	Invariant	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Invariant + Intersect	Movie – Year	IMDb1					
			IMDb3					
CMPNN	Original	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$					
	Movie – Director $i \in \{1, 3\}$		$\bigcup_i \text{IMDb}_i$					
	Augmentation	Movie – Year	IMDb1-4					
			IMDb1,3					
DHN	Invariant	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$					
	Movie – Director $i \in \{1, 3\}$		$\bigcup_i \text{IMDb}_i$					
	Augmentation	Movie – Year	IMDb1-4					
			IMDb1,3					
MAGNA	Invariant	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$					
	Movie – Director $i \in \{1, 3\}$		$\bigcup_i \text{IMDb}_i$					
	Augmentation	Movie – Year	IMDb1-4					
			IMDb1,3					
MAGNA	Invariant	Movie – Year	IMDb1					
			IMDb2					
	Movie – Director		IMDb3					
			IMDb4					
	Canonical $\bigcup_i \text{IMDb}_i$	Movie – Year $i \in \{1, 2, 3, 4\}$	$\bigcup_i \text{IMDb}_i$					
	Movie – Director $i \in \{1, 3\}$		$\bigcup_i \text{IMDb}_i$					
	Augmentation	Movie – Year	IMDb1-4					
			IMDb1,3					

Table 16: DBLP link prediction results

Model	Method	Variant	Precision	Recall	F1	Hits@1	Hits@3	MRR
RGCN	Original	DBLP1 DBLP2 DBLP3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$						
	Augmentation	DBLP1 DBLP2 DBLP3						
	Invariant	DBLP1 DBLP2 DBLP3						
MAGNN	Original	DBLP1 DBLP2 DBLP3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$						
	Augmentation	DBLP1 DBLP2 DBLP3						
	Invariant + Union	DBLP1 DBLP2 DBLP3						
	Invariant + Intersect	DBLP1 DBLP2 DBLP3						
CMPNN	Original	DBLP1 DBLP2 DBLP3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$						
	Augmentation	DBLP1 DBLP2 DBLP3						
	Invariant	DBLP1 DBLP2 DBLP3						
DHN	Original	DBLP1 DBLP2 DBLP3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$						
	Augmentation	DBLP1 DBLP2 DBLP3						
	Invariant	DBLP1 DBLP2 DBLP3						
MAGNA	Original	DBLP1 DBLP2 DBLP3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$						
	Augmentation	DBLP1 DBLP2 DBLP3						
	Invariant	DBLP1 DBLP2 DBLP3						

Table 17: DBLP link prediction scalability

Model	Method	Variant	Train Time (s)	Epochs	Parameter (MiB)	Peak GPU (MiB)
RGCN	Original	DBLP1 DBLP2 DBLP3				
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$				
	Augmentation	DBLP1-3				
	Invariant	DBLP1 DBLP2 DBLP3				
MAGNN	Original	DBLP1 DBLP2 DBLP3				
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$				
	Augmentation	DBLP1-3				
	Invariant + Union	DBLP1 DBLP2 DBLP3				
	Invariant + Intersect	DBLP1 DBLP2 DBLP3				
CMPNN	Original	DBLP1 DBLP2 DBLP3				
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$				
	Augmentation	DBLP1-3				
	Invariant	DBLP1 DBLP2 DBLP3				
DHN	Original	DBLP1 DBLP2 DBLP3				
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$				
	Augmentation	DBLP1-3				
	Invariant	DBLP1 DBLP2 DBLP3				
MAGNA	Original	DBLP1 DBLP2 DBLP3				
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$				
	Augmentation	DBLP1-3				
	Invariant	DBLP1 DBLP2 DBLP3				

Table 18: DBLP link prediction scalability

Model	Method	Variant	Train Time (s)	Epochs	Parameter (MiB)	Peak GPU (MiB)	
						Train	Infer.
RGCN	Original	DBLP1 DBLP2 DBLP3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$					
	Augmentation	DBLP1-3					
	Invariant	DBLP1 DBLP2 DBLP3					
MAGNN	Original	DBLP1 DBLP2 DBLP3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$					
	Augmentation	DBLP1-3					
	Invariant	DBLP1 DBLP2 DBLP3					
	Invariant + Union	DBLP1 DBLP2 DBLP3					
	Invariant + Intersect	DBLP1 DBLP2 DBLP3					
CMPNN	Original	DBLP1 DBLP2 DBLP3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$					
	Augmentation	DBLP1-3					
	Invariant	DBLP1 DBLP2 DBLP3					
DHN	Original	DBLP1 DBLP2 DBLP3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$					
	Augmentation	DBLP1-3					
	Invariant	DBLP1 DBLP2 DBLP3					
MAGNA	Original	DBLP1 DBLP2 DBLP3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{DBLP}_i$	$\bigcup_i \text{DBLP}_i$					
	Augmentation	DBLP1-3					
	Invariant	DBLP1 DBLP2 DBLP3					

Table 19: WordNet (WN18RR) link prediction results

Model	Method	Variant	Precision	Recall	F1	Hits@1	Hits@3	MRR
RGCN	Original	WN18RR1 WN18RR2 WN18RR3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$						
	Augmentation	WN18RR1 WN18RR2 WN18RR3						
	Invariant	WN18RR1 WN18RR2 WN18RR3						
CMPNN	Original	WN18RR1 WN18RR2 WN18RR3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$						
	Augmentation	WN18RR1 WN18RR2 WN18RR3						
	Invariant	WN18RR1 WN18RR2 WN18RR3						
DHN	Original	WN18RR1 WN18RR2 WN18RR3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$						
	Augmentation	WN18RR1 WN18RR2 WN18RR3						
	Invariant	WN18RR1 WN18RR2 WN18RR3						
MAGNA	Original	WN18RR1 WN18RR2 WN18RR3						
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$						
	Augmentation	WN18RR1 WN18RR2 WN18RR3						
	Invariant	WN18RR1 WN18RR2 WN18RR3						

Table 20: WordNet (WN18RR) link prediction invariance

Model	Method	Comparison	Kendall- τ	Kendall- τ @1	Kendall- τ @3
RGCN	Original	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3	1.000 ± 0.0000	1.000 ± 0.0000	1.000 ± 0.0000
	Augmentation	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Invariant	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
CMPNN	Original	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3	1.000 ± 0.0000	1.000 ± 0.0000	1.000 ± 0.0000
	Augmentation	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Invariant	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
DHN	Original	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3	1.000 ± 0.0000	1.000 ± 0.0000	1.000 ± 0.0000
	Augmentation	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Invariant	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
MAGNA	Original	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3	1.000 ± 0.0000	1.000 ± 0.0000	1.000 ± 0.0000
	Augmentation	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			
	Invariant	WN18RR1 vs. WN18RR2 WN18RR1 vs. WN18RR3 WN18RR2 vs. WN18RR3			

Table 21: WordNet (WN18RR) link prediction scalability

Model	Method	Variant	Train Time (s)	Epochs	Parameter (MiB)	Peak GPU (MiB)	
						Train	Infer.
RGCN	Original	WN18RR1 WN18RR2 WN18RR3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$					
	Augmentation	WN18RR1-3					
	Invariant	WN18RR1 WN18RR2 WN18RR3					
CMPNN	Original	WN18RR1 WN18RR2 WN18RR3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$					
	Augmentation	WN18RR1-3					
	Invariant	WN18RR1 WN18RR2 WN18RR3					
DHN	Original	WN18RR1 WN18RR2 WN18RR3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$					
	Augmentation	WN18RR1-3					
	Invariant	WN18RR1 WN18RR2 WN18RR3					
MAGNA	Original	WN18RR1 WN18RR2 WN18RR3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{WN18RR}_i$	$\bigcup_i \text{WN18RR}_i$					
	Augmentation	WN18RR1-3					
	Invariant	WN18RR1 WN18RR2 WN18RR3					

Table 22: Freebase node classification results

Model	Method	Variant	Accuracy	Precision	Recall	Micro-F1	Macro-F1
RGCN	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
SlotGAT	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
DHN	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
MAGNN	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant + Union	Freebase1 Freebase2 Freebase3					
	Invariant + Intersect	Freebase1 Freebase2 Freebase3					

Model	Method	Variant	Accuracy	Precision	Recall	Micro-F1	Macro-F1
PathNet	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
MAGNA	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
PAIN	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					
SeHGNN	Original	Freebase1 Freebase2 Freebase3					
	Canonical $\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase1 Freebase2 Freebase3					
	Invariant	Freebase1 Freebase2 Freebase3					

This section provides a more detailed discussion of the trade-offs among effectiveness, efficiency, and degree of invariance across the evaluated structural biases. The main results in Tables 25, 27, and 29 show that the invariant mappings generally preserve or improve task performance, while the invariance results in Tables 26, 28, and 30 show that they remove output-level variation across transformation-equivalent dataset variants. In the original baselines, Kendall- τ varies substantially across models and tasks, indicating that different structural biases have different degrees of natural invariance. After applying the invariant mapping, however, all invariant counterparts achieve Kendall- $\tau = 1$ with zero variance across the corresponding variant pairs. Since Kendall- τ measures agreement between output-score rankings across graph variants, this indicates that the invariant structural inputs induce identical output rankings across equivalent variants. Thus, the comparison between each baseline and its invariant counterpart should be interpreted primarily as a change in degree of invariance, rather than only as a conventional performance comparison.

Degree of invariance. The original baselines exhibit substantially different degrees of invariance. On IMDB node classification, Table 26 shows that SlotGAT is the most invariant original baseline, followed by MAGNN and SeHGNN, while RGCN is the least invariant. This suggests that type-aware and metapath-based structural biases can partially reduce sensitivity to transformation-equivalent graph variants, but they do not eliminate it. In contrast, RGCN*, SlotGAT*, MAGNN*+Union, MAGNN*+Intersect, SeHGNN*+RestrictedK, and SeHGNN*+FullK all achieve perfect Kendall- τ across IMDB variants. This shows that the invariant mapping removes the residual dependence on how the same information is represented in the input graph.

A similar pattern appears in link prediction. On DBLP, Table 28 shows that MAGNN and CMPNN have moderate output-ranking agreement across variants, whereas RGCN has much lower Kendall- τ , indicating stronger dependence on the particular graph representation produced by the mapping. After invariant mapping, MAGNN*+Union, MAGNN*+Intersect, RGCN*, and CMPNN* all achieve

Table 23: Freebase node classification invariance

Model	Method	Comparison	Kendall- τ	Model	Method	Comparison	Kendall- τ	Model	Method	Comparison	Kendall- τ
MAGNN	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	RGCN	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	PathNet	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Invariant + Union	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Invariant + Intersect	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3									
SeHGNN	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	SlotGAT	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	MAGNA	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3									

Model	Method	Comparison	Kendall- τ	Model	Method	Comparison	Kendall- τ	Model	Method	Comparison	Kendall- τ
RGCN	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	SlotGAT	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	MAGNA	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3									
SlotGAT	Original	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	DHN	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000	PAIN	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3	1.000 $\pm$ 0.000
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Augmentation	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3			Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3					
	Invariant	Freebase1 vs. Freebase2 Freebase1 vs. Freebase3 Freebase2 vs. Freebase3									

Table 24: Freebase node classification scalability

Model	Method	Variant	Train Time (s)	Epochs	Parameter (MiB)	Peak GPU (MiB)	
						Train	Infer.
RGCN	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
SlotGAT	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
DHN	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
MAGNN	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant + Union	Freebase 1 Freebase 2 Freebase 3					
	Invariant + Intersect	Freebase 1 Freebase 2 Freebase 3					
PathNet	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
MAGNA	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
PAIN	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					
SeHGNN	Original	Freebase 1 Freebase 2 Freebase 3					
	$\bigcup_{i \in \{1,2,3\}} \text{Freebase}_i$	$\bigcup_i \text{Freebase}_i$					
	Augmentation	Freebase 1-3					
	Invariant	Freebase 1 Freebase 2 Freebase 3					

Kendall- $\tau = 1$, including Kendall- $\tau@1$ and Kendall- $\tau@3$. Therefore, the invariant mapping stabilizes not only the full candidate ranking, but also the top-ranked predictions that are most relevant for link prediction metrics such as Hits@1 and Hits@3. On IMDB link prediction, Table 30 shows the same qualitative behavior: the original models show varying degrees of invariance, while the invariant counterparts produce identical rankings across equivalent variants. For DHN, the original homomorphism-pattern bias also exhibits only partial invariance. Although DHN aggregates over predefined rooted pattern occurrences and can capture higher-order structural information beyond local neighborhoods, these occurrences are still defined with respect to the particular graph representation supplied to the model. As a result, transformation-equivalent variants can induce different rooted pattern occurrences even when they encode the same semantic information.

Effectiveness. The effectiveness results show that improving the degree of invariance usually preserves, and in several cases improves, task performance. On IMDB node classification, Table 25 shows that RGCN benefits most from invariant graph-level augmentation: its Macro-F1 increases substantially after applying the invariant mapping. This suggests that the original local node-type-pair message-passing bias fails to consistently access transformation-equivalent evidence across variants, while the invariant graph-level input exposes this evidence in a stable form. MAGNN also improves under the Union mapping, indicating that a higher-recall invariant metapath set can provide useful semantic coverage. SlotGAT is already the strongest original classification baseline, and SlotGAT* remains close in performance while achieving perfect invariance. This indicates that SlotGAT’s node-type-aware aggregation is effective for IMDB classification, but high task performance alone does not guarantee invariance: the original SlotGAT still produces variant-dependent output rankings, whereas SlotGAT* removes this dependence through the invariant mapping.

SeHGNN shows a more nuanced pattern in Table 25. RestrictedK slightly improves over the original model, while FullK is less effective. This suggests that invariant metapath-feature construction is useful, but that expanding the feature space too aggressively can introduce redundant or noisy propagated channels. Thus, for feature-propagation models, the most effective invariant mapping may be a compact one that preserves the feature channels induced by transformation-equivalent propagation patterns without generating all possible channels up to length k .

For DBLP link prediction, Table 27 shows that MAGNN is already the strongest original baseline, which suggests that the Paper–Venue prediction task is well aligned with metapath-based structural evidence. However, RGCN* substantially improves over RGCN and approaches MAGNN-level ranking performance. This is an important result because it shows that invariant graph-level augmentation can make a simpler local message-passing architecture competitive with a more structured metapath-based method. CMPNN and CMPNN* have similar effectiveness on DBLP, suggesting that the original source-conditioned path propagation already captures much of the task-relevant evidence, while the invariant version mainly improves degree of invariance rather than raw performance.

On IMDB link prediction, Table 29 shows that the effectiveness trends depend on the target relation. RGCN* is especially strong on Movie–Link prediction, while MAGNN*+Union is strongest on Movie–Director prediction. CMPNN* improves over CMPNN in some settings, particularly for Movie–Director, but remains more task-dependent. These differences show that invariant mapping does not collapse the models into equivalent predictors. Instead, it equalizes their degree of invariance while preserving architecture-specific strengths and weaknesses.

For DHN, the invariant mapping primarily improves consistency rather than predictive effectiveness. On IMDB node classification, Table 25 shows that DHN* remains within the performance range of the original variants, while Table 29 shows that Movie–Director performance is largely preserved. Performance decreases slightly for IMDB Movie–Link and DBLP (Table 27), and remains weak on WordNet (Table 34). These results show that enforcing invariance does not necessarily improve predictive effectiveness or compensate for a structural bias that is poorly aligned with a particular task.

For MAGNA, when comparing the invariant mapping model (MAGNA*) with the models trained on the original variants (MAGNA), the performance stays around the same for all node classification and link prediction experiments. This further shows how enforcing invariance doesn’t necessarily improve performance, but it allows the model to learn a much more stable ranking logic that is highly robust to topological noise and neighborhood perturbations.

1320 PathNet [62] is focused on heterophily, so the algorithm is built around learning a better representation
1321 for a target node using sampled paths around that node. Because of that, the paper evaluates node
1322 classification, where the main question is whether path-based aggregation helps classify nodes when
1323 nearby nodes may have different labels. Link prediction would require edge-pair scoring, negative
1324 sampling and edge splits; it is outside the heterophily/node-label focus so we only performed the
1325 Node classification task.

1326 **Efficiency.** The efficiency results show that the cost of invariance is architecture-dependent. For
1327 IMDb node classification, Table 31 shows that graph-level mappings introduce moderate overhead.
1328 RGCN is the fastest original baseline, and although RGCN* increases training time, it remains
1329 substantially more efficient than path- and metapath-based methods. This makes RGCN* an attractive
1330 trade-off in settings where efficiency and invariance are both important.

1331 For DBLP link prediction, Table 38 shows a sharp efficiency separation between local message-
1332 passing models and path- or metapath-based models. RGCN and RGCN* train in tens of seconds,
1333 whereas MAGNN and MAGNN* variants require several hours, and CMPNN also incurs substantially
1334 higher runtime than RGCN. This indicates that the computational cost is driven less by the invariant
1335 mapping itself and more by the structural object that each architecture consumes. Local graph-
1336 level mappings remain efficient, while metapath-instance aggregation and source-conditioned path
1337 propagation are more expensive.

1338 For IMDb link prediction, Table 33 shows the same qualitative separation. RGCN is by far the
1339 fastest model: original RGCN trains in only a few seconds on Movie-Link and roughly one to two
1340 seconds on Movie-Director, while RGCN* remains similarly efficient and is even faster than RGCN
1341 on Movie-Link. In contrast, MAGNN requires hundreds of seconds, MAGNN*+Union increases
1342 this cost further, and CMPNN requires substantially longer training times, especially on Movie-
1343 Director. CMPNN* reduces runtime relative to CMPNN on several variants, but it remains much
1344 more expensive than RGCN*. These results show that invariant mappings can be highly efficient
1345 when the underlying structural object is compact, while path- and metapath-based instantiations carry
1346 larger computational costs.

1347 The invariant MAGNN variants further show that the size of the invariant structural object matters. In
1348 Table 31, MAGNN*+Union is more expensive than MAGNN*+Intersect on IMDb node classification
1349 because Union retains a larger set of metapaths and reconstruction paths. Table 33 shows the same
1350 pattern at a high level for link prediction: MAGNN*+Union is more expensive than original MAGNN,
1351 reflecting the additional cost of aggregating over the invariant metapath set. However, compact
1352 invariant constructions can still achieve the same perfect degree of invariance with lower runtime and
1353 competitive effectiveness. This suggests that high-recall invariant metapath coverage is not always
1354 necessary; a compact invariant reconstruction can sometimes provide the same invariance guarantee
1355 with better efficiency.

1356 The SeHGNN variants reinforce this conclusion. Table 31 shows that FullK is significantly more
1357 expensive than RestrictedK because it generates all feature channels up to length k on the invariant
1358 union graph. However, Table 25 shows that FullK does not consistently improve effectiveness and
1359 can underperform RestrictedK. Thus, for metapath-feature propagation, expanding the invariant
1360 feature space can increase computational cost without improving predictive performance. RestrictedK
1361 provides a better effectiveness-efficiency trade-off while still achieving perfect degree of invariance.

1362 **Cross-baseline trade-offs.** Overall, the original baselines occupy different points in the
1363 effectiveness-efficiency-invariance trade-off. RGCN is efficient but has low natural invariance,
1364 especially on DBLP link prediction, as shown in Table 28. SlotGAT is effective and relatively stable
1365 on IMDb classification, but still not invariant, as shown in Tables 25 and 26. MAGNN is highly
1366 effective on DBLP link prediction and moderately invariant in its original form, but is computationally
1367 expensive, as shown in Tables 27, 28, and 38. SeHGNN provides competitive classification perfor-
1368 mance, but its invariant feature construction must be controlled to avoid unnecessary expansions, as
1369 shown by the contrast between RestrictedK and FullK in Tables 25 and 31. CMPNN is naturally
1370 aligned with link prediction through source-conditioned propagation, but its computational cost is
1371 higher than local message-passing models and its original outputs remain variant-sensitive, as shown
1372 in Tables 27, 28, 38, 29, 30, and 33.

1373 After applying the invariant mappings, the main distinction among models changes. Their degree of
1374 invariance becomes identical, as all invariant counterparts achieve Kendall- $\tau = 1$ across equivalent
1375 variants in Tables 26, 28, and 30. The remaining differences are therefore primarily differences
1376 in effectiveness and efficiency, induced by the architecture-specific structural object consumed by
1377 each model. This supports the central claim of our framework: invariance should be imposed on
1378 the structural bias used by the model, rather than by replacing the architecture or modifying the
1379 learning objective. The invariant mapping replaces each model’s original structural object with an
1380 invariant counterpart, ensuring stable outputs across transformation-equivalent inputs while keeping
1381 the architecture, training protocol, and task objective fixed.

1382 **Summary.** The results support three conclusions. First, GNN architectures differ substantially in
1383 their natural degree of invariance, and strong task performance does not imply invariance. Second,
1384 applying the proposed invariant mapping raises the degree of invariance to perfect output-ranking
1385 agreement across equivalent variants for all evaluated architectures. Third, the effectiveness and
1386 efficiency of invariant models depend on how each architecture materializes structural bias: graph-
1387 level mappings are relatively efficient, metapath-set mappings can be effective but costly, and
1388 metapath-feature mappings require careful restriction to avoid unnecessary expansion.

Table 25: IMDb NC results: accuracy, precision, recall, micro-f1, and macro-f1 over 3 runs.

Model	Dataset	Accuracy	Precision	Recall	Micro-F1	Macro-F1
RGCN	IMDb1	0.548 $\pm$ 0.005	0.545 $\pm$ 0.005	0.546 $\pm$ 0.005	0.548 $\pm$ 0.005	0.545 $\pm$ 0.005
	IMDb2	0.563 $\pm$ 0.017	0.564 $\pm$ 0.014	0.566 $\pm$ 0.014	0.563 $\pm$ 0.017	0.562 $\pm$ 0.016
	IMDb3	0.576 $\pm$ 0.008	0.574 $\pm$ 0.009	0.572 $\pm$ 0.006	0.576 $\pm$ 0.008	0.573 $\pm$ 0.007
	IMDb4	0.541 $\pm$ 0.007	0.537 $\pm$ 0.008	0.537 $\pm$ 0.004	0.541 $\pm$ 0.007	0.536 $\pm$ 0.006
	Data Augmentation	0.5660 $\pm$ 0.0038	0.5650 $\pm$ 0.0038	0.5639 $\pm$ 0.0021	0.5660 $\pm$ 0.0038	0.5623 $\pm$ 0.0038
	backup Output Fusion	0.568 $\pm$ 0.016	0.566 $\pm$ 0.015	0.567 $\pm$ 0.014	0.568 $\pm$ 0.016	0.566 $\pm$ 0.0155
RGCN*	Output Fusion	0.540 $\pm$ 0.021	0.573 $\pm$ 0.029	0.508 $\pm$ 0.032	0.540 $\pm$ 0.021	0.482 $\pm$ 0.015
	IMDb1	0.645 $\pm$ 0.012	0.648 $\pm$ 0.016	0.640 $\pm$ 0.002	0.645 $\pm$ 0.012	0.647 $\pm$ 0.013
	IMDb2	0.645 $\pm$ 0.012	0.648 $\pm$ 0.016	0.640 $\pm$ 0.002	0.645 $\pm$ 0.012	0.647 $\pm$ 0.013
	IMDb3	0.645 $\pm$ 0.012	0.648 $\pm$ 0.016	0.640 $\pm$ 0.002	0.645 $\pm$ 0.012	0.647 $\pm$ 0.013
MAGNN	IMDb4	0.645 $\pm$ 0.012	0.648 $\pm$ 0.016	0.640 $\pm$ 0.002	0.645 $\pm$ 0.012	0.647 $\pm$ 0.013
	IMDb1	0.644 $\pm$ 0.003	0.642 $\pm$ 0.005	0.643 $\pm$ 0.007	0.644 $\pm$ 0.003	0.645 $\pm$ 0.004
	IMDb2	0.580 $\pm$ 0.028	0.582 $\pm$ 0.022	0.583 $\pm$ 0.015	0.580 $\pm$ 0.028	0.568 $\pm$ 0.041
	IMDb3	0.644 $\pm$ 0.009	0.642 $\pm$ 0.005	0.643 $\pm$ 0.007	0.644 $\pm$ 0.009	0.646 $\pm$ 0.009
	IMDb4	0.593 $\pm$ 0.012	0.592 $\pm$ 0.002	0.594 $\pm$ 0.007	0.593 $\pm$ 0.012	0.588 $\pm$ 0.013
	backup Output Fusion	0.641 $\pm$ 0.003	0.644 $\pm$ 0.001	0.643 $\pm$ 0.003	0.641 $\pm$ 0.003	0.6437 $\pm$ 0.003
MAGNN*	Output Fusion	0.592 $\pm$ 0.012	0.627 $\pm$ 0.008	0.581 $\pm$ 0.014	0.592 $\pm$ 0.012	0.581 $\pm$ 0.018
	Union	0.638 $\pm$ 0.005	0.643 $\pm$ 0.002	0.637 $\pm$ 0.007	0.638 $\pm$ 0.005	0.639 $\pm$ 0.005
MAGNN* + Union	IMDb1	0.647 $\pm$ 0.011	0.649 $\pm$ 0.009	0.647 $\pm$ 0.010	0.647 $\pm$ 0.011	0.648 $\pm$ 0.010
	IMDb2	0.647 $\pm$ 0.011	0.649 $\pm$ 0.009	0.647 $\pm$ 0.010	0.647 $\pm$ 0.011	0.648 $\pm$ 0.010
	IMDb3	0.647 $\pm$ 0.011	0.649 $\pm$ 0.009	0.647 $\pm$ 0.010	0.647 $\pm$ 0.011	0.648 $\pm$ 0.010
	IMDb4	0.647 $\pm$ 0.011	0.649 $\pm$ 0.009	0.647 $\pm$ 0.010	0.647 $\pm$ 0.011	0.648 $\pm$ 0.010
MAGNN* + Intersect	IMDb1	0.639 $\pm$ 0.005	0.645 $\pm$ 0.013	0.636 $\pm$ 0.006	0.639 $\pm$ 0.005	0.638 $\pm$ 0.006
	IMDb2	0.639 $\pm$ 0.005	0.645 $\pm$ 0.013	0.636 $\pm$ 0.006	0.639 $\pm$ 0.005	0.638 $\pm$ 0.006
	IMDb3	0.639 $\pm$ 0.005	0.645 $\pm$ 0.013	0.636 $\pm$ 0.006	0.639 $\pm$ 0.005	0.638 $\pm$ 0.006
	IMDb4	0.639 $\pm$ 0.005	0.68 $\pm$ 0.013	0.636 $\pm$ 0.006	0.639 $\pm$ 0.005	0.638 $\pm$ 0.006
SlotGAT	IMDb1	0.6571 $\pm$ 0.0060	0.6640 $\pm$ 0.0093	0.6572 $\pm$ 0.0059	0.6571 $\pm$ 0.0060	0.6584 $\pm$ 0.0055
	IMDb2	0.6404 $\pm$ 0.0176	0.6435 $\pm$ 0.0193	0.6403 $\pm$ 0.0192	0.6404 $\pm$ 0.0176	0.6396 $\pm$ 0.0188
	IMDb3	0.6499 $\pm$ 0.0041	0.6563 $\pm$ 0.0078	0.6490 $\pm$ 0.0065	0.6499 $\pm$ 0.0041	0.6497 $\pm$ 0.0040
	IMDb4	0.6392 $\pm$ 0.0121	0.6448 $\pm$ 0.0132	0.6396 $\pm$ 0.0096	0.6392 $\pm$ 0.0121	0.6380 $\pm$ 0.0132
	Output Fusion	0.5207 $\pm$ 0.0227	0.5225 $\pm$ 0.0234	0.5192 $\pm$ 0.0221	0.5207 $\pm$ 0.0227	0.5197 $\pm$ 0.0222
	IMDb1	0.6479 $\pm$ 0.0123	0.6542 $\pm$ 0.0118	0.6493 $\pm$ 0.0098	0.6479 $\pm$ 0.0123	0.6453 $\pm$ 0.0157
SlotGAT*	IMDb2	0.6479 $\pm$ 0.0123	0.6542 $\pm$ 0.0118	0.6493 $\pm$ 0.0098	0.6479 $\pm$ 0.0123	0.6453 $\pm$ 0.0157
	IMDb3	0.6479 $\pm$ 0.0123	0.6542 $\pm$ 0.0118	0.6493 $\pm$ 0.0098	0.6479 $\pm$ 0.0123	0.6453 $\pm$ 0.0157
	IMDb4	0.6479 $\pm$ 0.0123	0.6542 $\pm$ 0.0118	0.6493 $\pm$ 0.0098	0.6479 $\pm$ 0.0123	0.6453 $\pm$ 0.0157
	IMDb1	0.6304 $\pm$ 0.0068	0.6329 $\pm$ 0.0075	0.6298 $\pm$ 0.0060	0.6304 $\pm$ 0.0068	0.6300 $\pm$ 0.0068
SeHGNN	IMDb2	0.6419 $\pm$ 0.0054	0.6421 $\pm$ 0.0058	0.6434 $\pm$ 0.0063	0.6419 $\pm$ 0.0054	0.6426 $\pm$ 0.0059
	IMDb3	0.6404 $\pm$ 0.0113	0.6419 $\pm$ 0.0112	0.6412 $\pm$ 0.0111	0.6404 $\pm$ 0.0113	0.6410 $\pm$ 0.0114
	IMDb4	0.6308 $\pm$ 0.0069	0.6307 $\pm$ 0.0073	0.6344 $\pm$ 0.0064	0.6308 $\pm$ 0.0069	0.6312 $\pm$ 0.0071
	Output Fusion	0.6635 $\pm$ 0.0064	0.6650 $\pm$ 0.0077	0.6644 $\pm$ 0.0062	0.6635 $\pm$ 0.0064	0.6646 $\pm$ 0.0069
	Data Augmentation	0.4347 $\pm$ 0.0227	0.3934 $\pm$ 0.0392	0.4414 $\pm$ 0.0130	0.4347 $\pm$ 0.0227	0.3509 $\pm$ 0.0294
	IMDb1	0.6396 $\pm$ 0.0054	0.6433 $\pm$ 0.0059	0.6397 $\pm$ 0.0059	0.6396 $\pm$ 0.0054	0.6408 $\pm$ 0.0058
SeHGNN*+ RestrictedK	IMDb2	0.6396 $\pm$ 0.0054	0.6433 $\pm$ 0.0059	0.6397 $\pm$ 0.0059	0.6396 $\pm$ 0.0054	0.6408 $\pm$ 0.0058
	IMDb3	0.6396 $\pm$ 0.0054	0.6433 $\pm$ 0.0059	0.6397 $\pm$ 0.0059	0.6396 $\pm$ 0.0054	0.6408 $\pm$ 0.0058
	IMDb4	0.6396 $\pm$ 0.0054	0.6433 $\pm$ 0.0059	0.6397 $\pm$ 0.0059	0.6396 $\pm$ 0.0054	0.6408 $\pm$ 0.0058
	IMDb1	0.6280 $\pm$ 0.0085	0.6345 $\pm$ 0.0103	0.6275 $\pm$ 0.0077	0.6280 $\pm$ 0.0085	0.6294 $\pm$ 0.0087
SeHGNN*+ FullK	IMDb2	0.6280 $\pm$ 0.0085	0.6345 $\pm$ 0.0103	0.6275 $\pm$ 0.0077	0.6280 $\pm$ 0.0085	0.6294 $\pm$ 0.0087
	IMDb3	0.6280 $\pm$ 0.0085	0.6345 $\pm$ 0.0103	0.6275 $\pm$ 0.0077	0.6280 $\pm$ 0.0085	0.6294 $\pm$ 0.0087
	IMDb4	0.6280 $\pm$ 0.0085	0.6345 $\pm$ 0.0103	0.6275 $\pm$ 0.0077	0.6280 $\pm$ 0.0085	0.6294 $\pm$ 0.0087
	IMDb1	0.6220 $\pm$ 0.0251	0.6301 $\pm$ 0.0190	0.6223 $\pm$ 0.0261	0.6220 $\pm$ 0.0251	0.6182 $\pm$ 0.0294
DHN	IMDb2	0.5933 $\pm$ 0.0115	0.5965 $\pm$ 0.0107	0.5949 $\pm$ 0.0131	0.5933 $\pm$ 0.0115	0.5938 $\pm$ 0.0127
	IMDb3	0.6005 $\pm$ 0.0052	0.6098 $\pm$ 0.0074	0.6008 $\pm$ 0.0057	0.6005 $\pm$ 0.0052	0.5965 $\pm$ 0.0062
	IMDb4	0.5825 $\pm$ 0.0118	0.5956 $\pm$ 0.0046	0.5810 $\pm$ 0.0144	0.5825 $\pm$ 0.0118	0.5815 $\pm$ 0.0146
	Data Augmentation	0.5945 $\pm$ 0.0150	0.6038 $\pm$ 0.0169	0.5926 $\pm$ 0.0167	0.5945 $\pm$ 0.0150	0.5923 $\pm$ 0.0152
DHN*	IMDb1	0.5889 $\pm$ 0.0128	0.5918 $\pm$ 0.0107	0.5895 $\pm$ 0.0156	0.5889 $\pm$ 0.0128	0.5879 $\pm$ 0.0105
	IMDb2	0.5889 $\pm$ 0.0128	0.5918 $\pm$ 0.0107	0.5895 $\pm$ 0.0156	0.5889 $\pm$ 0.0128	0.5879 $\pm$ 0.0105
	IMDb3	0.5889 $\pm$ 0.0128	0.5918 $\pm$ 0.0107	0.5895 $\pm$ 0.0156	0.5889 $\pm$ 0.0128	0.5879 $\pm$ 0.0105
	IMDb4	0.5889 $\pm$ 0.0128	0.5918 $\pm$ 0.0107	0.5895 $\pm$ 0.0156	0.5889 $\pm$ 0.0128	0.5879 $\pm$ 0.0105
PathNet	IMDb1	0.5901 $\pm$ 0.0113	0.5969 $\pm$ 0.0123	0.5879 $\pm$ 0.0114	0.5901 $\pm$ 0.0113	0.5895 $\pm$ 0.0124
	IMDb2	0.5941 $\pm$ 0.0117	0.6010 $\pm$ 0.0143	0.5926 $\pm$ 0.0113	0.5941 $\pm$ 0.0117	0.5953 $\pm$ 0.0122
	IMDb3	0.5953 $\pm$ 0.0293	0.6011 $\pm$ 0.0299	0.5959 $\pm$ 0.0307	0.5953 $\pm$ 0.0293	0.5967 $\pm$ 0.0302
	IMDb4	0.5949 $\pm$ 0.0064	0.6006 $\pm$ 0.0079	0.5958 $\pm$ 0.0067	0.5949 $\pm$ 0.0064	0.5972 $\pm$ 0.0072
	backup Output Fusion	0.5957 $\pm$ 0.0093	0.6020 $\pm$ 0.0085	0.5950 $\pm$ 0.0103	0.5957 $\pm$ 0.0093	0.5968 $\pm$ 0.0097
	Output Fusion	0.5381 $\pm$ 0.0612	0.6142 $\pm$ 0.0193	0.5359 $\pm$ 0.0695	0.5381 $\pm$ 0.0612	0.5242 $\pm$ 0.0745
PathNet*	IMDb1	0.6096 $\pm$ 0.0048	0.6164 $\pm$ 0.0047	0.6103 $\pm$ 0.0048	0.6096 $\pm$ 0.0048	0.6126 $\pm$ 0.0047
	IMDb2	0.6096 $\pm$ 0.0048	0.6164 $\pm$ 0.0047	0.6103 $\pm$ 0.0048	0.6096 $\pm$ 0.0048	0.6126 $\pm$ 0.0047
	IMDb3	0.6096 $\pm$ 0.0048	0.6164 $\pm$ 0.0047	0.6103 $\pm$ 0.0048	0.6096 $\pm$ 0.0048	0.6126 $\pm$ 0.0047
	IMDb4	0.6096 $\pm$ 0.0048	0.6164 $\pm$ 0.0047	0.6103 $\pm$ 0.0048	0.6096 $\pm$ 0.0048	0.6126 $\pm$ 0.0047
MAGNA	IMDb1	0.5427 $\pm$ 0.0045	0.5373 $\pm$ 0.0057	0.5391 $\pm$ 0.0067	0.5427 $\pm$ 0.0045	0.5374 $\pm$ 0.0062
	IMDb2	0.5116 $\pm$ 0.0076	0.5073 $\pm$ 0.0067	0.5103 $\pm$ 0.0053	0.5116 $\pm$ 0.0076	0.5073 $\pm$ 0.0068
	IMDb3	0.5415 $\pm$ 0.0096	0.5381 $\pm$ 0.0091	0.5415 $\pm$ 0.0083	0.5415 $\pm$ 0.0096	0.5379 $\pm$ 0.0094
	IMDb4	0.5140 $\pm$ 0.0122	0.5097 $\pm$ 0.0107	0.5122 $\pm$ 0.0088	0.5140 $\pm$ 0.0122	0.5097 $\pm$ 0.0100
	Output Fusion	0.5391 $\pm$ 0.0107	0.5350 $\pm$ 0.0109	0.5378 $\pm$ 0.0105	0.5391 $\pm$ 0.0107	0.5351 $\pm$ 0.0111
	Data Augmentation	0.5075 $\pm$ 0.0125	0.5026 $\pm$ 0.0123	0.5058 $\pm$ 0.0108	0.5075 $\pm$ 0.0125	0.5027 $\pm$ 0.0125
MAGNA*	IMDb1	0.5459 $\pm$ 0.0080	0.5423 $\pm$ 0.0075	0.5468 $\pm$ 0.0081	0.5459 $\pm$ 0.0080	0.5420 $\pm$ 0.0080
	IMDb2	0.5459 $\pm$ 0.0080	0.5423 $\pm$ 0.0075	0.5468 $\pm$ 0.0081	0.5459 $\pm$ 0.0080	0.5420 $\pm$ 0.0080
	IMDb3	0.5459 $\pm$ 0.0080	0.5423 $\pm$ 0.0075	0.5468 $\pm$ 0.0081	0.5459 $\pm$ 0.0080	0.5420 $\pm$ 0.0080
	IMDb4	0.5459 $\pm$ 0.0080	0.5423 $\pm$ 0.0075	0.5468 $\pm$ 0.0081	0.5459 $\pm$ 0.0080	0.5420 $\pm$ 0.0080

Table 26: IMDb node classification invariance: kendall- τ over 3 runs.

Model	Comparison	Kendall- τ	Model	Comparison	Kendall- τ
RGCN	IMDb1 vs. IMDb2	0.4521 $\pm$ 0.0538	SlotGAT	IMDb1 vs. IMDb2	0.6885 $\pm$ 0.0442
	IMDb1 vs. IMDb3	0.3292 $\pm$ 0.0305		IMDb1 vs. IMDb3	0.7044 $\pm$ 0.0188
	IMDb1 vs. IMDb4	0.5841 $\pm$ 0.1772		IMDb1 vs. IMDb4	0.7682 $\pm$ 0.0090
	IMDb2 vs. IMDb3	0.4686 $\pm$ 0.1025		IMDb2 vs. IMDb3	0.6422 $\pm$ 0.0245
	IMDb2 vs. IMDb4	0.4329 $\pm$ 0.0329		IMDb2 vs. IMDb4	0.5848 $\pm$ 0.0472
	IMDb3 vs. IMDb4	0.2928 $\pm$ 0.0213		IMDb3 vs. IMDb4	0.6959 $\pm$ 0.0256
	Data Augmentation: IMDb1 vs. IMDb2	0.7509 $\pm$ 0.0721		IMDb1 vs. Output Fusion	0.8636 $\pm$ 0.0096
	Data Augmentation: IMDb1 vs. IMDb3	0.8022 $\pm$ 0.0392		IMDb2 vs. Output Fusion	0.8312 $\pm$ 0.0493
	Data Augmentation: IMDb1 vs. IMDb4	0.9327 $\pm$ 0.0151		IMDb3 vs. Output Fusion	0.7977 $\pm$ 0.0832
	Data Augmentation: IMDb2 vs. IMDb3	0.8753 $\pm$ 0.0342		IMDb4 vs. Output Fusion	0.7711 $\pm$ 0.0946
	Data Augmentation: IMDb2 vs. IMDb4	0.7698 $\pm$ 0.0534			
	Data Augmentation: IMDb3 vs. IMDb4	0.7759 $\pm$ 0.0526			
RGCN*	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000	SlotGAT*	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
MAGNN	IMDb1 vs. IMDb2	0.6847 $\pm$ 0.0304	SeHGNN	IMDb1 vs. IMDb2	0.6095 $\pm$ 0.0568
	IMDb1 vs. IMDb3	0.7079 $\pm$ 0.0803		IMDb1 vs. IMDb3	0.6390 $\pm$ 0.0161
	IMDb1 vs. IMDb4	0.6366 $\pm$ 0.0717		IMDb1 vs. IMDb4	0.6552 $\pm$ 0.0252
	IMDb2 vs. IMDb3	0.5707 $\pm$ 0.0424		IMDb2 vs. IMDb3	0.6148 $\pm$ 0.0485
	IMDb2 vs. IMDb4	0.6287 $\pm$ 0.0036		IMDb2 vs. IMDb4	0.6045 $\pm$ 0.0382
	IMDb3 vs. IMDb4	0.5845 $\pm$ 0.0898		IMDb3 vs. IMDb4	0.6462 $\pm$ 0.0255
	Output Fusion: IMDb1 vs. IMDb2	0.8445 $\pm$ 0.0081		IMDb1 vs. Output Fusion	0.7581 $\pm$ 0.0229
	Output Fusion: IMDb1 vs. IMDb3	0.9157 $\pm$ 0.0144		IMDb2 vs. Output Fusion	0.6951 $\pm$ 0.0492
	Output Fusion: IMDb1 vs. IMDb4	0.8565 $\pm$ 0.0056		IMDb3 vs. Output Fusion	0.7725 $\pm$ 0.0095
	Output Fusion: IMDb2 vs. IMDb3	0.8596 $\pm$ 0.0125		IMDb4 vs. Output Fusion	0.7719 $\pm$ 0.0333
	Output Fusion: IMDb2 vs. IMDb4	0.8083 $\pm$ 0.0125			
	Output Fusion: IMDb3 vs. IMDb4	0.8397 $\pm$ 0.0112			
	Data Augmentation: IMDb1 vs. IMDb2	-0.0558 $\pm$ 0.5741			
	Data Augmentation: IMDb1 vs. IMDb3	0.2448 $\pm$ 0.3645			
SeHGNN Augmentation	Data Augmentation: IMDb1 vs. IMDb4	0.2055 $\pm$ 0.3659			
	Data Augmentation: IMDb2 vs. IMDb3	0.5027 $\pm$ 0.3100			
	Data Augmentation: IMDb2 vs. IMDb4	-0.2358 $\pm$ 0.2102			
	Data Augmentation: IMDb3 vs. IMDb4	-0.3163 $\pm$ 0.3304			
MAGNN* + Union	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000	SeHGNN* + RestrictedK	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
MAGNN* + Intersect	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000	SeHGNN* + FullK	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
DHN	IMDb1 vs. IMDb2	0.8121 $\pm$ 0.0236	DHN*	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	0.8174 $\pm$ 0.0250		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	0.7589 $\pm$ 0.1295		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	0.7847 $\pm$ 0.0421		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	0.7563 $\pm$ 0.0917		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	0.7260 $\pm$ 0.1218		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
DHN Augmentation	Data Augmentation: IMDb1 vs. IMDb2	0.8724 $\pm$ 0.0741			
	Data Augmentation: IMDb1 vs. IMDb3	0.8929 $\pm$ 0.0630			
	Data Augmentation: IMDb1 vs. IMDb4	0.9261 $\pm$ 0.0426			
	Data Augmentation: IMDb2 vs. IMDb3	0.9221 $\pm$ 0.0531			
	Data Augmentation: IMDb2 vs. IMDb4	0.9123 $\pm$ 0.0566			
	Data Augmentation: IMDb3 vs. IMDb4	0.9200 $\pm$ 0.0578			
PathNet	IMDb1 vs. IMDb2	0.7908 $\pm$ 0.1322	PathNet*	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	0.7650 $\pm$ 0.0951		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	0.7780 $\pm$ 0.0917		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	0.7892 $\pm$ 0.0957		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	0.7448 $\pm$ 0.1908		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	0.7238 $\pm$ 0.1664		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
MAGNA	IMDb1 vs. IMDb2	0.6805 $\pm$ 0.0254	MAGNA*	IMDb1 vs. IMDb2	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb3	0.7172 $\pm$ 0.0107		IMDb1 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb1 vs. IMDb4	0.7475 $\pm$ 0.0423		IMDb1 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb3	0.7236 $\pm$ 0.0059		IMDb2 vs. IMDb3	1.0000 $\pm$ 0.0000
	IMDb2 vs. IMDb4	0.6964 $\pm$ 0.0205		IMDb2 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb3 vs. IMDb4	0.6598 $\pm$ 0.0140		IMDb3 vs. IMDb4	1.0000 $\pm$ 0.0000
	IMDb1 vs. Output Fusion	0.8379 $\pm$ 0.0112			
	IMDb2 vs. Output Fusion	0.8022 $\pm$ 0.0062			
	IMDb3 vs. Output Fusion	0.8198 $\pm$ 0.0060			
	IMDb4 vs. Output Fusion	0.8060 $\pm$ 0.0247			
	Data Augmentation: IMDb1 vs. IMDb2	0.8204 $\pm$ 0.0312			

Table 27: DBLP link prediction results over 3 runs.

Model	Variant	Precision	Recall	F1	Hits@1	Hits@3	MRR
MAGNN	DBLP1	0.7838 ± 0.1503	0.6353 ± 0.1135	0.6791 ± 0.0524	0.7811 ± 0.0447	0.9087 ± 0.0270	0.8575 ± 0.0311
	DBLP2	0.8576 ± 0.0427	0.6960 ± 0.0411	0.7662 ± 0.0130	0.7963 ± 0.0295	0.9165 ± 0.0138	0.8669 ± 0.0190
	DBLP3	0.7328 ± 0.0197	0.6943 ± 0.0213	0.7125 ± 0.0018	0.7629 ± 0.0196	0.9116 ± 0.0197	0.8477 ± 0.0124
	backup Output Fusion	0.8607 ± 0.0542	0.7387 ± 0.0152	0.7936 ± 0.0152	0.8250 ± 0.0096	0.9180 ± 0.0054	0.8837 ± 0.0061
	DBLP1-3	0.7365 ± 0.0787	0.9988 ± 0.0014	0.8462 ± 0.0528	0.9671 ± 0.0118	0.9993 ± 0.0012	0.9823 ± 0.0060
RGCN	DBLP1	0.2036 ± 0.0280	0.4501 ± 0.1017	0.2797 ± 0.0466	0.4930 ± 0.2710	0.7128 ± 0.1131	0.6508 ± 0.1814
	DBLP2	0.2472 ± 0.0466	0.6560 ± 0.1390	0.3525 ± 0.0455	0.5989 ± 0.2068	0.8341 ± 0.0318	0.7366 ± 0.1206
	DBLP3	0.2738 ± 0.1101	0.3316 ± 0.2320	0.2535 ± 0.0969	0.2657 ± 0.0879	0.6757 ± 0.0384	0.5148 ± 0.0593
	backup Output Fusion	0.2670 ± 0.0238	0.9169 ± 0.1155	0.4100 ± 0.0140	0.2582 ± 0.1167	0.6840 ± 0.1049	0.5126 ± 0.0919
	Output Fusion	0.2405 ± 0.0212	0.7071 ± 0.4131	0.3214 ± 0.1205	0.7528 ± 0.3486	0.8836 ± 0.1644	0.8347 ± 0.2332
	DBLP1-3	0.2305 ± 0.0201	0.7836 ± 0.2658	0.3529 ± 0.0544	0.7765 ± 0.2736	0.8295 ± 0.1875	0.8384 ± 0.1953
CMPNN	DBLP1	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4374 ± 0.0187	0.9111 ± 0.0083	0.6672 ± 0.0128
	DBLP2	0.2500 ± 0.0000	0.9997 ± 0.0004	0.3999 ± 0.0001	0.3621 ± 0.1457	0.8333 ± 0.1225	0.6068 ± 0.1114
	DBLP3	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4372 ± 0.0271	0.9112 ± 0.0096	0.6672 ± 0.0190
	backup Output Fusion	0.3654 ± 0.0290	0.7056 ± 0.0161	0.4810 ± 0.0140	0.4380 ± 0.1167	0.9049 ± 0.0194	0.6660 ± 0.0241
	DBLP1-3	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4529 ± 0.0177	0.9157 ± 0.0088	0.6771 ± 0.0134
MAGNN* + Union	DBLP1	0.8300 ± 0.0117	0.6998 ± 0.0309	0.7590 ± 0.0200	0.7749 ± 0.0159	0.9150 ± 0.0156	0.8539 ± 0.0126
	DBLP2	0.8300 ± 0.0117	0.6998 ± 0.0309	0.7590 ± 0.0200	0.7749 ± 0.0159	0.9150 ± 0.0156	0.8539 ± 0.0126
	DBLP3	0.8300 ± 0.0117	0.6998 ± 0.0309	0.7590 ± 0.0200	0.7749 ± 0.0159	0.9150 ± 0.0156	0.8539 ± 0.0126
MAGNN*	Union	0.9160 ± 0.0604	0.8143 ± 0.0289	0.8608 ± 0.0308	0.9244 ± 0.0333	0.9841 ± 0.0167	0.9552 ± 0.0222
MAGNN* + Intersect	DBLP1	0.7838 ± 0.1503	0.6353 ± 0.1135	0.6791 ± 0.0524	0.7811 ± 0.0447	0.9087 ± 0.0270	0.8575 ± 0.0311
	DBLP2	0.7838 ± 0.1503	0.6353 ± 0.1135	0.6791 ± 0.0524	0.7811 ± 0.0447	0.9087 ± 0.0270	0.8575 ± 0.0311
	DBLP3	0.7838 ± 0.1503	0.6353 ± 0.1135	0.6791 ± 0.0524	0.7811 ± 0.0447	0.9087 ± 0.0270	0.8575 ± 0.0311
RGCN*	DBLP1	0.3186 ± 0.0835	0.4719 ± 0.3487	0.3011 ± 0.0839	0.7654 ± 0.3312	0.8983 ± 0.1432	0.8465 ± 0.2167
	DBLP2	0.3186 ± 0.0835	0.4719 ± 0.3487	0.3011 ± 0.0839	0.7654 ± 0.3312	0.8983 ± 0.1432	0.8465 ± 0.2167
	DBLP3	0.3186 ± 0.0835	0.4719 ± 0.3487	0.3011 ± 0.0839	0.7654 ± 0.3312	0.8983 ± 0.1432	0.8465 ± 0.2167
CMPNN*	DBLP1	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4392 ± 0.0109	0.9085 ± 0.0117	0.6672 ± 0.0090
	DBLP2	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4392 ± 0.0109	0.9085 ± 0.0117	0.6672 ± 0.0090
	DBLP3	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4392 ± 0.0109	0.9085 ± 0.0117	0.6672 ± 0.0090
DHN	DBLP1	0.2360 ± 0.0059	0.8188 ± 0.0271	0.3662 ± 0.0056	0.4095 ± 0.0201	0.7910 ± 0.0131	0.6263 ± 0.0140
	DBLP2	0.2722 ± 0.0082	0.6785 ± 0.1355	0.3842 ± 0.0221	0.3814 ± 0.0432	0.7795 ± 0.0405	0.6065 ± 0.0341
	DBLP3	0.2952 ± 0.0623	0.7851 ± 0.1058	0.4186 ± 0.0422	0.4315 ± 0.0233	0.8368 ± 0.0234	0.6488 ± 0.0172
DHN*	DBLP1	0.2835 ± 0.0557	0.6438 ± 0.0667	0.3925 ± 0.0639	0.4234 ± 0.1050	0.7875 ± 0.0767	0.6304 ± 0.0764
	DBLP2	0.2835 ± 0.0557	0.6438 ± 0.0667	0.3925 ± 0.0639	0.4234 ± 0.1050	0.7875 ± 0.0767	0.6304 ± 0.0764
	DBLP3	0.2835 ± 0.0557	0.6438 ± 0.0667	0.3925 ± 0.0639	0.4234 ± 0.1050	0.7875 ± 0.0767	0.6304 ± 0.0764
MAGNA	DBLP1	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4835 ± 0.0055	0.9200 ± 0.0107	0.6955 ± 0.0055
	DBLP2	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4794 ± 0.0072	0.9179 ± 0.0027	0.6931 ± 0.0041
	DBLP3	0.2500 ± 0.0000	0.9999 ± 0.0002	0.4000 ± 0.0000	0.4725 ± 0.0032	0.9159 ± 0.0044	0.6895 ± 0.0020
	Output Fusion	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4855 ± 0.0004	0.9232 ± 0.0036	0.6977 ± 0.0005
	Augmentation	0.2594 ± 0.0071	0.9980 ± 0.0017	0.4117 ± 0.0088	0.5591 ± 0.1114	0.9369 ± 0.0223	0.7435 ± 0.0677
MAGNA*	DBLP1	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4786 ± 0.0052	0.9211 ± 0.0061	0.6929 ± 0.0036
	DBLP2	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4786 ± 0.0052	0.9211 ± 0.0061	0.6929 ± 0.0036
	DBLP3	0.2500 ± 0.0000	1.0000 ± 0.0000	0.4000 ± 0.0000	0.4786 ± 0.0052	0.9211 ± 0.0061	0.6929 ± 0.0036

Table 28: DBLP link prediction invariance: Kendall- τ over 3 runs. An em dash indicates that a three-run statistic is not estimable because one saved augmentation run assigns tied scores to every candidate within every query.

Model	Comparison	Kendall- τ	Kendall- $\tau@1$	Kendall- $\tau@3$
MAGNN	DBLP1 vs DBLP2	0.5393 ± 0.0660	0.6119 ± 0.0955	0.5028 ± 0.1080
	DBLP1 vs DBLP3	0.5178 ± 0.0538	0.6305 ± 0.0395	0.5193 ± 0.0922
	DBLP2 vs DBLP3	0.5263 ± 0.0651	0.6101 ± 0.0726	0.4846 ± 0.1148
	DBLP1-3: DBLP1 vs DBLP2	0.9264 ± 0.0271	0.9926 ± 0.0050	0.8943 ± 0.0551
	DBLP1-3: DBLP1 vs DBLP3	0.9247 ± 0.0254	0.9933 ± 0.0057	0.8886 ± 0.0508
	DBLP1-3: DBLP2 vs DBLP3	0.9372 ± 0.0254	0.9919 ± 0.0069	0.9103 ± 0.0347
MAGNN* + Union	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
MAGNN* + Intersect	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
RGCN	DBLP1 vs DBLP2	0.1752 ± 0.1324	0.1713 ± 0.1447	0.2097 ± 0.1369
	DBLP1 vs DBLP3	0.1263 ± 0.0718	0.0874 ± 0.0723	0.1935 ± 0.0582
	DBLP2 vs DBLP3	0.1120 ± 0.1489	0.1034 ± 0.1235	0.1850 ± 0.2211
	DBLP1-3: DBLP1 vs DBLP2	—	—	—
	DBLP1-3: DBLP1 vs DBLP3	—	—	—
	DBLP1-3: DBLP2 vs DBLP3	—	—	—
RGCN*	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CMPNN	DBLP1 vs DBLP2	0.3586 ± 0.0283	0.4002 ± 0.1750	0.2815 ± 0.0489
	DBLP1 vs DBLP3	0.7286 ± 0.0354	0.8347 ± 0.0742	0.7183 ± 0.0865
	DBLP2 vs DBLP3	0.4334 ± 0.0213	0.4045 ± 0.1847	0.2556 ± 0.1220
CMPNN*	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
DHN	DBLP1 vs DBLP2	0.1795 ± 0.0795	0.4560 ± 0.0725	0.0663 ± 0.0683
	DBLP1 vs DBLP3	0.2107 ± 0.0345	0.5611 ± 0.0187	0.2032 ± 0.0677
	DBLP2 vs DBLP3	0.4021 ± 0.1071	0.5386 ± 0.0721	0.2094 ± 0.1085
DHN*	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
MAGNA	DBLP1 vs DBLP2	0.5127 ± 0.0950	0.3627 ± 0.4607	0.6254 ± 0.1739
	DBLP1 vs DBLP3	0.5103 ± 0.0492	0.0703 ± 0.1459	0.5870 ± 0.1686
	DBLP2 vs DBLP3	0.4391 ± 0.0614	0.3922 ± 0.4165	0.6143 ± 0.1702
	DBLP1 vs Output Fusion	0.8668 ± 0.0233	0.8641 ± 0.0282	0.8200 ± 0.0271
	DBLP2 vs Output Fusion	0.8568 ± 0.0241	0.8766 ± 0.0312	0.8271 ± 0.0387
	DBLP3 vs Output Fusion	0.8832 ± 0.0201	0.8761 ± 0.0250	0.8391 ± 0.0190
	Data Augmentation: DBLP1 vs DBLP2	0.8278 ± 0.1413	0.8060 ± 0.1920	0.7071 ± 0.2028
	Data Augmentation: DBLP1 vs DBLP3	0.8937 ± 0.1166	0.6854 ± 0.2589	0.8743 ± 0.1188
	Data Augmentation: DBLP2 vs DBLP3	0.8824 ± 0.0854	0.6348 ± 0.3050	0.7699 ± 0.1423
MAGNA*	DBLP1 vs DBLP2	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP1 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	DBLP2 vs DBLP3	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

Table 29: IMDB link prediction results over 3 runs.

Model	Task	Variant	Precision	Recall	F1	Hits@1	Hits@3	MRR
CMPNN	Movie – Link	IMDb1	0.5259 ± 0.0016	1.0000 ± 0.0000	0.6893 ± 0.0014	0.6276 ± 0.0064	0.9944 ± 0.0023	0.8005 ± 0.0027
		IMDb2	0.5259 ± 0.0016	1.0000 ± 0.0000	0.6893 ± 0.0014	0.6296 ± 0.0133	0.9904 ± 0.0026	0.7996 ± 0.0066
		IMDb3	0.5259 ± 0.0016	1.0000 ± 0.0000	0.6893 ± 0.0014	0.6423 ± 0.0171	0.9952 ± 0.0010	0.8078 ± 0.0103
		IMDb4	0.5259 ± 0.0016	1.0000 ± 0.0000	0.6893 ± 0.0014	0.6320 ± 0.0119	0.9904 ± 0.0017	0.8022 ± 0.0068
		backup Output Fusion	0.5259 ± 0.0016	1.0000 ± 0.0000	0.6893 ± 0.0014	0.6280 ± 0.0128	0.9932 ± 0.0028	0.7995 ± 0.0061
		Data Augmentation	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	Movie – Director	IMDb1	0.4131 ± 0.0279	0.7552 ± 0.0471	0.5341 ± 0.0261	0.5383 ± 0.0212	0.9334 ± 0.0114	0.7285 ± 0.0148
		IMDb3	0.3783 ± 0.0138	0.7759 ± 0.0497	0.5086 ± 0.0164	0.4801 ± 0.0163	0.9111 ± 0.0102	0.6906 ± 0.0114
		backup Output Fusion	0.4211 ± 0.0397	0.7915 ± 0.0787	0.5466 ± 0.0318	0.5251 ± 0.0378	1.0000 ± 0.0000	0.7329 ± 0.0231
		Data Augmentation	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.1214 ± 0.0087	0.2982 ± 0.0279	0.2802 ± 0.0141
RGCN	Movie – Link	IMDb1	0.4955 ± 0.1396	0.7185 ± 0.2046	0.5849 ± 0.1637	0.5578 ± 0.2298	0.9514 ± 0.0259	0.7452 ± 0.1364
		IMDb2	0.6792 ± 0.1212	0.8668 ± 0.1131	0.7601 ± 0.1158	0.7767 ± 0.1579	0.9856 ± 0.0039	0.8781 ± 0.0859
		IMDb3	0.6627 ± 0.0550	0.7484 ± 0.0849	0.7028 ± 0.0684	0.7604 ± 0.0420	0.9817 ± 0.0030	0.8699 ± 0.0233
		IMDb4	0.6813 ± 0.0231	0.8872 ± 0.1275	0.7668 ± 0.0564	0.8214 ± 0.0421	0.9833 ± 0.0000	0.9038 ± 0.0221
		Data Augmentation	0.2292 ± 0.0859	0.8619 ± 0.1229	0.3544 ± 0.1147	0.3801 ± 0.1870	0.7806 ± 0.2061	0.5984 ± 0.1640
		backup Output Fusion	0.8849 ± 0.0263	0.9490 ± 0.0091	0.9156 ± 0.0148	0.9797 ± 0.0110	0.9992 ± 0.0006	0.9894 ± 0.0056
	Movie – Director	IMDb1	0.3113 ± 0.0287	0.3381 ± 0.1233	0.3097 ± 0.0549	0.3186 ± 0.0326	0.7835 ± 0.0355	0.5675 ± 0.0238
		IMDb3	0.2778 ± 0.0700	0.5662 ± 0.1468	0.3504 ± 0.0177	0.2444 ± 0.0488	0.6631 ± 0.0137	0.5024 ± 0.0305
		Data Augmentation	0.0556 ± 0.0050	0.6531 ± 0.2695	0.1008 ± 0.0053	0.0965 ± 0.0193	0.2225 ± 0.0399	0.2312 ± 0.0222
		backup Output Fusion	0.3660 ± 0.0130	0.8800 ± 0.0864	0.4070 ± 0.0072	0.3353 ± 0.0496	0.7863 ± 0.0415	0.5799 ± 0.0374
MAGNN	Movie – Link	IMDb1	0.2500 ± 0.0000	0.4876 ± 0.1244	0.3256 ± 0.0293	0.5152 ± 0.0165	0.5371 ± 0.0155	0.6404 ± 0.0119
		IMDb2	0.2500 ± 0.4988	0.4322 ± 0.3330	0.3167 ± 0.3000	0.4850 ± 0.0300	0.5050 ± 0.0300	0.6120 ± 0.0200
		IMDb3	0.2501 ± 0.0001	0.8102 ± 0.2050	0.3769 ± 0.0277	0.5056 ± 0.0429	0.5251 ± 0.0432	0.6327 ± 0.0324
		IMDb4	0.2501 ± 0.0002	0.8182 ± 0.1043	0.3820 ± 0.0109	0.5076 ± 0.0057	0.5303 ± 0.0057	0.6344 ± 0.0041
		backup Output Fusion	0.2501 ± 0.0000	0.9952 ± 0.0068	0.3953 ± 0.0000	0.5215 ± 0.0159	0.5374 ± 0.0203	0.6643 ± 0.0169
		Data Augmentation	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.4159 ± 0.0546	0.4330 ± 0.0525	0.4577 ± 0.0518
	Movie – Director	IMDb1	0.2496 ± 0.0008	0.5251 ± 0.2771	0.3078 ± 0.0875	0.4745 ± 0.0203	0.4896 ± 0.0175	0.6085 ± 0.0145
		IMDb3	0.2502 ± 0.0004	0.7608 ± 0.1717	0.3733 ± 0.0194	0.5068 ± 0.0261	0.5211 ± 0.0269	0.6322 ± 0.0196
		backup Output Fusion	0.2502 ± 0.0000	0.9992 ± 0.0011	0.4002 ± 0.0001	0.4777 ± 0.0186	0.4908 ± 0.0162	0.6105 ± 0.0133
		Data Augmentation	0.2497 ± 0.0003	0.7049 ± 0.1852	0.3651 ± 0.0284	0.4841 ± 0.0468	0.4956 ± 0.0423	0.6147 ± 0.0345
RGCN*	Movie – Link	IMDb1	0.7445 ± 0.0428	0.9769 ± 0.0156	0.8446 ± 0.0336	0.9043 ± 0.0318	0.9948 ± 0.0065	0.9489 ± 0.0169
		IMDb2	0.7445 ± 0.0428	0.9769 ± 0.0156	0.8446 ± 0.0336	0.9043 ± 0.0318	0.9948 ± 0.0065	0.9489 ± 0.0169
		IMDb3	0.7445 ± 0.0428	0.9769 ± 0.0156	0.8446 ± 0.0336	0.9043 ± 0.0318	0.9948 ± 0.0065	0.9489 ± 0.0169
		IMDb4	0.7445 ± 0.0428	0.9769 ± 0.0156	0.8446 ± 0.0336	0.9043 ± 0.0318	0.9948 ± 0.0065	0.9489 ± 0.0169
	Movie – Director	IMDb1	0.2952 ± 0.0224	0.3963 ± 0.1812	0.3158 ± 0.0553	0.3337 ± 0.0454	0.7165 ± 0.0272	0.5642 ± 0.0308
		IMDb3	0.2952 ± 0.0224	0.3963 ± 0.1812	0.3158 ± 0.0553	0.3337 ± 0.0454	0.7165 ± 0.0272	0.5642 ± 0.0308
CMPNN*	Movie – Link	IMDb1	0.5412 ± 0.0431	0.9981 ± 0.0376	0.7076 ± 0.0444	0.6600 ± 0.0728	0.9950 ± 0.0227	0.8228 ± 0.0464
		IMDb2	0.5412 ± 0.0431	0.9981 ± 0.0376	0.7076 ± 0.0444	0.6600 ± 0.0728	0.9950 ± 0.0227	0.8228 ± 0.0464
		IMDb3	0.5412 ± 0.0431	0.9981 ± 0.0376	0.7076 ± 0.0444	0.6600 ± 0.0728	0.9950 ± 0.0227	0.8228 ± 0.0464
		IMDb4	0.5412 ± 0.0431	0.9981 ± 0.0376	0.7076 ± 0.0444	0.6600 ± 0.0728	0.9950 ± 0.0227	0.8228 ± 0.0464
	Movie – Director	IMDb1	0.5873 ± 0.1904	0.6514 ± 0.1589	0.5948 ± 0.1725	0.6144 ± 0.2126	0.8025 ± 0.1383	0.7300 ± 0.1595
		IMDb3	0.5873 ± 0.1904	0.6514 ± 0.1589	0.5948 ± 0.1725	0.6144 ± 0.2126	0.8025 ± 0.1383	0.7300 ± 0.1595
MAGNN* + Union	Movie – Link	IMDb1	0.2500 ± 0.0000	0.8186 ± 0.1021	0.3819 ± 0.0113	0.5490 ± 0.0052	0.5678 ± 0.0062	0.6646 ± 0.0044
		IMDb2	0.2500 ± 0.0000	0.8186 ± 0.1021	0.3819 ± 0.0113	0.5490 ± 0.0052	0.5678 ± 0.0062	0.6646 ± 0.0044
		IMDb3	0.2500 ± 0.0000	0.8186 ± 0.1021	0.3819 ± 0.0113	0.5490 ± 0.0052	0.5678 ± 0.0062	0.6646 ± 0.0044
		IMDb4	0.2500 ± 0.0000	0.8186 ± 0.1021	0.3819 ± 0.0113	0.5490 ± 0.0052	0.5678 ± 0.0062	0.6646 ± 0.0044
	Movie – Director	IMDb1	0.2500 ± 0.0000	0.7951 ± 0.1184	0.3789 ± 0.0133	0.8090 ± 0.1470	0.8182 ± 0.1409	0.8584 ± 0.1092
		IMDb3	0.2500 ± 0.0000	0.7951 ± 0.1184	0.3789 ± 0.0133	0.8090 ± 0.1470	0.8182 ± 0.1409	0.8584 ± 0.1092
MAGNN*	Movie – Link	Union	0.2500 ± 0.0000	0.4617 ± 0.1600	0.3137 ± 0.0476	0.6232 ± 0.0628	0.6411 ± 0.0611	0.7206 ± 0.0467
	Movie – Director	Union	0.2497 ± 0.0003	0.5570 ± 0.0524	0.3441 ± 0.0108	0.4809 ± 0.0186	0.4964 ± 0.0157	0.6135 ± 0.0132
DHN	Movie–Link	IMDb1	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0283 ± 0.0091	0.1073 ± 0.0179	0.1455 ± 0.0120
		IMDb2	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0395 ± 0.0020	0.1280 ± 0.0026	0.1632 ± 0.0016
		IMDb3	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0435 ± 0.0025	0.1264 ± 0.0065	0.1638 ± 0.0023
		IMDb4	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0335 ± 0.0035	0.1132 ± 0.0151	0.1519 ± 0.0063
		Data Augmentation	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0040 ± 0.0030	0.0131 ± 0.0076	0.0758 ± 0.0057
		Movie–Director	IMDb1	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.1220 ± 0.0026	0.2895 ± 0.0026
	IMDb3	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.1232 ± 0.0029	0.2839 ± 0.0045	0.2620 ± 0.0016	
	Data Augmentation	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.1220 ± 0.0042	0.2887 ± 0.0009	0.2628 ± 0.0023	
DHN*	Movie–Link	IMDb1	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0295 ± 0.0111	0.0985 ± 0.0181	0.1405 ± 0.0177
		IMDb2	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0295 ± 0.0111	0.0985 ± 0.0181	0.1405 ± 0.0177
		IMDb3	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0295 ± 0.0111	0.0985 ± 0.0181	0.1405 ± 0.0177
		IMDb4	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.0295 ± 0.0111	0.0985 ± 0.0181	0.1405 ± 0.0177
	Movie–Director	IMDb1	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.1232 ± 0.0035	0.2867 ± 0.0015	0.2630 ± 0.0015
		IMDb3	0.0500 ± 0.0000	1.0000 ± 0.0000	0.0952 ± 0.0000	0.1232 ± 0.0035	0.2867 ± 0.0015	0.2630 ± 0.0015
MAGNA	Movie–Link	IMDb1	0.0625 ± 0.0024	1.0000 ± 0.0000	0.1176 ± 0.0043	0.9988 ± 0.0010	1.0000 ± 0.0000	0.9994 ± 0.0005
		IMDb2	0.0645 ± 0.0037	1.0000 ± 0.0000	0.1212 ± 0.0065	0.9952 ± 0.0026	0.9996 ± 0.0006	0.9973 ± 0.0015
		IMDb3	0.0692 ± 0.0056	1.0000 ± 0.0000	0.1294 ± 0.0098	0.9952 ± 0.0034	1.0000 ± 0.0000	0.9976 ± 0.0017
		IMDb4	0.0694 ± 0.0016	1.0000 ± 0.0000	0.1298 ± 0.0029	0.9964 ± 0.0017	1.0000 ± 0.0000	0.9981 ± 0.0009
		Output Fusion	0.0603 ± 0.0003	1.0000 ± 0.0000	0.1629 ± 0.0073	0.9996 ± 0.0001	1.0000 ± 0.0000	0.9998 ± 0.0001
		Data Augmentation	0.0887 ± 0.0043	1.0000 ± 0.0000	0.1138 ± 0.0005	0.9996 ± 0.0006	1.0000 ± 0.0000	0.9998 ± 0.0003
	Movie–Director	IMDb1	0.0479 ± 0.0004	0.9047 ± 0.0080	0.0910 ± 0.0007	0.0793 ± 0.0108	0.2061 ± 0.0150	0.2144 ± 0.0091
		IMDb3	0.0488 ± 0.0003	0.9298 ± 0.0049	0.0928 ± 0.0005	0.0490 ± 0.0035	0.1587 ± 0.0054	0.1791 ± 0.0033
		Output Fusion	0.0489 ± 0.0002	0.9430 ± 0.0025	0.0930 ± 0.0003	0.0614 ± 0.0011	0.1671 ± 0.0082	0.1901 ± 0.0020
		Data Augmentation	0.0505 ± 0.0009	0.5967 ± 0.0769	0.0930 ± 0.0025	0.0554 ± 0.0055	0.1625 ± 0.0010	0.1868 ± 0.0048
MAGNA*	Movie–Link	IMDb1	0.0696 ± 0.0063	1.0000 ± 0.0000	0.1302 ± 0.0109	0.9988 ± 0.0010	1.0000 ± 0.0000	0.9994 ± 0.0005
		IMDb2	0.0696 ± 0.0063	1.0000 ± 0.0000	0.1302 ± 0.0109	0.9988 ± 0.0010	1.0000 ± 0.0000	0.9994 ± 0.0005
		IMDb3	0.0696 ± 0.0063	1.0000 ± 0.0000	0.1302 ± 0.0109	0.9988 ± 0.0010	1.0000 ± 0.0000	0.9994 ± 0.0005
		IMDb4	0.0696 ± 0.0063	1.0000 ± 0.0000	0.1302 ± 0.0109	0.9988 ± 0.0010	1.0000 ± 0.0000	0.9994 ± 0.0005
	Movie–Director	IMDb1	0.0481 ± 0.0006	0.7978 ± 0.0307	0.0908 ± 0.0010	0.0758 ± 0.0020	0.1994 ± 0.0083	0.2134 ± 0.0020
		IMDb3	0.0481 ± 0.0006	0.7978 ± 0.0307	0.0908 ± 0.0010	0.0758 ± 0.0020	0.1994 ± 0.0083	0.2134 ± 0.0020
		IMDb2	0.0481 ± 0.0006	0.7978 ± 0.0307	0.0908 ± 0.0010	0.0758 ± 0.0020	0.1994 ± 0.0083	0.2134 ± 0.0020
		IMDb4	0.0481 ± 0.0006	0.7978 ± 0.0307	0.0908 ± 0.0010	0.0758 ± 0.0020	0.1994 ± 0.0083	0.2134 ± 0.0020

Table 30: IMDb link prediction invariance: Kendall- τ over 3 runs.

Model	Task	Comparison	Kendall- τ	Kendall- τ @1	Kendall- τ @3
CMPNN	Movie – Link	IMDb1 vs IMDb2	0.5048 $\pm$ 0.0048	0.6404 $\pm$ 0.0096	0.4631 $\pm$ 0.0098
		IMDb1 vs IMDb3	0.6111 $\pm$ 0.0830	0.7460 $\pm$ 0.0706	0.6124 $\pm$ 0.1087
		IMDb1 vs IMDb4	0.5045 $\pm$ 0.0060	0.6407 $\pm$ 0.0122	0.4718 $\pm$ 0.0109
		IMDb2 vs IMDb3	0.5124 $\pm$ 0.0048	0.6487 $\pm$ 0.0167	0.4665 $\pm$ 0.0246
		IMDb2 vs IMDb4	0.5195 $\pm$ 0.0239	0.6659 $\pm$ 0.0431	0.5085 $\pm$ 0.0711
		IMDb3 vs IMDb4	0.4968 $\pm$ 0.0090	0.6388 $\pm$ 0.0010	0.4689 $\pm$ 0.0088
CMPNN*	Movie – Link	IMDb1 vs IMDb3	0.6224 $\pm$ 0.0285	0.6870 $\pm$ 0.0514	0.5779 $\pm$ 0.0604
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
RGCN	Movie – Link	IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	0.3423 $\pm$ 0.0612	0.1184 $\pm$ 0.0242	0.3524 $\pm$ 0.1136
		IMDb1 vs IMDb3	0.3951 $\pm$ 0.0731	0.2061 $\pm$ 0.0567	0.4781 $\pm$ 0.2582
		IMDb1 vs IMDb4	0.3976 $\pm$ 0.0888	0.1814 $\pm$ 0.0223	0.5127 $\pm$ 0.2413
		IMDb2 vs IMDb3	0.4336 $\pm$ 0.0561	0.1817 $\pm$ 0.0135	0.4848 $\pm$ 0.1313
		IMDb2 vs IMDb4	0.4469 $\pm$ 0.0537	0.1389 $\pm$ 0.0152	0.6666 $\pm$ 0.2536
		IMDb3 vs IMDb4	0.4596 $\pm$ 0.0171	0.1702 $\pm$ 0.0085	0.7476 $\pm$ 0.0839
		Data Augmentation: IMDb1 vs IMDb2	0.5630 $\pm$ 0.1297	−0.1310 $\pm$ 0.3750	0.0080 $\pm$ 0.2224
		Data Augmentation: IMDb1 vs IMDb3	0.6550 $\pm$ 0.0831	0.0422 $\pm$ 0.3939	0.1343 $\pm$ 0.2435
		Data Augmentation: IMDb1 vs IMDb4	0.7749 $\pm$ 0.0867	0.2673 $\pm$ 0.3426	0.3073 $\pm$ 0.2155
		Data Augmentation: IMDb2 vs IMDb3	0.7375 $\pm$ 0.0976	0.2169 $\pm$ 0.2960	0.2718 $\pm$ 0.2015
		Data Augmentation: IMDb2 vs IMDb4	0.6628 $\pm$ 0.0716	0.1382 $\pm$ 0.2923	0.1925 $\pm$ 0.1720
		Data Augmentation: IMDb3 vs IMDb4	0.5938 $\pm$ 0.0994	−0.0708 $\pm$ 0.3662	0.0454 $\pm$ 0.2267
		IMDb1 vs IMDb3	0.1653 $\pm$ 0.1757	−0.0530 $\pm$ 0.2353	0.1088 $\pm$ 0.2550
RGCN*	Movie – Director	Data Augmentation: IMDb1 vs IMDb3	0.5336 $\pm$ 0.2632	−0.0024 $\pm$ 0.2982	0.1097 $\pm$ 0.2227
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	0.3423 $\pm$ 0.0611	0.1184 $\pm$ 0.0242	0.3524 $\pm$ 0.1135
		IMDb1 vs IMDb3	0.2133 $\pm$ 0.1059	0.0850 $\pm$ 0.1177	0.0832 $\pm$ 0.1230
		IMDb1 vs IMDb4	0.1361 $\pm$ 0.0595	0.0384 $\pm$ 0.0494	0.0408 $\pm$ 0.0535
		IMDb2 vs IMDb3	0.3524 $\pm$ 0.1135	0.1816 $\pm$ 0.0134	0.4847 $\pm$ 0.1313
		IMDb2 vs IMDb4	0.4468 $\pm$ 0.0537	0.1389 $\pm$ 0.0151	0.6666 $\pm$ 0.2535
		IMDb3 vs IMDb4	0.3789 $\pm$ 0.0208	0.1537 $\pm$ 0.0329	0.1524 $\pm$ 0.0315
MAGNN	Movie – Link	Data Augmentation: IMDb1 vs IMDb2	0.4583 $\pm$ 0.2903	0.4099 $\pm$ 0.0775	−0.3965 $\pm$ 0.1039
		Data Augmentation: IMDb1 vs IMDb3	0.6590 $\pm$ 0.1922	0.6384 $\pm$ 0.2158	−0.0829 $\pm$ 0.2801
		Data Augmentation: IMDb1 vs IMDb4	0.6807 $\pm$ 0.1806	0.6691 $\pm$ 0.0346	−0.0373 $\pm$ 0.0735
		Data Augmentation: IMDb2 vs IMDb3	0.6264 $\pm$ 0.2099	0.4725 $\pm$ 0.1161	−0.3167 $\pm$ 0.1571
		Data Augmentation: IMDb2 vs IMDb4	0.5808 $\pm$ 0.2575	0.4992 $\pm$ 0.0715	−0.2743 $\pm$ 0.0851
		Data Augmentation: IMDb3 vs IMDb4	0.6139 $\pm$ 0.2168	0.5570 $\pm$ 0.0268	−0.1948 $\pm$ 0.0302
		IMDb1 vs IMDb3	0.2674 $\pm$ 0.2263	0.1424 $\pm$ 0.1950	0.1421 $\pm$ 0.1951
		Data Augmentation: IMDb1 vs IMDb3	0.9565 $\pm$ 0.0089	0.9617 $\pm$ 0.0349	0.4497 $\pm$ 0.0830
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
MAGNN* + Union	Movie – Link	IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DHN	Movie–Link	IMDb1 vs IMDb3	0.1988 $\pm$ 0.0265	0.0778 $\pm$ 0.0423	0.0440 $\pm$ 0.0340
		IMDb1 vs IMDb4	0.2915 $\pm$ 0.0425	0.0974 $\pm$ 0.0692	0.0477 $\pm$ 0.0318
		IMDb2 vs IMDb3	0.4025 $\pm$ 0.1418	0.0683 $\pm$ 0.0540	0.1749 $\pm$ 0.0458
		IMDb2 vs IMDb4	0.7353 $\pm$ 0.0490	0.6408 $\pm$ 0.0169	0.8072 $\pm$ 0.0332
		IMDb3 vs IMDb4	0.4571 $\pm$ 0.0261	0.2926 $\pm$ 0.0510	0.3114 $\pm$ 0.0150
		IMDb3 vs IMDb4	0.3376 $\pm$ 0.0327	0.0884 $\pm$ 0.0427	0.2012 $\pm$ 0.0272
		Data Augmentation: IMDb1 vs IMDb2	0.7541 $\pm$ 0.0582	−0.0718 $\pm$ 0.2425	0.1054 $\pm$ 0.1527
		Data Augmentation: IMDb1 vs IMDb3	0.7715 $\pm$ 0.0514	−0.0359 $\pm$ 0.2116	0.1339 $\pm$ 0.1357
		Data Augmentation: IMDb1 vs IMDb4	0.8786 $\pm$ 0.0321	0.4107 $\pm$ 0.2630	0.4614 $\pm$ 0.1495
		Data Augmentation: IMDb2 vs IMDb3	0.9078 $\pm$ 0.0187	0.5837 $\pm$ 0.0638	0.5880 $\pm$ 0.0644
		Data Augmentation: IMDb2 vs IMDb4	0.8113 $\pm$ 0.0407	0.1619 $\pm$ 0.1543	0.2560 $\pm$ 0.1342
		Data Augmentation: IMDb3 vs IMDb4	0.7658 $\pm$ 0.0524	−0.0311 $\pm$ 0.1996	0.1293 $\pm$ 0.1467
		IMDb1 vs IMDb3	0.9026 $\pm$ 0.0325	0.9469 $\pm$ 0.0336	0.9591 $\pm$ 0.0187
		Data Augmentation: IMDb1 vs IMDb3	0.9917 $\pm$ 0.0024	0.9793 $\pm$ 0.0050	0.9763 $\pm$ 0.0061
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
DHN*	Movie–Link	IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb2 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb3 vs IMDb4	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb2	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
		IMDb1 vs IMDb3	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
MAGNA	Movie–Link	IMDb1 vs IMDb4	0.3682 $\pm$ 0.0563	0.9940 $\pm$ 0.0020	0.7121 $\pm$ 0.0090
		IMDb1 vs IMDb3	0.3145 $\pm$ 0.0116	0.9940 $\pm$ 0.0035	0.6887 $\pm$ 0.0037
		IMDb1 vs IMDb4	0.3109 $\pm$ 0.0119	0.9952 $\pm$ 0.0026	0.6986 $\pm$ 0.0033
		IMDb2 vs IMDb3	0.2987 $\pm$ 0.0132	0.9904 $\pm$ 0.0026	0.6925 $\pm$ 0.0023
		IMDb2 vs IMDb4	0.2878 $\pm$ 0.0147	0.9916 $\pm$ 0.0010	0.6909 $\pm$ 0.0148
		IMDb3 vs IMDb4	0.3057 $\pm$ 0.0159	0.9916 $\pm$ 0.0045	0.6988 $\pm$ 0.0100
		IMDb1 vs Output Fusion	0.7308 $\pm$ 0.1549	0.9992 $\pm$ 0.0011	0.8137 $\pm$ 0.0770
		IMDb2 vs Output Fusion	0.6067 $\pm$ 0.1506	0.9948 $\pm$ 0.0023	0.7733 $\pm$ 0.0443
		IMDb3 vs Output Fusion	0.4237 $\pm$ 0.0502	0.9948 $\pm$ 0.0039	0.7180 $\pm$ 0.0105
		IMDb1 vs IMDb2	0.3682 $\pm$ 0.0563	0.9940 $\pm$ 0.0020	0.7121 $\pm$ 0.0090
		IMDb1 vs IMDb3	0.3145 $\pm$ 0.0116	0.9940 $\pm$ 0.0035	0.6887 $\pm$ 0.003

Table 31: IMDB node classification results: training time and epochs over 3 runs.

Model	Variant	Train Time (s)	Epochs	Model Parameter (MiB)	Peak Train GPU (MiB)	Peak Inference GPU (MiB)
RGCN	IMDb1	5.05 $\pm$ 1.49	33.00 $\pm$ 7.79	8.79 $\pm$ 0.00	170.46 $\pm$ 0.00	152.87 $\pm$ 0.00
	IMDb2	6.04 $\pm$ 0.71	41.00 $\pm$ 4.90	8.79 $\pm$ 0.00	170.46 $\pm$ 0.00	152.87 $\pm$ 0.00
	IMDb3	5.36 $\pm$ 1.04	37.00 $\pm$ 7.12	8.79 $\pm$ 0.00	170.46 $\pm$ 0.00	152.87 $\pm$ 0.00
	IMDb4	3.97 $\pm$ 0.07	27.33 $\pm$ 0.47	8.79 $\pm$ 0.00	170.46 $\pm$ 0.00	152.87 $\pm$ 0.00
	Data Augmentation	5.52 $\pm$ 0.15	24.67 $\pm$ 1.15	8.79	152.59 $\pm$ 0.00	137.85 $\pm$ 0.00
	backup Output Fusion	20.4 $\pm$ 2.1	138.3 $\pm$ 12.9	35.16 $\pm$ 0.00	681.84 $\pm$ 0.00	611.48 $\pm$ 0.00
RGCN*	Output Fusion	2.14 $\pm$ 0.98	89.33 $\pm$ 1.89	8.79 $\pm$ 0.00	199.37 $\pm$ 0.00	159.32 $\pm$ 0.00
	IMDb1	15.54 $\pm$ 4.35	25.67 $\pm$ 2.11	3.22 $\pm$ 0.00	531.27 $\pm$ 0.00	283.90 $\pm$ 0.00
	IMDb2	15.54 $\pm$ 4.35	25.67 $\pm$ 2.11	3.22 $\pm$ 0.00	531.27 $\pm$ 0.00	283.90 $\pm$ 0.00
	IMDb3	15.54 $\pm$ 4.35	25.67 $\pm$ 2.11	3.22 $\pm$ 0.00	531.27 $\pm$ 0.00	283.90 $\pm$ 0.00
	IMDb4	15.54 $\pm$ 4.35	25.67 $\pm$ 2.11	3.22 $\pm$ 0.00	531.27 $\pm$ 0.00	283.90 $\pm$ 0.00
MAGNN	IMDb1	30.00 $\pm$ 1.24	34.67 $\pm$ 0.58	5.14 $\pm$ 0.00	5560.29 $\pm$ 1.24	2283.70 $\pm$ 0.16
	IMDb2	29.30 $\pm$ 0.10	34.00 $\pm$ 0.00	5.14 $\pm$ 0.00	5687.94 $\pm$ 2.83	2299.87 $\pm$ 0.38
	IMDb3	36.93 $\pm$ 0.44	34.67 $\pm$ 0.58	5.14 $\pm$ 0.00	6059.79 $\pm$ 1.67	2360.28 $\pm$ 0.92
	IMDb4	22.45 $\pm$ 0.99	34.67 $\pm$ 0.58	5.14 $\pm$ 0.00	3141.08 $\pm$ 1.12	1258.66 $\pm$ 0.24
	backup Output Fusion	118.7 $\pm$ 1.7	138.0 $\pm$ 1.4	20.56 $\pm$ 0.00	20449.10 $\pm$ 3.69	8202.51 $\pm$ 1.04
	Output Fusion	113.00 $\pm$ 1.52	124.00 $\pm$ 0.00	5.13 $\pm$ 0.00	9577.79 $\pm$ 3.73	5879.15 $\pm$ 3.19
MAGNN	Union	15.57 $\pm$ 0.79	34.00 $\pm$ 0.00	5.13 $\pm$ 0.00	5401.97 $\pm$ 1.49	2312.55 $\pm$ 0.61
MAGNN* + Union	IMDb1	54.06 $\pm$ 5.85	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14917.20 $\pm$ 1.94	4436.88 $\pm$ 0.97
	IMDb2	51.59 $\pm$ 5.65	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14917.20 $\pm$ 1.94	4436.88 $\pm$ 0.97
	IMDb3	49.82 $\pm$ 0.82	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14917.20 $\pm$ 1.94	4436.88 $\pm$ 0.97
	IMDb4	49.11 $\pm$ 1.69	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14917.20 $\pm$ 1.94	4436.88 $\pm$ 0.97
MAGNN* + Intersect	IMDb1	29.92 $\pm$ 0.82	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14921.14 $\pm$ 2.58	4436.32 $\pm$ 0.61
	IMDb2	23.75 $\pm$ 0.07	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14921.14 $\pm$ 2.58	4436.32 $\pm$ 0.61
	IMDb3	31.39 $\pm$ 0.47	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14921.14 $\pm$ 2.58	4436.32 $\pm$ 0.61
	IMDb4	22.92 $\pm$ 1.25	17.67 $\pm$ 0.47	3.05 $\pm$ 0.00	14921.14 $\pm$ 2.58	4436.32 $\pm$ 0.61
SlotGAT	IMDb1	7.67 $\pm$ 0.91	50.0 $\pm$ 1.6	12.40	1448.67 $\pm$ 69.61	1100.46 $\pm$ 58.78
	IMDb2	6.80 $\pm$ 0.06	48.3 $\pm$ 0.5	12.40	1496.29 $\pm$ 1.21	1142.13 $\pm$ 0.35
	IMDb3	9.99 $\pm$ 2.70	71.3 $\pm$ 19.4	12.40	1496.36 $\pm$ 1.28	1142.34 $\pm$ 0.44
	IMDb4	9.48 $\pm$ 1.86	68.3 $\pm$ 13.9	12.40	1498.27 $\pm$ 1.09	1141.72 $\pm$ 0.76
	Output Fusion	33.94 $\pm$ 5.53	237.9 $\pm$ 35.4	49.60	1498.27 $\pm$ 1.09	1142.34 $\pm$ 0.44
SlotGAT*	IMDb1	7.39 $\pm$ 0.21	49.0 $\pm$ 0.8	12.40	1468.85 $\pm$ 72.64	1111.25 $\pm$ 57.57
	IMDb2	7.39 $\pm$ 0.21	49.0 $\pm$ 0.8	12.40	1468.85 $\pm$ 72.64	1111.25 $\pm$ 57.57
	IMDb3	7.39 $\pm$ 0.21	49.0 $\pm$ 0.8	12.40	1468.85 $\pm$ 72.64	1111.25 $\pm$ 57.57
	IMDb4	7.39 $\pm$ 0.21	49.0 $\pm$ 0.8	12.40	1468.85 $\pm$ 72.64	1111.25 $\pm$ 57.57
SeHGNN	IMDb1	59.33 $\pm$ 12.25	100.00 $\pm$ 18.06	243.16	3103.34 $\pm$ 81.96	2985.24 $\pm$ 1.86
	IMDb2	42.14 $\pm$ 4.19	88.00 $\pm$ 5.72	243.16	2315.61 $\pm$ 0.39	2093.22 $\pm$ 1.53
	IMDb3	41.27 $\pm$ 4.55	82.00 $\pm$ 8.04	243.16	2486.04 $\pm$ 0.61	2341.16 $\pm$ 0.06
	IMDb4	45.80 $\pm$ 0.44	90.67 $\pm$ 4.11	243.16	2462.28 $\pm$ 0.11	2279.05 $\pm$ 0.04
	Output Fusion	188.54 $\pm$ 21.43	360.67 $\pm$ 35.93	972.64	3103.34 $\pm$ 81.96	2985.24 $\pm$ 1.86
	Data Augmentation	209.64 $\pm$ 2.32	51.00 $\pm$ 0.00	433.33 $\pm$ 0.00	9072.85 $\pm$ 0.12	5120.30 $\pm$ 1.47
SeHGNN* + RestrictedK	IMDb1	101.40 $\pm$ 32.40	112.33 $\pm$ 26.66	385.07	4956.22 $\pm$ 175.20	4734.18 $\pm$ 0.15
	IMDb2	101.40 $\pm$ 32.40	112.33 $\pm$ 26.66	385.07	4956.22 $\pm$ 175.20	4734.18 $\pm$ 0.15
	IMDb3	101.40 $\pm$ 32.40	112.33 $\pm$ 26.66	385.07	4956.22 $\pm$ 175.20	4734.18 $\pm$ 0.15
	IMDb4	101.40 $\pm$ 32.40	112.33 $\pm$ 26.66	385.07	4956.22 $\pm$ 175.20	4734.18 $\pm$ 0.15
SeHGNN* + FullK	IMDb1	214.87 $\pm$ 38.20	110.00 $\pm$ 21.23	777.46	9952.37 $\pm$ 355.97	9572.85 $\pm$ 0.12
	IMDb2	214.87 $\pm$ 38.20	110.00 $\pm$ 21.23	777.46	9952.37 $\pm$ 355.97	9572.85 $\pm$ 0.12
	IMDb3	214.87 $\pm$ 38.20	110.00 $\pm$ 21.23	777.46	9952.37 $\pm$ 355.97	9572.85 $\pm$ 0.12
	IMDb4	214.87 $\pm$ 38.20	110.00 $\pm$ 21.23	777.46	9952.37 $\pm$ 355.97	9572.85 $\pm$ 0.12
DHN	IMDb1	1.47 $\pm$ 0.18	52.67 $\pm$ 2.89			
	IMDb2	1.47 $\pm$ 0.09	56.33 $\pm$ 3.21			
	IMDb3	1.39 $\pm$ 0.04	53.00 $\pm$ 1.73			
	IMDb4	1.27 $\pm$ 0.14	56.67 $\pm$ 6.43			
	Data Augmentation	11.53 $\pm$ 4.24	57.00 $\pm$ 22.54	2.40 $\pm$ 0.00	2128.36 $\pm$ 0.00	1251.90 $\pm$ 0.00
DHN*	IMDb1	5.48 $\pm$ 0.80	61.00 $\pm$ 7.81			
	IMDb2	5.48 $\pm$ 0.80	61.00 $\pm$ 7.81			
	IMDb3	5.48 $\pm$ 0.80	61.00 $\pm$ 7.81			
	IMDb4	5.48 $\pm$ 0.80	61.00 $\pm$ 7.81			
PathNet	IMDb1	17.9 $\pm$ 0.8	110.7 $\pm$ 5.2	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb2	19.9 $\pm$ 5.3	132.7 $\pm$ 35.6	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb3	19.5 $\pm$ 5.3	130.7 $\pm$ 34.9	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb4	16.2 $\pm$ 0.2	108.3 $\pm$ 1.2	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	backup Output Fusion	73.6 $\pm$ 11.1	482.3 $\pm$ 75.4	3.72 $\pm$ 0.00	18812.12 $\pm$ 0.12	5147.12 $\pm$ 0.00
	Output Fusion	31.93 $\pm$ 0.37	124.00 $\pm$ 0.00	0.93 $\pm$ 0.00	6658.25 $\pm$ 0.00	5636.09 $\pm$ 0.00
PathNet*	IMDb1	9.2 $\pm$ 3.1	47.0 $\pm$ 0.8	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb2	9.2 $\pm$ 3.1	47.0 $\pm$ 0.8	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb3	9.2 $\pm$ 3.1	47.0 $\pm$ 0.8	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
	IMDb4	9.2 $\pm$ 3.1	47.0 $\pm$ 0.8	0.93 $\pm$ 0.00	4703.03 $\pm$ 0.06	1286.78 $\pm$ 0.00
MAGNA	IMDb1	15.06 $\pm$ 0.84	98.00 $\pm$ 1.41	16.12	955.03 $\pm$ 18.93	292.34 $\pm$ 11.32
	IMDb2	14.61 $\pm$ 0.43	96.67 $\pm$ 1.70	16.12	968.42 $\pm$ 0.00	300.34 $\pm$ 0.00
	IMDb3	14.87 $\pm$ 0.47	97.67 $\pm$ 1.89	16.12	968.42 $\pm$ 0.00	300.34 $\pm$ 0.00
	IMDb4	14.66 $\pm$ 0.87	96.00 $\pm$ 4.24	16.12	968.42 $\pm$ 0.00	300.34 $\pm$ 0.00
	Output Fusion	59.20 $\pm$ 2.61	388.34 $\pm$ 9.24	64.48	968.42 $\pm$ 0.00	300.34 $\pm$ 0.00
	Data Augmentation	17.70 $\pm$ 1.65	53.00 $\pm$ 5.10	16.14	970.29 $\pm$ 7.65	387.93 $\pm$ 0.00
MAGNA*	IMDb1	14.37 $\pm$ 0.97	87.00 $\pm$ 8.64	16.12	964.15 $\pm$ 18.99	296.39 $\pm$ 11.32
	IMDb2	14.37 $\pm$ 0.97	87.00 $\pm$ 8.64	16.12	964.15 $\pm$ 18.99	296.39 $\pm$ 11.32
	IMDb3	14.37 $\pm$ 0.97	87.00 $\pm$ 8.64	16.12	964.15 $\pm$ 18.99	296.39 $\pm$ 11.32
	IMDb4	14.37 $\pm$ 0.97	87.00 $\pm$ 8.64	16.12	964.15 $\pm$ 18.99	296.39 $\pm$ 11.32

Table 32: DBLP link prediction training time and epochs over 3 runs.

Model	Variant	Train Time (s)	Epochs	Model Parameter (MiB)	Peak Train GPU (MiB)	Peak Inference GPU (MiB)
MAGNN	DBLP1	14959.50 $\pm$ 858.68	8.33 $\pm$ 0.47	9.01 $\pm$ 0.00	259.67 $\pm$ 8.07	97.84 $\pm$ 1.68
	DBLP2	14751.78 $\pm$ 2772.78	9.67 $\pm$ 1.70	9.01 $\pm$ 0.00	269.74 $\pm$ 9.66	98.88 $\pm$ 2.50
	DBLP3	15798.15 $\pm$ 2746.73	12.00 $\pm$ 2.16	9.01 $\pm$ 0.00	269.28 $\pm$ 8.03	97.62 $\pm$ 1.87
	backup Output Fusion	45509.43 $\pm$ 3996.27	30.00 $\pm$ 2.79	27.03 $\pm$ 0.00	798.69 $\pm$ 14.93	294.34 $\pm$ 3.55
	DBLP1-3	34190.79 $\pm$ 4727.69	35.00 $\pm$ 3.46	9.00 $\pm$ 0.00	101.06 $\pm$ 125.39	41.22 $\pm$ 6.01
RGCN	DBLP1	29.50 $\pm$ 4.01	23.33 $\pm$ 3.40	23.36 $\pm$ 0.00	2538.92 $\pm$ 0.00	1980.66 $\pm$ 0.00
	DBLP2	23.90 $\pm$ 4.34	20.67 $\pm$ 4.11	23.36 $\pm$ 0.00	2325.03 $\pm$ 0.00	1806.98 $\pm$ 0.00
	DBLP3	34.17 $\pm$ 16.16	29.00 $\pm$ 12.83	23.36 $\pm$ 0.00	2389.03 $\pm$ 0.00	1859.61 $\pm$ 0.00
	DBLP1-3	88.88 $\pm$ 39.25	35.00 $\pm$ 15.52	20.86	1840.53 $\pm$ 0.45	1313.43 $\pm$ 0.18
	backup Output Fusion	87.57 $\pm$ 17.21	73.00 $\pm$ 13.89	70.08 $\pm$ 0.00	7252.98 $\pm$ 0.00	5647.25 $\pm$ 0.00
	Output Fusion	111.61 $\pm$ 5.55	69.00 $\pm$ 12.25	23.36 $\pm$ 0.00	2568.99 $\pm$ 0.58	1970.16 $\pm$ 28.98
CMPNN	DBLP1	11293.71 $\pm$ 23.66	6.00 $\pm$ 0.00	0.27 $\pm$ 0.00	2427.38 $\pm$ 0.14	803.80 $\pm$ 0.05
	DBLP2	5058.47 $\pm$ 683.07	6.67 $\pm$ 0.94	0.27 $\pm$ 0.00	3318.41 $\pm$ 0.00	1101.75 $\pm$ 0.00
	DBLP3	4618.16 $\pm$ 3.97	6.00 $\pm$ 0.00	0.27 $\pm$ 0.00	3319.64 $\pm$ 0.00	1102.07 $\pm$ 0.00
	backup Output Fusion	20970.34 $\pm$ 683.49	18.67 $\pm$ 0.94	0.81 $\pm$ 0.00	9065.43 $\pm$ 0.14	3007.62 $\pm$ 0.05
	DBLP1-3	125.05 $\pm$ 0.06	6.00 $\pm$ 0.00	0.22	9072.69 $\pm$ 0.00	7043.98 $\pm$ 0.00
MAGNN*	Union	11492.89 $\pm$ 2613.02	8.00 $\pm$ 1.63	9.00 $\pm$ 0.00	1069.14 $\pm$ 5.02	352.39 $\pm$ 1.97
MAGNN* + Union	DBLP1	16031.08 $\pm$ 1648.58	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	245.69 $\pm$ 13.28	95.78 $\pm$ 2.77
	DBLP2	10268.90 $\pm$ 2645.72	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	245.69 $\pm$ 13.28	95.78 $\pm$ 2.77
	DBLP3	16206.74 $\pm$ 1624.89	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	245.69 $\pm$ 13.28	95.78 $\pm$ 2.77
MAGNN* + Intersect	DBLP1	15731.08 $\pm$ 2379.51	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	246.04 $\pm$ 13.09	95.75 $\pm$ 3.08
	DBLP2	13941.65 $\pm$ 1817.32	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	246.04 $\pm$ 13.09	95.75 $\pm$ 3.08
	DBLP3	12866.96 $\pm$ 3962.26	10.00 $\pm$ 1.63	8.71 $\pm$ 0.00	246.04 $\pm$ 13.09	95.75 $\pm$ 3.08
RGCN*	DBLP2	26.69 $\pm$ 2.42	22.00 $\pm$ 2.16	23.36 $\pm$ 0.00	2555.56 $\pm$ 0.49	1247.57 $\pm$ 0.51
	DBLP2	26.69 $\pm$ 2.42	22.00 $\pm$ 2.16	23.36 $\pm$ 0.00	2555.56 $\pm$ 0.49	1247.57 $\pm$ 0.51
	DBLP2	26.69 $\pm$ 2.42	22.00 $\pm$ 2.16	23.36 $\pm$ 0.00	2555.56 $\pm$ 0.49	1247.57 $\pm$ 0.51
CMPNN*	DBLP1	6056.74 $\pm$ 770.49	6.00 $\pm$ 0.00	0.20 $\pm$ 0.00	1653.76 $\pm$ 0.00	745.86 $\pm$ 0.00
	DBLP2	6056.74 $\pm$ 770.49	6.00 $\pm$ 0.00	0.20 $\pm$ 0.00	1653.76 $\pm$ 0.00	745.86 $\pm$ 0.00
	DBLP3	6056.74 $\pm$ 770.49	6.00 $\pm$ 0.00	0.20 $\pm$ 0.00	1653.76 $\pm$ 0.00	745.86 $\pm$ 0.00
DHN	DBLP1	1.45 $\pm$ 0.25	13.00 $\pm$ 2.16			
	DBLP2	3.04 $\pm$ 1.14	36.00 $\pm$ 12.75			
	DBLP3	4.88 $\pm$ 2.34	58.00 $\pm$ 30.10			
DHN*	DBLP1	3.52 $\pm$ 2.53	34.67 $\pm$ 26.71			
	DBLP2	3.52 $\pm$ 2.53	34.67 $\pm$ 26.71			
	DBLP3	3.52 $\pm$ 2.53	34.67 $\pm$ 26.71			
MAGNA	DBLP1	312.04 $\pm$ 39.72	32.67 $\pm$ 4.99	24.04	1649.12 $\pm$ 42.06	469.63 $\pm$ 18.83
	DBLP2	411.56 $\pm$ 101.94	41.00 $\pm$ 9.93	24.04	1673.02 $\pm$ 0.36	481.31 $\pm$ 0.87
	DBLP3	259.18 $\pm$ 20.43	25.00 $\pm$ 1.41	24.04	1673.97 $\pm$ 0.30	481.37 $\pm$ 0.48
	Output Fusion	982.78 $\pm$ 162.09	98.67 $\pm$ 16.33	72.12	1673.97 $\pm$ 0.30	481.37 $\pm$ 0.48
	Data Augmentation	1111.73 $\pm$ 214.22	44.00 $\pm$ 8.49	24.05	2190.05 $\pm$ 0.05	581.08 $\pm$ 0.05
MAGNA*	DBLP1	298.43 $\pm$ 119.10	31.00 $\pm$ 13.14	24.05	1603.26 $\pm$ 41.78	469.98 $\pm$ 18.97
	DBLP2	298.43 $\pm$ 119.10	31.00 $\pm$ 13.14	24.05	1603.26 $\pm$ 41.78	469.98 $\pm$ 18.97
	DBLP3	298.43 $\pm$ 119.10	31.00 $\pm$ 13.14	24.05	1603.26 $\pm$ 41.78	469.98 $\pm$ 18.97

Table 33: IMDb link prediction training time and epochs over 3 runs.

Model	Task	Variant	Train Time (s)	Epochs	Model Parameter (MiB)	Peak Train GPU (MiB)	Peak Inference GPU (MiB)
CMPNN	Movie – Link	IMDb1	1179.417 $\pm$ 4503.3313	15.667 $\pm$ 6.1283	0.33 $\pm$ 0.00	1920.20 $\pm$ 0.00	399.93 $\pm$ 0.00
		IMDb2	1647.248 $\pm$ 4503.3313	18.126 $\pm$ 6.1283	0.24 $\pm$ 0.00	1052.66 $\pm$ 0.00	363.74 $\pm$ 0.00
		IMDb3	11879.4178 $\pm$ 4503.3313	15.6667 $\pm$ 6.1283	0.24 $\pm$ 0.00	1053.82 $\pm$ 0.00	364.16 $\pm$ 0.00
		IMDb4	11879.4178 $\pm$ 4503.3313	15.6667 $\pm$ 6.1283	0.24 $\pm$ 0.00	1053.82 $\pm$ 0.00	364.16 $\pm$ 0.00
		backup Output Fusion	26585.5006 $\pm$ 9006.6626	65.1264 $\pm$ 12.2566	1.05 $\pm$ 0.00	5080.50 $\pm$ 0.00	1491.99 $\pm$ 0.00
		Data Augmentation	17.45 $\pm$ 0.02	6.00 $\pm$ 0.00	0.33	2926.55 $\pm$ 0.00	1027.73 $\pm$ 0.00
	Movie – Director	IMDb1	7724.0143 $\pm$ 1.2438	7.6667 $\pm$ 1.2472	0.33 $\pm$ 0.00	2480.55 $\pm$ 0.00	511.19 $\pm$ 0.00
		IMDb3	8077.2431 $\pm$ 2.1862	8.5718 $\pm$ 0.0526	0.33 $\pm$ 0.00	2480.55 $\pm$ 0.00	511.19 $\pm$ 0.00
		backup Output Fusion	15801.2574 $\pm$ 2.5153	16.2385 $\pm$ 1.2483	0.66 $\pm$ 0.00	4961.10 $\pm$ 0.00	1022.38 $\pm$ 0.00
		Data Augmentation	18.57 $\pm$ 2.91	9.00 $\pm$ 1.41	0.33	3751.04 $\pm$ 0.00	1312.93 $\pm$ 0.00
RGCN	Movie – Link	IMDb1	3.50 $\pm$ 0.67	40.67 $\pm$ 10.14	18.29 $\pm$ 0.00	511.59 $\pm$ 0.00	365.57 $\pm$ 0.00
		IMDb2	4.47 $\pm$ 0.57	52.67 $\pm$ 6.65	18.29 $\pm$ 0.00	511.59 $\pm$ 0.00	365.57 $\pm$ 0.00
		IMDb3	3.54 $\pm$ 0.28	42.67 $\pm$ 4.50	18.29 $\pm$ 0.00	511.59 $\pm$ 0.00	365.57 $\pm$ 0.00
		IMDb4	3.92 $\pm$ 1.22	48.67 $\pm$ 15.17	18.29 $\pm$ 0.00	511.59 $\pm$ 0.00	365.57 $\pm$ 0.00
		Data Augmentation	10.53 $\pm$ 5.62	28.33 $\pm$ 13.58	14.08	293.17 $\pm$ 0.67	211.35 $\pm$ 0.62
		backup Output Fusion	19.5 $\pm$ 1.2	184.7 $\pm$ 17.9	73.16 $\pm$ 0.00	2046.36 $\pm$ 0.00	1462.28 $\pm$ 0.00
	Movie – Director	IMDb1	1.58 $\pm$ 0.37	19.33 $\pm$ 0.47	18.29 $\pm$ 0.00	444.81 $\pm$ 0.00	312.21 $\pm$ 0.00
		IMDb3	1.16 $\pm$ 0.11	17.33 $\pm$ 1.25	18.29 $\pm$ 0.00	444.81 $\pm$ 0.00	312.21 $\pm$ 0.00
		Data Augmentation	4.80 $\pm$ 1.03	19.33 $\pm$ 2.89	15.79	357.19 $\pm$ 0.00	254.46 $\pm$ 0.00
MAGNN	Movie – Link	IMDb1	317.72 $\pm$ 5.60	12.00 $\pm$ 0.82	5.28 $\pm$ 0.00	867.12 $\pm$ 0.00	260.84 $\pm$ 0.00
		IMDb2	295.57 $\pm$ 6.92	11.00 $\pm$ 0.00	5.28 $\pm$ 0.00	884.73 $\pm$ 0.00	259.84 $\pm$ 0.00
		IMDb3	312.30 $\pm$ 19.55	11.00 $\pm$ 0.00	5.28 $\pm$ 0.00	873.24 $\pm$ 0.00	259.67 $\pm$ 0.00
		IMDb4	326.11 $\pm$ 7.53	11.00 $\pm$ 0.00	5.28 $\pm$ 0.00	871.01 $\pm$ 0.00	260.03 $\pm$ 0.00
		backup Output Fusion	1222.6 $\pm$ 7.3	44.0 $\pm$ 0.0	21.12 $\pm$ 0.00	3496.10 $\pm$ 0.00	1040.38 $\pm$ 0.00
		Data Augmentation	1692.65 $\pm$ 267.08	45.33 $\pm$ 2.31	5.27 $\pm$ 0.00	1260.91 $\pm$ 8.92	592.22 $\pm$ 0.00
	Movie – Director	IMDb1	501.58 $\pm$ 22.04	13.67 $\pm$ 0.94	5.30 $\pm$ 0.00	859.46 $\pm$ 0.00	323.28 $\pm$ 0.00
		IMDb3	756.98 $\pm$ 246.68	17.67 $\pm$ 5.91	5.30 $\pm$ 0.00	901.14 $\pm$ 0.00	322.77 $\pm$ 0.00
		backup Output Fusion	982.2 $\pm$ 95.5	23.7 $\pm$ 2.4	10.60 $\pm$ 0.00	1760.60 $\pm$ 0.00	646.05 $\pm$ 0.00
RGCN*	Movie – Link	Data Augmentation	983.97 $\pm$ 132.14	24.00 $\pm$ 2.00	5.27 $\pm$ 0.00	1040.89 $\pm$ 1.92	438.86 $\pm$ 0.00
		IMDb1	2.06 $\pm$ 0.71	19.67 $\pm$ 2.62	18.30 $\pm$ 0.00	576.74 $\pm$ 1.02	265.32 $\pm$ 0.00
		IMDb2	2.06 $\pm$ 0.71	19.67 $\pm$ 2.62	18.30 $\pm$ 0.00	576.74 $\pm$ 1.02	265.32 $\pm$ 0.00
		IMDb3	2.06 $\pm$ 0.71	19.67 $\pm$ 2.62	18.30 $\pm$ 0.00	576.74 $\pm$ 1.02	265.32 $\pm$ 0.00
		IMDb4	2.06 $\pm$ 0.71	19.67 $\pm$ 2.62	18.30 $\pm$ 0.00	576.74 $\pm$ 1.02	265.32 $\pm$ 0.00
	Movie – Director	IMDb1	2.01 $\pm$ 0.30	19.67 $\pm$ 0.94	18.30 $\pm$ 0.00	738.92 $\pm$ 0.73	346.91 $\pm$ 0.22
		IMDb3	2.01 $\pm$ 0.30	19.67 $\pm$ 0.94	18.30 $\pm$ 0.00	738.92 $\pm$ 0.73	346.91 $\pm$ 0.22
CMPNN*	Movie – Link	IMDb1	1425.00 $\pm$ 421.47	8.00 $\pm$ 2.36	0.33 $\pm$ 0.00	1922.92 $\pm$ 0.00	399.86 $\pm$ 0.00
		IMDb2	1425.00 $\pm$ 443.42	8.00 $\pm$ 2.36	0.33 $\pm$ 0.00	1922.92 $\pm$ 0.00	399.86 $\pm$ 0.00
		IMDb3	1425.00 $\pm$ 430.81	8.00 $\pm$ 2.36	0.33 $\pm$ 0.00	1922.92 $\pm$ 0.00	399.86 $\pm$ 0.00
		IMDb4	1425.00 $\pm$ 432.80	8.00 $\pm$ 2.36	0.33 $\pm$ 0.00	1922.92 $\pm$ 0.00	399.86 $\pm$ 0.00
	Movie – Director	IMDb1	4800.00 $\pm$ 724.90	16.00 $\pm$ 2.49	0.33 $\pm$ 0.00	2502.76 $\pm$ 0.00	512.54 $\pm$ 0.00
		IMDb3	4800.00 $\pm$ 542.59	16.00 $\pm$ 2.49	0.33 $\pm$ 0.00	2502.76 $\pm$ 0.00	512.54 $\pm$ 0.00
	Movie – Link	Union	217.95 $\pm$ 13.73	11.67 $\pm$ 0.47	5.27 $\pm$ 0.00	284.38 $\pm$ 1.49	184.01 $\pm$ 0.00
	Movie – Director	Union	381.29 $\pm$ 19.45	13.33 $\pm$ 0.47	5.27 $\pm$ 0.00	835.70 $\pm$ 1.98	305.87 $\pm$ 0.01
MAGNN* + Union	Movie – Link	IMDb1	444.13 $\pm$ 5.30	11.67 $\pm$ 0.47	5.32 $\pm$ 0.00	357.85 $\pm$ 0.00	190.99 $\pm$ 0.00
		IMDb2	444.13 $\pm$ 5.30	11.67 $\pm$ 0.47	5.32 $\pm$ 0.00	357.85 $\pm$ 0.00	190.99 $\pm$ 0.00
		IMDb3	444.13 $\pm$ 5.30	11.67 $\pm$ 0.47	5.32 $\pm$ 0.00	357.85 $\pm$ 0.00	190.99 $\pm$ 0.00
		IMDb4	444.13 $\pm$ 5.30	11.67 $\pm$ 0.47	5.32 $\pm$ 0.00	357.85 $\pm$ 0.00	190.99 $\pm$ 0.00
	Movie – Director	IMDb1	1088.55 $\pm$ 200.58	14.00 $\pm$ 2.16	5.35 $\pm$ 0.00	962.84 $\pm$ 0.00	338.73 $\pm$ 0.00
		IMDb3	1088.55 $\pm$ 200.58	14.00 $\pm$ 2.16	5.35 $\pm$ 0.00	962.84 $\pm$ 0.00	338.73 $\pm$ 0.00
DHN	Movie-Link	IMDb1	0.33 $\pm$ 0.17	16.00 $\pm$ 0.00			
		IMDb2	0.33 $\pm$ 0.16	16.00 $\pm$ 0.00			
		IMDb3	0.32 $\pm$ 0.16	16.00 $\pm$ 0.00			
		IMDb4	0.33 $\pm$ 0.17	16.00 $\pm$ 0.00			
		Data Augmentation	2.49 $\pm$ 0.24	16.00 $\pm$ 0.00	3.21 $\pm$ 0.00	338.37 $\pm$ 0.00	74.79 $\pm$ 0.00
	Movie-Director	IMDb1	0.44 $\pm$ 0.16	20.00 $\pm$ 0.00			
		IMDb3	0.39 $\pm$ 0.16	17.33 $\pm$ 1.89			
		Data Augmentation	2.31 $\pm$ 0.50	18.00 $\pm$ 0.00	4.07 $\pm$ 0.00	421.38 $\pm$ 0.00	95.89 $\pm$ 0.00
DHN*	Movie-Link	IMDb1	0.38 $\pm$ 0.16	16.67 $\pm$ 0.47			
		IMDb2	0.38 $\pm$ 0.16	16.67 $\pm$ 0.47			
		IMDb3	0.38 $\pm$ 0.16	16.67 $\pm$ 0.47			
		IMDb4	0.38 $\pm$ 0.16	16.67 $\pm$ 0.47			
	Movie-Director	IMDb1	0.51 $\pm$ 0.17	20.00 $\pm$ 0.00			
		IMDb3	0.51 $\pm$ 0.17	20.00 $\pm$ 0.00			
MAGNA	Movie-Link	IMDb1	12.36 $\pm$ 4.48	14.33 $\pm$ 3.30	13.44	802.26 $\pm$ 28.04	230.45 $\pm$ 15.10
		IMDb2	12.29 $\pm$ 2.39	14.67 $\pm$ 2.05	13.44	822.09 $\pm$ 0.00	241.03 $\pm$ 0.00
		IMDb3	15.49 $\pm$ 3.43	18.67 $\pm$ 2.62	13.44	822.09 $\pm$ 0.00	241.03 $\pm$ 0.00
		IMDb4	16.41 $\pm$ 2.86	20.00 $\pm$ 1.41	13.44	822.09 $\pm$ 0.00	241.03 $\pm$ 0.00
		Output Fusion	56.55 $\pm$ 13.16	67.67 $\pm$ 9.38	53.76	822.09 $\pm$ 0.00	241.03 $\pm$ 0.00
		Data Augmentation	39.48 $\pm$ 6.85	19.33 $\pm$ 3.30	13.45	1028.51 $\pm$ 0.00	287.54 $\pm$ 0.00
	Movie-Director	IMDb1	26.60 $\pm$ 2.09	27.00 $\pm$ 2.94	16.12	1002.46 $\pm$ 33.32	285.70 $\pm$ 17.61
		IMDb3	15.01 $\pm$ 2.23	15.00 $\pm$ 1.41	16.12	1025.78 $\pm$ 0.35	298.17 $\pm$ 0.02
		Output Fusion	41.61 $\pm$ 4.32	42.00 $\pm$ 4.35	32.24	1025.78 $\pm$ 0.35	298.17 $\pm$ 0.02
MAGNA*	Movie-Link	Data Augmentation	27.14 $\pm$ 6.08	22.33 $\pm$ 5.44	16.14	1297.12 $\pm$ 0.01	341.78 $\pm$ 0.01
		IMDb1	16.46 $\pm$ 2.50	19.67 $\pm$ 2.49	13.44	805.72 $\pm$ 28.04	232.01 $\pm$ 15.10
		IMDb2	16.46 $\pm$ 2.50	19.67 $\pm$ 2.49	13.44	805.72 $\pm$ 28.04	232.01 $\pm$ 15.10
		IMDb3	16.46 $\pm$ 2.50	19.67 $\pm$ 2.49	13.44	805.72 $\pm$ 28.04	232.01 $\pm$ 15.10
		IMDb4	16.46 $\pm$ 2.50	19.67 $\pm$ 2.49	13.44	805.72 $\pm$ 28.04	232.01 $\pm$ 15.10
	Movie-Director	IMDb1	28.69 $\pm$ 4.06	28.67 $\pm$ 2.05	16.12	1008.04 $\pm$ 33.25	288.20 $\pm$ 17.63
		IMDb3	28.69 $\pm$ 4.06	28.67 $\pm$ 2.05	16.12	1008.04 $\pm$ 33.25	288.20 $\pm$ 17.63

Table 34: Wordnet link prediction results over 3 runs.

Model	Variant	Accuracy	Precision	Recall	F1	Hits@1	Hits@3	MRR
RGCN	No Changes	0.9871 $\pm$ 0.0001	0.9705 $\pm$ 0.0014	0.6783 $\pm$ 0.0021	0.7565 $\pm$ 0.0020	0.2813 $\pm$ 0.0024	0.3299 $\pm$ 0.0015	0.3123 $\pm$ 0.0018
	All Inverse Edges	0.9995 $\pm$ 0.0001	0.9888 $\pm$ 0.0032	0.9997 $\pm$ 0.0001	0.9942 $\pm$ 0.0017	0.7330 $\pm$ 0.0082	0.9299 $\pm$ 0.0034	0.8357 $\pm$ 0.0057
	Universal Edges	0.9991 $\pm$ 0.0001	0.9772 $\pm$ 0.0034	0.9995 $\pm$ 0.0001	0.9881 $\pm$ 0.0018	0.6659 $\pm$ 0.0066	0.8816 $\pm$ 0.0084	0.7824 $\pm$ 0.0055
	No Changes + All Inverse + Universal	0.9889 $\pm$ 0.0002	0.9576 $\pm$ 0.0086	0.7351 $\pm$ 0.0107	0.8090 $\pm$ 0.0077	0.3318 $\pm$ 0.0158	0.3897 $\pm$ 0.0065	0.3722 $\pm$ 0.0105
RGCN*	Universal Edges	0.9991 $\pm$ 0.0001	0.9772 $\pm$ 0.0034	0.9995 $\pm$ 0.0001	0.9881 $\pm$ 0.0018	0.6659 $\pm$ 0.0066	0.8816 $\pm$ 0.0084	0.7824 $\pm$ 0.0055
CMPNN	Unchanged	0.9972 $\pm$ 0.0062	0.3430 $\pm$ 0.0310	0.5104 $\pm$ 0.0534	0.3315 $\pm$ 0.0304	0.3561 $\pm$ 0.0187	0.3473 $\pm$ 0.0215	
	All Inverse Edges	0.9821 $\pm$ 0.0110	0.4033 $\pm$ 0.0545	0.5718 $\pm$ 0.0893	0.2884 $\pm$ 0.0233	0.3437 $\pm$ 0.0295	0.3214 $\pm$ 0.0192	
	Universal Edges	0.9981 $\pm$ 0.0082	0.3302 $\pm$ 0.0376	0.4963 $\pm$ 0.0643	0.2983 $\pm$ 0.0145	0.3563 $\pm$ 0.0165	0.3317 $\pm$ 0.0116	
	Output Fusion	0.9982 $\pm$ 0.0085	0.3497 $\pm$ 0.0410	0.5180 $\pm$ 0.0690	0.3511 $\pm$ 0.0227	0.3660 $\pm$ 0.0216	0.3631 $\pm$ 0.0174	
CMPNN*	Universal Edges	0.9981 $\pm$ 0.0082	0.3302 $\pm$ 0.0376	0.4963 $\pm$ 0.0643	0.2983 $\pm$ 0.0145	0.3563 $\pm$ 0.0165	0.3317 $\pm$ 0.0116	
DHN	No Changes	0.2469 $\pm$ 0.1143	0.5038 $\pm$ 0.0020	0.5385 $\pm$ 0.0314	0.2041 $\pm$ 0.0752	0.0021 $\pm$ 0.0006	0.0058 $\pm$ 0.0006	0.0077 $\pm$ 0.0002
	All Inverse Edges	0.8308 $\pm$ 0.0024	0.5435 $\pm$ 0.0011	0.8379 $\pm$ 0.0065	0.5349 $\pm$ 0.0023	0.0168 $\pm$ 0.0025	0.0346 $\pm$ 0.0045	0.0329 $\pm$ 0.0035
	Universal Edges	0.0290 $\pm$ 0.0003	0.5044 $\pm$ 0.0002	0.5022 $\pm$ 0.0000	0.0289 $\pm$ 0.0003	0.0000 $\pm$ 0.0000	0.0018 $\pm$ 0.0008	0.0026 $\pm$ 0.0003
	No Changes + All Inverse + Universal	0.5563 $\pm$ 0.3865	0.0424 $\pm$ 0.0186	0.5808 $\pm$ 0.3420	0.0698 $\pm$ 0.0249	0.0036 $\pm$ 0.0011	0.0071 $\pm$ 0.0007	0.0082 $\pm$ 0.0005
DHN*	Universal Edges	0.0290 $\pm$ 0.0003	0.5044 $\pm$ 0.0002	0.5022 $\pm$ 0.0000	0.0289 $\pm$ 0.0003	0.0000 $\pm$ 0.0000	0.0018 $\pm$ 0.0008	0.0026 $\pm$ 0.0003
MAGNA	No Changes	0.9877 $\pm$ 0.0001	0.8365 $\pm$ 0.0051	0.8480 $\pm$ 0.0063	0.8422 $\pm$ 0.0054	0.3470 $\pm$ 0.0024	0.4483 $\pm$ 0.0031	0.4143 $\pm$ 0.0015
	All Inverse Edges	0.9903 $\pm$ 0.0001	0.9082 $\pm$ 0.0023	0.8187 $\pm$ 0.0012	0.8577 $\pm$ 0.0033	0.4292 $\pm$ 0.0041	0.4860 $\pm$ 0.0036	0.4725 $\pm$ 0.0046
	Output Fusion	0.9911 $\pm$ 0.0002	0.9204 $\pm$ 0.0124	0.8327 $\pm$ 0.0024	0.8712 $\pm$ 0.0035	0.8089 $\pm$ 0.0152	0.6907 $\pm$ 0.0234	0.8555 $\pm$ 0.0173
	Data Augmentation	0.9902 $\pm$ 0.0003	0.9143 $\pm$ 0.0068	0.8113 $\pm$ 0.0080	0.8550 $\pm$ 0.0053	0.4217 $\pm$ 0.0023	0.4818 $\pm$ 0.0031	0.4658 $\pm$ 0.0020
MAGNA*	Universal Edges	0.9903 $\pm$ 0.0001	0.9082 $\pm$ 0.0023	0.8187 $\pm$ 0.0012	0.8577 $\pm$ 0.0033	0.4292 $\pm$ 0.0041	0.4860 $\pm$ 0.0036	0.4725 $\pm$ 0.0046

Table 35: Wordnet link prediction training time and epochs over 3 runs.

Model	Variant	Train Time (s)	Epochs	Model Parameter (MiB)	Peak Train GPU (MiB)	Peak Inference GPU (MiB)
RGCN	No Changes	643.17 $\pm$ 33.99	1590.00 $\pm$ 87.18	37.80	1841.72 $\pm$ 1.25	396.61 $\pm$ 1.30
	All Inverse Edges	721.85 $\pm$ 88.68	1613.33 $\pm$ 200.08	37.80	2540.95 $\pm$ 0.00	577.40 $\pm$ 0.04
	Universal Edges	1564.31 $\pm$ 321.31	2046.67 $\pm$ 395.01	37.80	8495.99 $\pm$ 1.05	2324.61 $\pm$ 0.41
	No Changes + All Inverse + Universal	10599.55 $\pm$ 3622.09	1623.33 $\pm$ 539.01	37.80	14216.17 $\pm$ 21.99	14161.50 $\pm$ 22.27
RGCN*	Universal Edges	3843.0 $\pm$ 35.5	3000.0 $\pm$ 0.0			
CMPNN	Unchanged	4016.7153 $\pm$ 531.1746	15.0000 $\pm$ 1.9355	3.6318	55770.7100 $\pm$ 0.0000	3309.0034 $\pm$ 0.0000
	All Inverse Edges	3345.6854 $\pm$ 4.6919	6.0000 $\pm$ 0.0000	3.6318	55806.4414 $\pm$ 0.0000	3319.8770 $\pm$ 0.0000
	Universal Edges	38120.1463 $\pm$ 13926.2455	8.0000 $\pm$ 1.3228	3.6318	56099.1963 $\pm$ 0.0000	3430.3281 $\pm$ 0.0000
	Output Fusion	45482.5470 $\pm$ 13936.3727	29.0000 $\pm$ 2.3444	10.8955	56099.1963 $\pm$ 0.0000	10059.2085 $\pm$ 0.0000
CMPNN*	Universal Edges	38120.1463 $\pm$ 13926.2455	8.0000 $\pm$ 1.3228	3.6318	56099.1963 $\pm$ 0.0000	3430.3281 $\pm$ 0.0000
DHN	No Changes	1.19 $\pm$ 0.38	14.00 $\pm$ 0.00			
	All Inverse Edges	21.27 $\pm$ 0.52	200.00 $\pm$ 0.00			
	Universal Edges	7.52 $\pm$ 0.52	39.33 $\pm$ 3.30			
	No Changes + All Inverse + Universal	189.05 $\pm$ 38.53	14.00 $\pm$ 2.65	11.25 $\pm$ 0.00	2854.28 $\pm$ 0.00	295.81 $\pm$ 3.10
DHN*	Universal Edges	7.52 $\pm$ 0.52	39.33 $\pm$ 3.30			
MAGNA	No Changes	8668.73 $\pm$ 1546.30	100.00 $\pm$ 0.00	38.56	2418.93 $\pm$ 10.87	661.83 $\pm$ 13.63
	All Inverse Edges	13686.67 $\pm$ 2143.50	100.00 $\pm$ 0.00	38.56	2598.70 $\pm$ 3.76	745.53 $\pm$ 0.05
	Output Fusion	22355.40 $\pm$ 3689.80	200.00 $\pm$ 0.00	77.12	2598.70 $\pm$ 3.76	745.53 $\pm$ 0.05
	Data Augmentation	25222.10 $\pm$ 1670.57	73.33 $\pm$ 6.60	38.56	2513.75 $\pm$ 567.06	753.21 $\pm$ 30.42
MAGNA*	Universal Edges	13686.67 $\pm$ 2143.50	100.00 $\pm$ 0.00	38.56	2598.70 $\pm$ 3.76	745.53 $\pm$ 0.05

Table 36: Wordnet link prediction invariance: Kendall- τ over 3 runs. Joint 3-var denotes one shared model trained on No Changes, All Inverse Edges, and Universal.

Model	Comparison	Kendall- τ	Kendall- τ @1	Kendall- τ @3
RGCN	No Changes vs All Inverse Edges	0.1363 ± 0.0154	0.1096 ± 0.0042	0.0308 ± 0.0670
	No Changes vs Universal Edges	0.1434 ± 0.0213	0.1049 ± 0.0042	0.0272 ± 0.0018
	All Inverse Edges vs Universal Edges	0.2919 ± 0.0246	0.6920 ± 0.0158	0.3677 ± 0.1295
	Joint 3-var: No Changes vs All Inverse Edges	0.5200 ± 0.0129	0.4869 ± 0.0285	0.2174 ± 0.0446
	Joint 3-var: No Changes vs Universal	0.4317 ± 0.0096	0.4151 ± 0.0270	0.1186 ± 0.0392
RGCN*	Joint 3-var: All Inverse Edges vs Universal	0.7798 ± 0.0027	0.6766 ± 0.0087	0.4341 ± 0.0189
	No Changes vs All Inverse Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	No Changes vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CMPNN	All Inverse Edges vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	Universal Edges vs All Inverse Edges	0.1402 ± 0.0623	0.7428 ± 0.0381	0.7294 ± 0.0393
	Universal Edges vs Unchanged	0.1376 ± 0.1054	0.7051 ± 0.0590	0.6987 ± 0.0621
	All Inverse Edges vs Unchanged	0.2811 ± 0.0683	0.8311 ± 0.0603	0.7781 ± 0.0298
	Output Fusion vs Universal Edges	0.4116 ± 0.0839	0.8080 ± 0.0486	0.8057 ± 0.0507
CMPNN*	Output Fusion vs All Inverse Edges	0.3407 ± 0.0653	0.8456 ± 0.0492	0.7726 ± 0.0346
	Output Fusion vs Unchanged	0.3795 ± 0.0869	0.8264 ± 0.0597	0.7951 ± 0.0460
	No Changes vs All Inverse Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
DHN	No Changes vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	All Inverse Edges vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	No Changes vs All Inverse Edges	0.0548 ± 0.0171	0.5510 ± 0.0039	0.4835 ± 0.0047
	No Changes vs Universal Edges	0.1070 ± 0.2470	0.6190 ± 0.0052	0.6487 ± 0.0109
	All Inverse Edges vs Universal Edges	0.0589 ± 0.0079	0.4255 ± 0.0097	0.4340 ± 0.0111
DHN*	Joint 3-var: No Changes vs All Inverse Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	Joint 3-var: No Changes vs Universal Edges	0.2298 ± 0.1720	-0.4732 ± 0.1214	-0.2603 ± 0.0985
	Joint 3-var: All Inverse Edges vs Universal Edges	0.2298 ± 0.1720	-0.4732 ± 0.1214	-0.2603 ± 0.0985
MAGNA	No Changes vs All Inverse Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	No Changes vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	All Inverse Edges vs Universal Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
	No Changes vs All Inverse Edges	0.3078 ± 0.0132	0.7807 ± 0.0139	0.7559 ± 0.0032
MAGNA*	No Changes vs Output Fusion	0.6610 ± 0.0052	0.8669 ± 0.0172	0.6871 ± 0.0093
	All Inverse Edges vs Output Fusion	0.6214 ± 0.0074	0.8599 ± 0.0141	0.6943 ± 0.0062
	Data Augmentation: No Changes vs All Inverse Edges	0.9609 ± 0.0028	0.9740 ± 0.0043	0.9646 ± 0.0030
MAGNA*	No Changes vs All Inverse Edges	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

Table 37: Freebase node classification results: accuracy, precision, recall, micro-f1 and macro-f1 over 3 runs.

Model	Variant	Accuracy	Precision	Recall	Micro-F1	Macro-F1
RGCN	Unchanged	0.6864 $\pm$ 0.0022	0.6304 $\pm$ 0.0075	0.5364 $\pm$ 0.0049	0.6864 $\pm$ 0.0022	0.5543 $\pm$ 0.0050
	Exact 2	0.6956 $\pm$ 0.0047	0.6427 $\pm$ 0.0138	0.5492 $\pm$ 0.0063	0.6956 $\pm$ 0.0047	0.5757 $\pm$ 0.0071
	Unchanged + Exact 2	0.6950 $\pm$ 0.0042	0.6251 $\pm$ 0.0033	0.5366 $\pm$ 0.0041	0.6950 $\pm$ 0.0042	0.5663 $\pm$ 0.0055
	Output Fusion	0.6973 $\pm$ 0.0051	0.6456 $\pm$ 0.0072	0.5439 $\pm$ 0.0069	0.6973 $\pm$ 0.0051	0.5683 $\pm$ 0.0101
RGCN*	Universal	0.6956 $\pm$ 0.0047	0.6427 $\pm$ 0.0138	0.5492 $\pm$ 0.0063	0.6956 $\pm$ 0.0047	0.5757 $\pm$ 0.0071
MAGNN	Unchanged	0.5227 $\pm$ 0.0011	0.3170 $\pm$ 0.0172	0.1783 $\pm$ 0.0029	0.5227 $\pm$ 0.0011	0.1602 $\pm$ 0.0049
	Exact 2	0.5217 $\pm$ 0.0019	0.3011 $\pm$ 0.0538	0.1781 $\pm$ 0.0041	0.5217 $\pm$ 0.0019	0.1599 $\pm$ 0.0075
	backup Output Fusion	0.5227 $\pm$ 0.0006	0.3060 $\pm$ 0.0052	0.1768 $\pm$ 0.0014	0.5227 $\pm$ 0.0006	0.1573 $\pm$ 0.0023
	Output Fusion	0.3855 $\pm$ 0.0044	0.1092 $\pm$ 0.0001	0.1499 $\pm$ 0.0005	0.3855 $\pm$ 0.0044	0.1224 $\pm$ 0.0007
MAGNN* Union	Unchanged	0.6000 $\pm$ 0.0057	0.5857 $\pm$ 0.0515	0.3013 $\pm$ 0.0171	0.6000 $\pm$ 0.0057	0.3266 $\pm$ 0.0193
	Exact 2	0.6000 $\pm$ 0.0057	0.5857 $\pm$ 0.0515	0.3013 $\pm$ 0.0171	0.6000 $\pm$ 0.0057	0.3266 $\pm$ 0.0193
	backup Output Fusion	0.6000 $\pm$ 0.0057	0.5857 $\pm$ 0.0515	0.3013 $\pm$ 0.0171	0.6000 $\pm$ 0.0057	0.3266 $\pm$ 0.0193
	Output Fusion	0.6105 $\pm$ 0.0026	0.5991 $\pm$ 0.0124	0.3208 $\pm$ 0.0093	0.6105 $\pm$ 0.0026	0.3535 $\pm$ 0.0139
MAGNN* Intersection	Unchanged	0.6151 $\pm$ 0.0025	0.5871 $\pm$ 0.1001	0.3230 $\pm$ 0.0161	0.6151 $\pm$ 0.0025	0.3528 $\pm$ 0.0204
	Exact 2	0.6135 $\pm$ 0.0011	0.6169 $\pm$ 0.0852	0.3190 $\pm$ 0.0104	0.6135 $\pm$ 0.0011	0.3521 $\pm$ 0.0195
	backup Output Fusion	0.6143 $\pm$ 0.0013	0.5930 $\pm$ 0.0952	0.3212 $\pm$ 0.0134	0.6143 $\pm$ 0.0013	0.3528 $\pm$ 0.0204
	Output Fusion	0.6076 $\pm$ 0.0062	0.5738 $\pm$ 0.0932	0.3232 $\pm$ 0.0102	0.6076 $\pm$ 0.0062	0.3567 $\pm$ 0.0162
SlotGAT	Unchanged	0.6530 $\pm$ 0.0014	0.5724 $\pm$ 0.0220	0.4313 $\pm$ 0.0135	0.6530 $\pm$ 0.0014	0.4567 $\pm$ 0.0114
	Exact 2	0.6394 $\pm$ 0.0128	0.5319 $\pm$ 0.0266	0.4032 $\pm$ 0.0375	0.6394 $\pm$ 0.0128	0.4320 $\pm$ 0.0325
	Output Fusion	0.6579 $\pm$ 0.0062	0.5317 $\pm$ 0.0490	0.4207 $\pm$ 0.0247	0.6579 $\pm$ 0.0062	0.4471 $\pm$ 0.0250
SlotGAT*	Universal	0.6394 $\pm$ 0.0128	0.5319 $\pm$ 0.0266	0.4032 $\pm$ 0.0375	0.6394 $\pm$ 0.0128	0.4320 $\pm$ 0.0325
SeHGNN	Unchanged	0.6321 $\pm$ 0.0114	0.5746 $\pm$ 0.0315	0.4089 $\pm$ 0.0334	0.6321 $\pm$ 0.0114	0.4473 $\pm$ 0.0283
	Exact 2	0.6269 $\pm$ 0.0086	0.5454 $\pm$ 0.0280	0.4211 $\pm$ 0.0287	0.6269 $\pm$ 0.0086	0.4497 $\pm$ 0.0284
	Output Fusion	0.6373 $\pm$ 0.0120	0.5665 $\pm$ 0.0404	0.4211 $\pm$ 0.0327	0.6373 $\pm$ 0.0120	0.4576 $\pm$ 0.0298
SeHGNN* + fullK	Unchanged	0.6269 $\pm$ 0.0086	0.5454 $\pm$ 0.0280	0.4211 $\pm$ 0.0287	0.6269 $\pm$ 0.0086	0.4497 $\pm$ 0.0284
	Exact 2	0.6269 $\pm$ 0.0086	0.5454 $\pm$ 0.0280	0.4211 $\pm$ 0.0287	0.6269 $\pm$ 0.0086	0.4497 $\pm$ 0.0284
	Output Fusion	0.6269 $\pm$ 0.0086	0.5454 $\pm$ 0.0280	0.4211 $\pm$ 0.0287	0.6269 $\pm$ 0.0086	0.4497 $\pm$ 0.0284
SeHGNN* + restrictedK	Unchanged	0.5644 $\pm$ 0.0136	0.4184 $\pm$ 0.0925	0.3410 $\pm$ 0.0221	0.5644 $\pm$ 0.0136	0.3396 $\pm$ 0.0233
	Exact 2	0.5644 $\pm$ 0.0136	0.4184 $\pm$ 0.0925	0.3410 $\pm$ 0.0221	0.5644 $\pm$ 0.0136	0.3396 $\pm$ 0.0233
	Output Fusion	0.5644 $\pm$ 0.0136	0.4184 $\pm$ 0.0925	0.3410 $\pm$ 0.0221	0.5644 $\pm$ 0.0136	0.3396 $\pm$ 0.0233
PathNet	Unchanged	0.5508 $\pm$ 0.0062	0.3347 $\pm$ 0.0356	0.3324 $\pm$ 0.0260	0.5508 $\pm$ 0.0062	0.2973 $\pm$ 0.0044
	Exact 2	0.5414 $\pm$ 0.0083	0.3953 $\pm$ 0.0697	0.3071 $\pm$ 0.0275	0.5414 $\pm$ 0.0083	0.2877 $\pm$ 0.0112
	backup Output Fusion	0.5502 $\pm$ 0.0072	0.3402 $\pm$ 0.0306	0.3266 $\pm$ 0.0284	0.5502 $\pm$ 0.0072	0.2958 $\pm$ 0.0093
	Output Fusion	0.5133 $\pm$ 0.0263	0.2317 $\pm$ 0.1554	0.1755 $\pm$ 0.0240	0.5133 $\pm$ 0.0263	0.1492 $\pm$ 0.0424
PathNet* Union	Unchanged	0.5470 $\pm$ 0.0137	0.3909 $\pm$ 0.0894	0.3195 $\pm$ 0.0330	0.5470 $\pm$ 0.0137	0.2896 $\pm$ 0.0159
	Exact 2	0.5470 $\pm$ 0.0137	0.3909 $\pm$ 0.0894	0.3195 $\pm$ 0.0330	0.5470 $\pm$ 0.0137	0.2896 $\pm$ 0.0159
	backup Output Fusion	0.5470 $\pm$ 0.0137	0.3909 $\pm$ 0.0894	0.3195 $\pm$ 0.0330	0.5470 $\pm$ 0.0137	0.2896 $\pm$ 0.0159
	Output Fusion	0.5202 $\pm$ 0.0289	0.2075 $\pm$ 0.1180	0.1812 $\pm$ 0.0278	0.5202 $\pm$ 0.0289	0.1543 $\pm$ 0.0452
PathNet* Intersection	Unchanged	0.5424 $\pm$ 0.0126	0.3211 $\pm$ 0.0455	0.3277 $\pm$ 0.0564	0.5424 $\pm$ 0.0126	0.2844 $\pm$ 0.0209
	Exact 2	0.5424 $\pm$ 0.0126	0.3211 $\pm$ 0.0455	0.3277 $\pm$ 0.0564	0.5424 $\pm$ 0.0126	0.2844 $\pm$ 0.0209
	backup Output Fusion	0.5424 $\pm$ 0.0126	0.3211 $\pm$ 0.0455	0.3277 $\pm$ 0.0564	0.5424 $\pm$ 0.0126	0.2844 $\pm$ 0.0209
	Output Fusion	0.5188 $\pm$ 0.0284	0.2003 $\pm$ 0.1160	0.1805 $\pm$ 0.0277	0.5188 $\pm$ 0.0284	0.1537 $\pm$ 0.0456
DHN	Unchanged	0.5097 $\pm$ 0.0187	0.1393 $\pm$ 0.0655	0.1806 $\pm$ 0.0368	0.5097 $\pm$ 0.0187	0.1504 $\pm$ 0.0550
	Exact 2	0.4938 $\pm$ 0.0063	0.1704 $\pm$ 0.0619	0.1626 $\pm$ 0.0047	0.4938 $\pm$ 0.0063	0.1377 $\pm$ 0.0082
DHN*	Universal	0.4938 $\pm$ 0.0063	0.1704 $\pm$ 0.0619	0.1626 $\pm$ 0.0047	0.4938 $\pm$ 0.0063	0.1377 $\pm$ 0.0082
MAGNA	Unchanged	0.6137 $\pm$ 0.0119	0.5066 $\pm$ 0.0643	0.4142 $\pm$ 0.0397	0.6137 $\pm$ 0.0119	0.4370 $\pm$ 0.0314
	Exact 2	0.6124 $\pm$ 0.0067	0.4915 $\pm$ 0.0092	0.4080 $\pm$ 0.0208	0.6124 $\pm$ 0.0067	0.4267 $\pm$ 0.0136
	Output Fusion	0.6237 $\pm$ 0.0112	0.5245 $\pm$ 0.0046	0.3976 $\pm$ 0.0070	0.6237 $\pm$ 0.0112	0.4243 $\pm$ 0.0083
	Data Augmentation	0.5853 $\pm$ 0.0115	0.4896 $\pm$ 0.0436	0.3908 $\pm$ 0.0161	0.5853 $\pm$ 0.0115	0.4065 $\pm$ 0.0177
MAGNA*	Universal	0.6124 $\pm$ 0.0067	0.4915 $\pm$ 0.0092	0.4080 $\pm$ 0.0208	0.6124 $\pm$ 0.0067	0.4267 $\pm$ 0.0136

Table 38: Freebase node classification training time and epochs over 3 runs.

Model	Variant	Train Time (s)	Epochs	Model Parameter (MiB)	Peak Train GPU (MiB)	Peak Inference GPU (MiB)
RGCN	Unchanged	22.58 $\pm$ 0.94	100.00 $\pm$ 0.00	89.06	6962.87 $\pm$ 20.02	412.72 $\pm$ 0.57
	Exact 2	135.96 $\pm$ 0.33	100.00 $\pm$ 0.00	89.06	23543.05 $\pm$ 0.00	13934.76 $\pm$ 0.00
	Unchanged + Exact 2	1322.16 $\pm$ 114.47	92.33 $\pm$ 8.02	44.53	5135.09 $\pm$ 0.00	4422.94 $\pm$ 1223.96
	Output Fusion	158.54 $\pm$ 1.27	200.00 $\pm$ 0.00	178.12	23543.05 $\pm$ 0.00	13934.76 $\pm$ 0.00
RGCN*	Universal	14959.50 $\pm$ 858.68	8.33 $\pm$ 0.47	89.06	23543.05 $\pm$ 0.00	13934.76 $\pm$ 0.00
MAGNN	Unchanged	36.48 $\pm$ 5.18	91.00 $\pm$ 12.73	44.39 $\pm$ 0.00	8725.42 $\pm$ 0.27	5402.47 $\pm$ 1.12
	Exact 2	25.83 $\pm$ 1.10	75.67 $\pm$ 3.09	44.39 $\pm$ 0.00	8405.02 $\pm$ 1.14	5342.81 $\pm$ 0.00
	backup Output Fusion	62.31 $\pm$ 6.03	166.67 $\pm$ 14.82	88.79 $\pm$ 0.00	8725.42 $\pm$ 0.27	10745.28 $\pm$ 1.12
	Output Fusion	28.80 $\pm$ 1.84	30.67 $\pm$ 1.89	44.80 $\pm$ 0.00	17638.17 $\pm$ 1.89	8615.35 $\pm$ 0.68
MAGNN* Union	Unchanged	991.84 $\pm$ 406.98	7.33 $\pm$ 0.58	44.27 $\pm$ 0.00	298.38 $\pm$ 1.11	319.05 $\pm$ 1.76
	Exact 2	924.77 $\pm$ 397.22	7.33 $\pm$ 0.58	44.27 $\pm$ 0.00	298.38 $\pm$ 1.11	319.05 $\pm$ 1.76
	backup Output Fusion	1916.61 $\pm$ 796.60	14.67 $\pm$ 1.15	88.54 $\pm$ 0.00	298.38 $\pm$ 1.11	638.09 $\pm$ 3.51
	Output Fusion	972.70 $\pm$ 57.58	6.33 $\pm$ 0.47	44.30 $\pm$ 0.00	287.73 $\pm$ 0.01	191.25 $\pm$ 0.00
MAGNN* Intersection	Unchanged	766.23 $\pm$ 378.25	8.33 $\pm$ 1.53	44.26 $\pm$ 0.00	294.49 $\pm$ 1.25	317.06 $\pm$ 2.99
	Exact 2	704.13 $\pm$ 435.14	9.00 $\pm$ 1.00	44.26 $\pm$ 0.00	294.49 $\pm$ 1.25	317.80 $\pm$ 2.78
	backup Output Fusion	1470.36 $\pm$ 806.31	17.33 $\pm$ 2.31	88.52 $\pm$ 0.00	294.49 $\pm$ 1.25	634.86 $\pm$ 5.63
	Output Fusion	2156.64 $\pm$ 885.03	8.33 $\pm$ 3.21	44.26 $\pm$ 0.00	287.58 $\pm$ 0.01	191.22 $\pm$ 0.00
SlotGAT	Unchanged	74.88 $\pm$ 74.01	71.0 $\pm$ 6.4	123.22	29312.95 $\pm$ 1326.19	22075.14 $\pm$ 1326.65
	Exact 2	3363.55 $\pm$ 216.99	95.7 $\pm$ 6.1	123.22	65226.18 $\pm$ 0.33	42091.24 $\pm$ 0.03
	Output Fusion	3438.43 $\pm$ 291.00	166.7 $\pm$ 12.5	246.44	65226.18 $\pm$ 0.33	42091.24 $\pm$ 0.03
SlotGAT*	Universal	3363.55 $\pm$ 216.99	95.7 $\pm$ 6.1	123.22	65226.18 $\pm$ 0.33	42091.24 $\pm$ 0.03
SeHGNN	Unchanged	129.34 $\pm$ 1.36	85.67 $\pm$ 3.79	576.17	12037.14 $\pm$ 1.46	8711.16 $\pm$ 2.59
	Exact 2	332.87 $\pm$ 18.17	86.67 $\pm$ 3.79	576.17	14253.64 $\pm$ 0.56	13141.16 $\pm$ 3.19
	Output Fusion	462.22 $\pm$ 19.49	172.33 $\pm$ 7.57	1152.34	14253.64 $\pm$ 0.56	21852.33 $\pm$ 2.82
SeHGNN* + fullK	Universal	333.25 $\pm$ 11.36	86.67 $\pm$ 3.79	576.17	14253.95 $\pm$ 1.91	13142.60 $\pm$ 6.51
PathNet	Unchanged	0.5508 $\pm$ 0.0062	0.3347 $\pm$ 0.0356	0.3324 $\pm$ 0.0260	0.5508 $\pm$ 0.0062	0.2973 $\pm$ 0.0044
	Exact 2	0.5414 $\pm$ 0.0083	0.3953 $\pm$ 0.0697	0.3071 $\pm$ 0.0275	0.5414 $\pm$ 0.0083	0.2877 $\pm$ 0.0112
	backup Output Fusion	0.5502 $\pm$ 0.0072	0.3402 $\pm$ 0.0306	0.3266 $\pm$ 0.0284	0.5502 $\pm$ 0.0072	0.2958 $\pm$ 0.0093
	Output Fusion	6.44 $\pm$ 0.16	9.67 $\pm$ 0.47	44.19 $\pm$ 0.00	7561.79 $\pm$ 0.05	2238.50 $\pm$ 0.48
PathNet* Union	Unchanged	0.5408 $\pm$ 0.0110	0.4255 $\pm$ 0.0502	0.3077 $\pm$ 0.0354	0.5408 $\pm$ 0.0110	0.2868 $\pm$ 0.0214
	Exact 2	0.5403 $\pm$ 0.0091	0.4302 $\pm$ 0.0519	0.3088 $\pm$ 0.0275	0.5403 $\pm$ 0.0091	0.2866 $\pm$ 0.0184
	backup Output Fusion	0.5395 $\pm$ 0.0108	0.4296 $\pm$ 0.0519	0.3061 $\pm$ 0.0346	0.5395 $\pm$ 0.0108	0.2854 $\pm$ 0.0209
	Output Fusion	6.20 $\pm$ 0.13	9.67 $\pm$ 0.47	44.18 $\pm$ 0.00	7561.79 $\pm$ 0.05	2238.50 $\pm$ 0.48
PathNet* Intersection	Unchanged	0.5416 $\pm$ 0.0100	0.4357 $\pm$ 0.0458	0.3104 $\pm$ 0.0242	0.5416 $\pm$ 0.0100	0.2911 $\pm$ 0.0126
	Exact 2	0.5414 $\pm$ 0.0083	0.3953 $\pm$ 0.0697	0.3071 $\pm$ 0.0275	0.5414 $\pm$ 0.0083	0.2877 $\pm$ 0.0112
	backup Output Fusion	0.5443 $\pm$ 0.0103	0.4025 $\pm$ 0.0634	0.3104 $\pm$ 0.0266	0.5443 $\pm$ 0.0103	0.2909 $\pm$ 0.0113
	Output Fusion	6.03 $\pm$ 0.07	9.67 $\pm$ 0.47	44.18 $\pm$ 0.00	7561.79 $\pm$ 0.05	2238.50 $\pm$ 0.48
DHN	Unchanged	5.45 $\pm$ 0.47	200.00 $\pm$ 0.00			
	Exact 2	233.28 $\pm$ 0.27	200.00 $\pm$ 0.00			
DHN*	Universal	233.28 $\pm$ 0.27	200.00 $\pm$ 0.00			
MAGNA	Unchanged	80.67 $\pm$ 1.74	32.67 $\pm$ 0.94	44.29	5908.55 $\pm$ 2070.95	1007.18 $\pm$ 1.99
	Exact 2	650.45 $\pm$ 35.65	36.00 $\pm$ 1.63	44.29	24274.14 $\pm$ 0.48	11208.44 $\pm$ 0.49
	Output Fusion	731.12 $\pm$ 37.39	68.67 $\pm$ 2.57	88.58	24274.14 $\pm$ 0.48	11208.44 $\pm$ 0.49
	Data Augmentation	861.41 $\pm$ 39.12	36.00 $\pm$ 1.63	44.29	22516.39 $\pm$ 700.63	9289.95 $\pm$ 0.01
MAGNA*	Universal	650.45 $\pm$ 35.65	36.00 $\pm$ 1.63	44.29	24274.14 $\pm$ 0.48	11208.44 $\pm$ 0.49

Table 39: Freebase node classification invariance: Kendall- τ over 3 runs.

Model	Comparison	Kendall- τ
RGCN	Unchanged vs Exact 2	0.7622 ± 0.0243
	Unchanged + Exact 2: Unchanged vs Exact 2	0.9810 ± 0.0029
	Unchanged vs Output Fusion	0.8839 ± 0.0134
	Exact 2 vs Output Fusion	0.8786 ± 0.0220
RGCN*	Unchanged vs Exact 2	1.0000 ± 0.0000
MAGNN	Unchanged vs. Exact 2	0.7447 ± 0.0337
	Unchanged vs. backup Output Fusion	0.8769 ± 0.0342
	Exact 2 vs. backup Output Fusion	0.8678 ± 0.0635
	Unchanged vs. Output Fusion	0.9981 ± 0.0009
	Exact 2 vs. Output Fusion	0.9981 ± 0.0011
MAGNN* Union	Unchanged vs. Exact 2	1.0000 ± 0.0000
	Unchanged vs. backup Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. backup Output Fusion	1.0000 ± 0.0000
	Unchanged vs. Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. Output Fusion	1.0000 ± 0.0000
MAGNN* Intersection	Unchanged vs. Exact 2	0.9031 ± 0.1678
	Unchanged vs. backup Output Fusion	0.9383 ± 0.1069
	Exact 2 vs. backup Output Fusion	0.9648 ± 0.0609
	Unchanged vs. Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. Output Fusion	1.0000 ± 0.0000
SlotGAT	Unchanged vs Exact 2	0.4248 ± 0.0339
	Unchanged vs Output Fusion	0.7130 ± 0.0265
	Exact 2 vs Output Fusion	0.7026 ± 0.0092
SlotGAT*	Unchanged vs Exact 2	1.0000 ± 0.0000
SeHGNN	Unchanged vs Exact 2	0.7255 ± 0.0143
	Unchanged vs Output Fusion	0.8622 ± 0.0036
	Exact 2 vs Output Fusion	0.8633 ± 0.0134
SeHGNN* + fullK	Unchanged vs Exact 2	1.0000 ± 0.0000
SeHGNN* + restrictedK	Unchanged vs Exact 2	1.0000 ± 0.0000
PathNet	Unchanged vs. Exact 2	0.8859 ± 0.0280
	Unchanged vs. backup Output Fusion	0.9436 ± 0.0136
	Exact 2 vs. backup Output Fusion	0.9424 ± 0.0145
	Unchanged vs. Output Fusion	0.9871 ± 0.0022
	Exact 2 vs. Output Fusion	0.9861 ± 0.0033
PathNet* Union	Unchanged vs. Exact 2	1.0000 ± 0.0000
	Unchanged vs. backup Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. backup Output Fusion	1.0000 ± 0.0000
	Unchanged vs. Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. Output Fusion	1.0000 ± 0.0000
PathNet* Intersection	Unchanged vs. Exact 2	1.0000 ± 0.0000
	Unchanged vs. backup Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. backup Output Fusion	1.0000 ± 0.0000
	Unchanged vs. Output Fusion	1.0000 ± 0.0000
	Exact 2 vs. Output Fusion	1.0000 ± 0.0000
DHN	Unchanged vs Exact 2	0.6615 ± 0.1064
DHN*	Unchanged vs Exact 2	1.0000 ± 0.0000
MAGNA	Unchanged vs Exact 2	0.6702 ± 0.0304
	Unchanged vs Output Fusion	0.8252 ± 0.0077
	Exact 2 vs Output Fusion	0.8089 ± 0.0220
	Data Augmentation: Unchanged vs Exact 2	0.8502 ± 0.0162
MAGNA*	Unchanged vs Exact 2	1.0000 ± 0.0000